

4.2. Immunization

1. Complete Freund's adjuvant.
2. Incomplete Freund's adjuvant.

5. Method

5.1. Coupling of Peptides to Carrier Proteins

Several carrier proteins have been used successfully including bovine serum albumin, ovalbumin, thyroglobulin, and keyhole limpet hemocyanin (KLH). A recent review of anti-peptide antibodies (16) lists almost two hundred examples, and one carrier clearly predominates. Keyhole limpet hemocyanin has long been the immunologist's choice as a general antigen to produce humoral antibodies, and it is frequently favored as a carrier for synthetic peptides. Hemocyanins are giant molecules, up to 9,000,000 mol wt, with quite repetitious structures. Many moles of peptide can be reacted per mole of KLH. All the other carriers have, however, been used with good results.

The methods available for coupling synthetic peptides to carriers are also quite varied, but can be classified depending upon the functional group of the peptide that is employed for the reaction. Most, although not all, peptides used as immunogens are synthesized with free amino and carboxy termini, and these are most commonly employed for coupling. This is not only because there are several methods worked out for primary amines and carboxylic groups, but also because it is probably not a good thing to modify amino acids within a sequence since it will make the structure less like that of the parent molecule. Glutaraldehyde is the most commonly used reagent for coupling peptides via

amine groups. Most papers refer to the method reported by Kagan and Glick (17), but I have found this reference rather difficult to find, so I present my method below, which is a minor modification of theirs.

This makes enough immunogen for a series of three injections into two rabbits.

1. Dialyze the KLH against PBS (normally 1 mL of KLH solution against 2 L of PBS for at least 2 h) without sodium azide, and then measure the protein concentration. A quick way of doing this is to dilute a small aliquot into 50 mM sodium borate buffer, pH 9.0, and measure the optical density of the solution at 280 nm. A 1.0 mg/mL solution of hemocyanin in this buffer gives 1.4 AU. Do not measure at neutral pH, since hemocyanin gives significant light scattering unless it is dissociated by acid or alkaline buffers.
2. Dissolve 7.5 mg of peptide in 100 mM sodium phosphate buffer, pH 8.0, mix with 7.5 mg of KLH, and make up to 0.5 mL with buffer.
3. Add 5 μ L of 25% glutaraldehyde, mix well, and leave for 15 min at room temperature. The mixture may appear cloudy during this period.
4. Add a further 2.5 μ L of glutaraldehyde and leave for 15 min.
5. Add 100 μ L of 1 M glycine, pH 6.0, and leave for 10 min to quench the glutaraldehyde. The reaction mixture will change to yellow to brown color. Store in aliquots at -20°C until use.

Other types of coupling reactions employ EDIC (1-ethyl-3-(3-dimethylaminopropyl)carbodiimide), which reacts with amine and carboxylic acid groups (18), MBS (*m*-maleimidobenzoic acid *N*-hydroxysuccin-

imide ester), which reacts with free sulfhydryl groups (19), and *bis*-diazobenzidine, which reacts with tyrosine residues (20).

5.2. Immunization

1. Immunization protocols are very much the acquired preference of the experimenter. I suggest the following (for rabbits), since I have found it simple and reliable. I also use the same schedule for mice and rats, but employ about one tenth the amount of immunogen for mice and one fifth for rats.
2. First take a preimmune bleed. From a medium-sized rabbit (3–4 kg) about 20–25 mL of *serum* and from a mouse 50–150 μ L would be ideal. Do not forget that this is the one thing that you will never be able to obtain again once you have immunized.
3. Day 1. Mix one third of the immunogen (200 μ L from the glutaraldehyde protocol) with 0.8 mL of PBS and 1.0 mL of *complete* Freund's adjuvant to a stiff paste, and inject each animal at multiple sites (about 2–4) subcutaneously with a total of 1.0 mL of the mixture.
4. Day 14. Repeat as for day 1, but use *incomplete* Freund's adjuvant.
5. Day 28. Repeat as for day 14.
6. Day 35. Test bleed. 10–20 mL of *serum* for rabbits, 50–100 μ L for mice.
7. Day 49. Test bleed.

6. Notes

1. I recommend ELISA for screening for anti-peptide antibodies, since the plates can be coated with

peptide alone, thereby allowing the anti-peptide titer to be measured without any contribution from anti-carrier (usually a lot), or anti-crosslinker (sometimes present) response. The general procedure as applied to screening for hybridomas has been published (*see* vol. 1 of this series and ref. 21). For measuring serum titers, I normally measure in quadruplicate, starting at a dilution of serum of 1/50 and going down in five steps of fourfold dilution to 1/12,800. This will be a good range for most antisera, although high titer ones may need further dilutions. With most peptides, 50% of color development should be reached at a dilution of about 1/3200, although this is a considerable generalization. The anti-peptide antibodies can now be tested for their ability to recognize the protein of interest by immunoprecipitation or western blotting (*see* vol. 1 and this volume). Some will work in one or both assays, some in neither. The more peptides that have been used, the more likely it is that anti-protein reactive antibodies will be obtained.

2. Beware that complete Freund's adjuvant will cause just as good immune response in humans as it will in mice or rabbits. Wear gloves and eye protection whenever handling it.
3. Several companies sell KLH as a lyophilized powder. I have heard from others that this is virtually impossible to dissolve without obvious large aggregates. To avoid this problem, I would recommend buying it as a solution or slurry.

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Chapter 25

Production of Antibodies Using Proteins in Gel Bands

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1. Introduction

A number of methods for preparing proteins as antigens have been described (1). These include solubilization of protein samples in buffered solutions (2), solubilization of nitrocellulose filters to which proteins have been adsorbed (3), and emulsification of protein bands in polyacrylamide gels for direct injections (4–8). The latter technique can be used to immunize mice or rabbits for production of antisera or to immunize mice for production of monoclonal antibodies (9–11). This approach is particularly advantageous when protein purification by other means is not practical, as in the case of proteins insoluble without detergent. A further advantage of this method is an enhancement of the immune response, since polyacrylamide helps to retain the antigen in the animal and so acts as an adjuvant (7). The

use of the protein directly in the gel band (without elution) is also helpful when only small amounts of protein are available. For instance, in this laboratory, we routinely immunize mice with 5–10 μg total protein using this method; we have not determined the lower limit of total protein that can be used to immunize rabbits. Since polyacrylamide is also highly immunogenic, however, it is necessary in some cases to affinity-purify the desired antibodies from the resulting antiserum or to produce hybridomas that can be screened selectively for the production of specific antibodies, to obtain the desired reagent.

2. Materials

1. Gel electrophoresis apparatus; acid-urea polyacrylamide gel or SDS-polyacrylamide gel.
2. Staining solution: 0.1% Coomassie Brilliant Blue-R (Sigma B-0630) in 50% (v/v) methanol/10% (v/v) acetic acid.
3. Destaining solution: 5% (v/v) methanol/7% (v/v) acetic acid.
4. 2% (v/v) glutaraldehyde (Sigma G-6257).
5. Transilluminator.
6. Sharp razor blades.
7. Conical plastic centrifuge tubes and ethanol.
8. Lyophilizer and dry ice.
9. Plastic, disposable syringes (3- and 1-mL).
10. 18-Gage needles.
11. Spatula and weighing paper.
12. Freund's complete and Freund's incomplete adjuvants (Gibco Laboratories, Grand Island, NY).
13. Phosphate-buffered saline solution (PBS) (50 mM sodium phosphate, pH 7.25/150 mM sodium chloride).

14. Microemulsifying needle, 18-gage (Becton Dickinson, Rutherford, NJ).
15. Female Balb-c mice, 7–8 wk old, or New Zealand white rabbits.

3. Method

1. Following electrophoresis (*see* Note 1 in section 4), the gel is stained by gentle agitation in several volumes of staining solution for 30 min. The gel is partially destained by gentle agitation in several changes of destaining solution for 30–45 min. Proteins in the gel are then crosslinked by immersing the gel with gentle shaking in 2% glutaraldehyde for 45–60 min (12). This step minimizes loss of proteins during subsequent destaining steps and enhances the immunological response by polymerizing the proteins. The gel is then completely destained, usually overnight.
2. The gel is viewed on a transilluminator, and the bands of interest are cut out with a razor blade. The gel pieces are pushed to the bottom of a conical plastic centrifuge tube with a spatula and pulverized. The samples in the tubes are frozen in dry ice and lyophilized.
3. To prepare the dried polyacrylamide pieces for injection, a small portion of the dried material is lifted out of the tube with a spatula and placed on a small square of weighing paper. In dry climates it is useful to first wipe the outside of the tube with ethanol to reduce static electricity. The material is then gently tapped into the top of a 3-mL syringe to which is attached the microemulsifying needle (Fig. 1A). Keeping the syringe horizontal, 200 μ L of PBS solution is carefully introduced to the barrel of

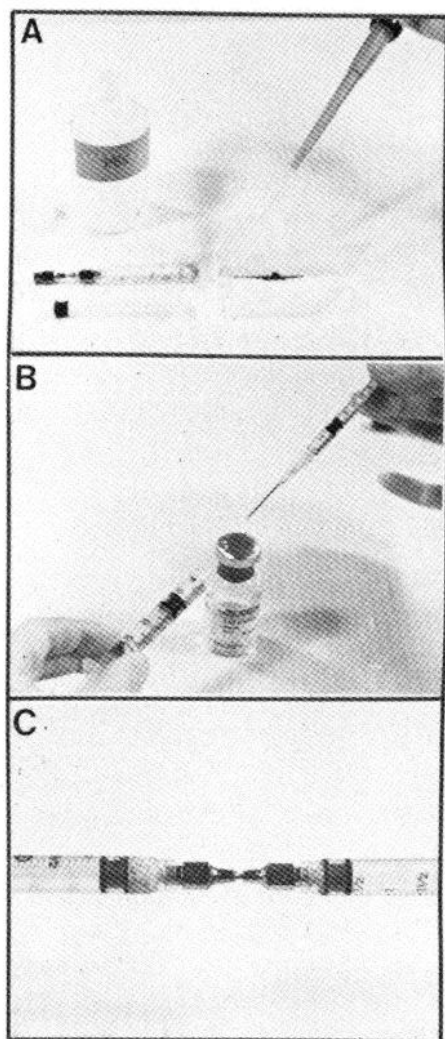


Fig. 1. Preparation of emulsion for immunizations. To prepare proteins in gel bands for injections, an emulsion of Freund's adjuvant and dried polyacrylamide pieces is prepared. (A) Dried polyacrylamide is resuspended in 200 μ L of PBS solution in the barrel of a 3-mL syringe to which is attached a microemulsifying needle. (B). Freund's adjuvant is transferred into the barrel of a second 3-mL syringe. (C). An emulsion is formed by mixing the contents of the two syringes through the microemulsifying needle.

the syringe, and the plunger is inserted. Next, 200 μ L of Freund's adjuvant is drawn into a 1-mL tuberculin syringe and transferred into the needle end of a second 3-mL syringe (Fig. 1B). This syringe is then attached to the free end of the microemulsifying needle. The two plungers are pushed alternatively to mix the components of the two syringes (Fig. 1C). These will form an emulsion within 15 min; it is generally extremely difficult to mix the material any further.

4. This mixture is injected intraperitoneally or subcutaneously into a female Balb-c mouse, or subcutaneously into the back of the neck of a rabbit (*see* refs. 13 and 14). Since the emulsion is very viscous, it is best to use 18-gage needles and to anesthetize the animals. For mice, subsequent injections are administered after 2 wk and after 3 more wk. If monoclonal antibodies are desired, the animals are sacrificed 3–4 d later, and the spleen cells are fused with myeloma cells (13 and *see* vol. 1 of this series). The immunization schedule for rabbits calls for subsequent injections after 1 mo or when serum titers start to diminish. Antiserum is obtained from either tail bleeds or eye bleeds from the immunized mice, or from ear bleeds from immunized rabbits. The antibodies are assayed by any of the standard techniques (*see* vol. 1 in this series).

4. Notes

1. We have produced antisera to protein bands in acetic acid–urea gels (15), Triton–acetic acid–urea gels (16,17), or SDS–polyacrylamide gels (18). In our experience, antibodies produced to proteins in one denaturing gel system will cross-react to those

same proteins fractionated in another denaturing gel system and will usually cross-react with the native protein. We have consistently obtained antibodies from animals immunized by these procedures.

2. It is extremely important that all glutaraldehyde be removed from the gel during the destaining washes, since any residual glutaraldehyde will be toxic to the animal. Residual glutaraldehyde can easily be detected by smell. It is equally important to remove all acetic acid during lyophilization. Monoacrylamide is also toxic, whereas polyacrylamide is not. We do observe, however, that approximately 50 mm² of polyacrylamide per injection is the maximum that a mouse can tolerate.
3. Freund's complete adjuvant is used for the initial immunization; Freund's incomplete adjuvant is used for all subsequent injections. The mycobacteria in complete adjuvant enhance the immune response by provoking a local inflammation. Additional doses of mycobacteria may be toxic.

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Chapter 26

Purification of Immunoglobulins Using Protein A–Sepharose

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1. Introduction

Most immunochemical techniques used in molecular biology rely on a specific class of antibodies, immunoglobulin G (IgG). In mammals, IgG antibodies are produced during the secondary humoral response and contribute about 80% of the serum immunoglobulin. Factors affecting the experimental production of antibodies have been discussed in detail (1). In this chapter we describe a method for the rapid and efficient purification of IgG to near homogeneity from rabbit serum. This is achieved by affinity chromatography using protein A.

Protein A is a cell wall component of *Staphylococcus aureus* (Cowan 1 strain). This protein specifically binds the crystallizable fragment (Fc) region of IgG from many mammals (2). Because of this, protein A can be

used to separate IgG from other serum proteins. The principal disadvantage of the technique is that it is not suitable for purifying IgG from all sources. The affinity of protein A for IgG varies among species and subclasses. In addition, other classes of immunoglobulin from some species may bind to this ligand, although in practice this is rarely a major problem. The interaction between protein A and immunoglobulins from several important species is summarized in Table 1.

There are obvious advantages to using purified IgG rather than crude serum as a source of antibodies, since other proteins present in antisera may interfere in many immunological methods. For example, contaminating proteins reduce the amount of antibody that can be immobilized on a solid support in enzyme-linked immunosorbant assays (*see* vol. 1 of this series, Chapter 38). This restricts the detection limit of the assay. In addition to such general protein effects, specific contaminants may interfere in several techniques. Digestion of proteins by proteases present in antisera may confound the identification of immunoprecipitated polypeptides. Alternatively, measurement of enzyme activity in immuno-inactivation studies is often hindered by contaminating enzymes, NADH oxidase in particular, that directly interfere with the enzyme assay. These examples indicate that the specific problems associated with using crude sera vary depending on the application. Most of these problems are overcome by using purified IgG.

2. Materials

1. Rabbit antiserum: The production of suitable antisera has been described in a previous chapter (*see* vol. 1 in this series, Chapter 32).
2. Protein A-Sepharose Cl-4B supplied commercially as freeze-dried beads.

Table 1
Affinity of Protein A for Immunoglobulins from Various Species^a

Species	Serum IgG level, mg/mL	Binding		
		Strong	Weak	Unreactive
Rabbit	5	IgG		
Human	12	IgG1 ^b , 2, 4 IgA2 IgM (some)		IgG3
Guinea pig	6	IgG1, 2 ^b		
Mouse	7	IgG2a, 2b, 3	IgG1 ^b IgM	
Pig	18	IgG IgM (some) IgA (some)		
Goat		IgG2	IgG1	
Sheep		IgG2		IgG1
Dog	9	IgG IgM (some) IgA (some)		
Rat	16	IgG1, 2c	IgG2b	IgG2a
Cow	20	IgG2		IgG1

^aThe values for serum IgG levels are approximate, and significant variation may occur among individuals. Immunoglobulins failing to bind to protein A at pH 8.0 are described as unreactive. Immunoglobulins that bind to protein A, but are released above pH 6.0, are considered weakly bound. Strong binding indicates retention below pH 6.0 (data from refs. 2-7).

^bDenotes the major IgG subclass.

3. Buffers

0.1M Sodium phosphate, pH 8.0 (1 L).

0.1M Sodium citrate, pH 3.5 (100 mL).

0.1M Sodium citrate, pH 3.0 (100 mL).

1.0M Tris-HCl, pH 8.5 (50 mL).

The pH of the above solutions are adjusted at 20°C. These buffers are stable, but are susceptible to microbial contamination. They should be stored at 4°C and discarded if contamination occurs.

4. Saturated ammonium sulfate: add 80 g of $(\text{NH}_4)_2\text{SO}_4$ (enzyme grade—low in heavy metals) to 100 mL of water and stir overnight at 25°C. The solution, which should contain some undissolved crystals, is adjusted to pH 7.8 using NH_4OH (see Note 1 in section 4). This solution is stable for several months at room temperature. Do not store at 4°C.
5. Phosphate buffered saline (2 L): 10 mM NaH_2PO_4 , 150 mM NaCl adjusted to pH 7.2 using NaOH . This solution is stable and may be stored at 4°C. It is susceptible to microbial contamination, however, and is usually made as required.

3. Methods

3.1. Preparation of Protein A-Sepharose Column

1. Add 1.5 g of protein A-Sepharose to 10 mL of phosphate buffer, pH 8.0, and allow the resin to swell at room temperature for 30 min. All subsequent steps are performed at 4°C.
2. Pour the resin into a small column (0.9 x 15 cm) and wash with phosphate buffer, pH 8.0, at 30 mL/h for about 30 min, until the resin has settled. The total volume should be about 5 mL.
3. Wash the column with 20 mL of citrate buffer, pH 3.0, to elute any bound material, and re-equilibrate the column using 50 mL of phosphate buffer, pH 8.0. The column is now ready for use (see Note 2 in section 4).

3.2. Purification of IgG

1. To 10 mL of serum at room temperature slowly add 5 mL of saturated ammonium sulfate, while stir-

- ring constantly. Continue stirring for 1 h (*see* Note 3 in section 4), then centrifuge the mixture at 15,000g for 15 min at 25°C.
2. Discard the supernatant and dissolve the pellet in 5 mL of phosphate buffer, pH 8.0. Perform all subsequent steps at 4°C.
 3. Dialyze the sample against 500 mL of phosphate buffer, pH 8.0, for 3 h to remove residual ammonium sulfate, then apply the dialyzed sample to the protein A–Sepharose column at 20 mL/h and discard the effluent. Most serum proteins pass straight through this column.
 4. Wash the column with 50 mL of phosphate buffer, pH 8.0, to ensure complete removal of unbound proteins.
 5. Elute the IgG from the column with 20 mL of citrate buffer, pH 3.5. Collect the eluate as 1-mL fractions in tubes containing 0.4 mL Tris-HCl, pH 8.5, to neutralize the buffer and prevent denaturation of IgG.
 6. Combine the fractions containing the bulk of the IgG ($A_{280} > 0.2$) and dialyze twice, each time for at least 2 h against 1 L phosphate buffered saline. Divide the purified IgG into 1-mL aliquots and store at -20°C (*see* Note 4 in section 4).
 7. Remove residual protein from the protein A–Sepharose by washing with 20 mL of citrate buffer, pH 3.0, and immediately re-equilibrate the column with 40 mL of phosphate buffer, pH 8.0 (*see* Note 5 in section 4).

4. Notes

1. When adjusting the pH of the saturated ammonium sulfate, monitor the change using a suitable narrow-range pH paper, then use an electrode to

check the final pH by diluting an aliquot of the solution about 20-fold in water. Do not use a combination glass electrode to measure directly the pH of the saturated solution, since the high salt and junction potentials may cause reading errors of greater than 1 pH unit.

2. If the protein A-Sepharose column is not to be used immediately, it should be stored at 4°C in phosphate buffer containing 0.05% (w/v) NaN_3 .
3. The initial ammonium sulfate precipitation step in the purification of IgG is not essential. This procedure does extend the life of the protein A-Sepharose column, however, by removing lipids and proteases present in serum. If steps 1 and 2 and the dialysis of step 3 in section 3.2. are omitted, the serum should be diluted with an equal volume of phosphate-buffered saline before being applied to the protein A-Sepharose column (*see* step 3).
4. Avoid repeated cycles of freezing and thawing by dividing the purified IgG into smaller aliquots if necessary. For transport, IgG may be lyophilized. The powder is stable at room temperature for several days, but should be returned to -20°C when convenient.
5. To prevent denaturation of protein A, the column should be washed and re-equilibrated with phosphate buffer as soon as possible after use. The column should be stored as described in note 2. If washed and stored correctly, this column may be used at least 50 times without noticeable deterioration in its ability to bind IgG.
6. An initial indication of the purity of the IgG preparation may be obtained by measuring its absorbance at 280 and 250 nm. Although the exact value may vary between antibody clones, in general $A_{280}/$

A_{250} for mammalian IgG is about 2.5–3.0. In contrast, for other serum proteins this ratio is about 1.0–1.5. A more rigorous test of purity is provided by SDS-polyacrylamide gel electrophoresis of both reduced and nonreduced samples using the system described by Laemmli (see vol. 1 of this series, chapter 6). Typical results are shown in Fig. 1.

Provided the preparation is pure, the quantity of IgG may be estimated from its absorbance at 280 nm. A solution of IgG containing 1 mg/mL has an absorbance, in a 1-cm light path, of about 1.4.

7. With care this method can be used to purify IgG from sources other than rabbit. The amount of rabbit IgG purified, however, approaches the capacity of the affinity column. Other species may have higher serum IgG concentrations (Table 1). For such species, the volumes of antiserum and ammonium sulfate should be reduced proportionally. All other conditions should remain the same. Antisera in which most IgG binds poorly, or not at all, such as those from rat, sheep, and goat, should be purified by other methods (3).
8. The affinity of protein A for IgG subclasses, if present, may vary (Table 1). This variation has been exploited to identify and purify mouse IgG subclasses, which can be separated by selective desorption from the column (4).

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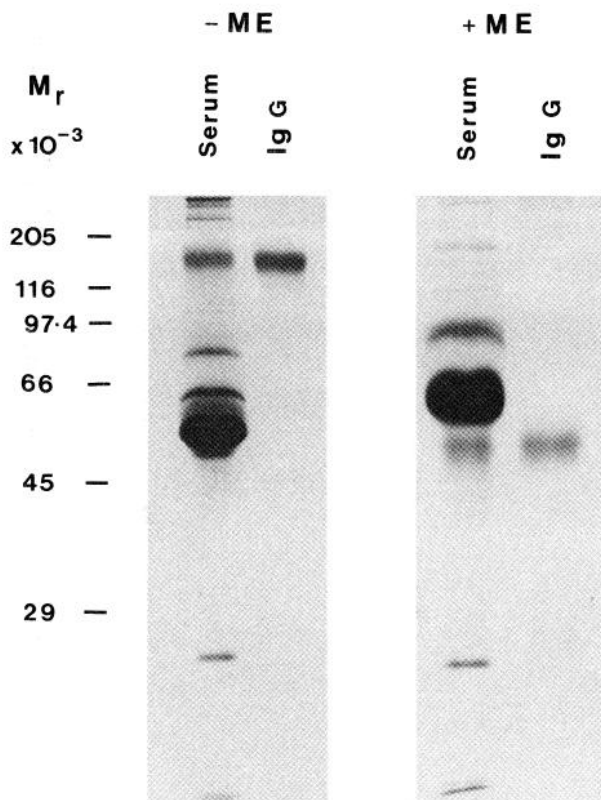


Fig. 1. Analysis of purified immunoglobulin G by SDS-polyacrylamide gel electrophoresis. Samples of crude rabbit serum and purified IgG were fractionated by SDS-polyacrylamide gel electrophoresis both before and after chemical reduction using 2-mercaptoethanol, and are labeled -ME and +ME, respectively. The molecular weight standards used are myosin (205 kdalton), β -galactosidase (116 kdalton), phosphorylase b (97.4 kdalton), bovine serum albumin (66 kdalton), ovalbumin (45 kdalton), and carbonic anhydrase (29 kdalton). The polypeptides were stained using Coomassie Blue.

Native IgG has a molecular mass of 150 kdalton and contains two heavy chains (50 kdalton) and two light chains (about 25 kdalton) held together by disulfide bonds. Following reduction by 2-mercaptoethanol, the heavy and light chains can be resolved. The light chain is often diffuse and stains poorly. No contaminants are visible in the purified IgG preparation.

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Chapter 27

Antibody–Enzyme Conjugate Formation

G. B. Wisdom

1. Introduction

The ideal antibody–enzyme conjugate retains all the immunoreactivity and enzyme activity of its components, has defined and appropriate proportions, and is stable. There are many methods of labeling immunoglobulins with enzymes, but none meet all these criteria (1). Functional groups in the protein, such as amino, thiol, or saccharide groups, are exploited, and in nearly all cases bifunctional reagents are used to link the enzyme to the immunoglobulin G (IgG). The three most commonly used methods are presented here with minor modifications to the originally described procedures.

1.1. Alkaline Phosphatase Labeling Using Glutaraldehyde

The chemistry of the homobifunctional reagent, glutaraldehyde, is complex. It reacts with amino groups of proteins, and when two proteins are mixed in its pres-

ence, stable conjugates are obtained without the formation of Schiff bases. This method (2) can be applied to most proteins, but self-coupling is a problem unless the proteins are at appropriate concentrations.

1.2. Horseradish Peroxidase Labeling Using Periodate Oxidation

Horseradish peroxidase labeling using periodate oxidation (3) is unusual in that an exogenous cross-linking reagent is not used. The peroxidase is a glycoprotein, and reaction of its saccharide residues with sodium periodate results in the formation of aldehyde groups. (The enzyme has few amino groups, and self-coupling can be avoided.) When IgG is added, the modified enzyme reacts with the immunoglobulin's amino groups to form Schiff bases. These are then reduced to give a stable conjugate.

1.3. β -Galactosidase Labeling Using *m*-Maleimidobenzoyl-*N*-hydroxysuccinimide Ester (MBS)

MBS is a heterobifunctional reagent, and it is of particular value when one of the proteins involved has no free thiol groups, e. g., IgG. In this method (4), the IgG is first modified by allowing the *N*-hydroxysuccinimide ester of MBS to react with amino groups in the protein. After the removal of unused reagent, the enzyme is added and the modified IgG reacts with thiol groups in the β -galactosidase via its maleimide groups to form thioether links.

The method employing alkaline phosphatase and glutaraldehyde is the easiest to carry out and gives use-

ful conjugates, although the yields are relatively low; the other two methods give very active conjugates with similar levels of activity (5). The strength of the signal produced by a conjugate, however, will depend on the substrate and detection system used, as well as the specific activities of its components and the method of preparation. In all cases the conjugates formed are somewhat heterogeneous and their molecular weights may exceed 10^6 . Removal of unconjugated IgG is important in most applications; this is usually achieved by gel filtration.

Antibody–enzyme conjugates can be used in several related techniques. The most common of these is the ELISA (1,2; and *see* vol. 1 of this series), and, in the "sandwich" forms of this technique, these conjugates are used for the measurement of antibodies and macromolecular antigens. The conjugates can also be applied to the detection of antigens or antibodies in tissue slices or cell layers and to the measurement of substances dotted or blotted on nitrocellulose or other membranes (*see* Chapters 31–34 in this volume).

2. Materials

Ideally, if a specific antibody conjugate is required, the IgG should be purified by immunoaffinity chromatography unless it is derived from a hybridoma and contamination by other proteins is minimal. For many purposes, however, the total IgG fraction of an antiserum may be used.

The enzymes should be free of other proteins and interfering substances. Several manufacturers provide suitable enzymes at very high specific activities, e. g., Boehringer Mannheim Corp., Sigma Chemical Co.

2.1. Preparation of the IgG Fraction of Serum

1. Ammonium sulfate $[(\text{NH}_4)_2\text{SO}_4]$.
2. 20 mM Sodium phosphate buffer, pH 7.2, containing 0.15M NaCl (PBS).
3. 5 mM Sodium phosphate buffer, pH 6.5.
4. DE-52 cellulose (Whatman Ltd., Maidstone, UK) or similar DEAE ion exchanger

2.2. Alkaline Phosphatase Labeling Using Glutaraldehyde

1. Alkaline phosphatase from calf intestinal mucosa, 1000 U/mg or greater. (This is often available as an ammonium sulfate suspension.)
2. PBS.
3. Glutaraldehyde.
4. 50 mM Tris-HCl buffer, pH 7.5, containing 1 mM MgCl_2 , 0.02% NaN_3 , and 2% bovine serum albumin (BSA).

2.3. Horseradish Peroxidase Labeling Using Periodate Oxidation

1. Horseradish peroxidase, 250 purpurogallin units/mg or greater.
2. 0.1M sodium periodate (NaIO_4).
3. 1 mM Sodium acetate buffer, pH 4.4.
4. 10mM Sodium carbonate buffer, pH 9.5.
5. 0.2M Sodium carbonate buffer, pH 9.5.
6. Sodium borohydride (NaBH_4), 4 mg/mL.
7. Ultrogel AcA34 (LKB Produkter AB, Bromma, Sweden) or similar gel filtration medium.
8. PBS.
9. BSA.

2.4. β -Galactosidase Labeling Using MBS

1. β -Galactosidase from *Escherichia coli*, 600 U/mg or greater.
2. 0.1M sodium phosphate buffer, pH 7.0, containing 50 mM NaCl.
3. MBS.
4. Dioxan.
5. Sephadex G-25 (Pharmacia Fine Chemicals AB, Uppsala, Sweden) or equivalent gel filtration medium.
6. 10 mM Sodium phosphate buffer, pH 7.0, containing 50 mM NaCl and 10 mM MgCl_2 .
7. 2-Mercaptoethanol.
8. DEAE-Sepharose (Pharmacia).
9. 10 mM Tris-HCl buffer, pH 7.0, containing 10 mM MgCl_2 and 10 mM 2-mercaptoethanol.
10. Item 9 containing 0.5M NaCl.
11. Item 9 containing 3% BSA and 0.6% NaN_3 .

3. Methods

3.1. Preparation of the IgG Fraction of Serum

1. Stir serum gently at room temperature and add $(\text{NH}_4)_2\text{SO}_4$ slowly to give a concentration of 230 g/L. Adjust the pH to 7.8 if necessary and continue stirring for about 1 h.
2. Centrifuge the suspension at 1500g for 15 min at room temperature. Discard the supernatant.
3. Dissolve the pellet in sufficient PBS to give a volume equal to that of the original serum.
4. Repeat steps 1 and 2 in this section.
5. Dissolve the pellet in the minimal amount of PBS and dialyze against 5 mM sodium phosphate buffer, pH 6.5, at 4°C.

6. Add the dialyzed solution to a column of DE-52 (about 5 mL for each mL of serum) equilibrated at room temperature with the phosphate buffer and elute with the same buffer.
7. Pool the fractions with high A_{280} , concentrate by salt precipitation or ultrafiltration, and store the IgG at -20°C or at 4°C in the presence of an antimicrobial agent.

3.2. Alkaline Phosphatase Labeling Using Glutaraldehyde

1. Centrifuge a portion of the ammonium sulfate suspension, containing 1.5 mg of enzyme, at about 2000g for 10 min at 4°C ; discard the supernatant.
2. Add 0.5 mg of IgG in 100 μL of PBS.
3. Dialyze the mixture against PBS overnight at 4°C .
4. Add 5% glutaraldehyde (about 10 μL) to give a final concentration of 0.2% (v/v), and stir the mixture for 2 h at room temperature.
5. Dilute the volume to 1 mL and dialyze, as in step 3 in this section.
6. Dilute the dialyzed solution to 10 mL with 50 mM Tris-HCl buffer, pH 7.5, containing 1 mM MgCl_2 , 0.02% NaN_3 , and 2% BSA; store at 4°C .

3.3. Horseradish Peroxidase Labeling Using Periodate Oxidation

1. Dissolve 2 mg of enzyme in 500 μL of water.
2. Add 100 μL of freshly prepared 0.1M NaIO_4 and mix for 20 min at room temperature.
3. Dialyze the modified enzyme against 1 mM sodium acetate buffer, pH 4.4, overnight at 4°C .
4. Dissolve 4 mg of IgG in 500 μL of 10 mM sodium carbonate buffer, pH 9.5.

5. Adjust the pH of the dialyzed solution to 9.0–9.5 by adding 10 μ L of 0.2M sodium carbonate buffer, pH 9.5, and immediately add the IgG solution. Stir the mixture for 2 h at room temperature.
6. Add 50 μ L of freshly prepared NaBH₄ solution (4 mg/mL), and stir the mixture occasionally over a period of 2 h at 4°C.
7. Fractionate the conjugate mixture on a column of Ultrogel AcA34 (1.5 x 85 cm) in PBS. Determine the A_{280} and A_{403} .
8. Pool the fractions in the first peak (both A280 and A403 peaks coincide), add BSA to a final concentration of 5 mg/mL, and store aliquots at –20°C.

3.4. β -Galactosidase Labeling Using MBS

1. Dissolve 1.5 mg of IgG in 1.5 mL of 0.1M sodium phosphate buffer, pH 7.0, containing 50 mM NaCl.
2. Add 0.32 mg of MBS in 15 μ L of dioxan, mix, and incubate for 1 h at 30°C.
3. Fractionate the mixture on a column of Sephadex G-25 (0.9 x 30 cm), equilibrated with 10 mM sodium phosphate buffer, pH 7.0, containing 50 mM NaCl and 10 mM MgCl₂, and elute with the same buffer. Collect 0.5-mL fractions. Measure the A_{280} and pool the fractions in the first peak (about 3 mL).
4. Add 1.5 mg of enzyme to the modified IgG solution, mix, and incubate for 1 h at 30°C.
5. Stop the reaction by adding 1M 2-mercaptoethanol to give a final concentration of 10 mM (about 30 μ L).
6. Fractionate the mixture on a column of DEAE-Sephrose (0.9 x 15 cm) equilibrated with 10 mM Tris-HCl buffer, pH 7.0, containing 10 mM MgCl₂ and 10 mM mercaptoethanol. Elute with the Tris buffer (50 mL), followed by Tris buffer containing

0.5M NaCl (50 mL). Collect 3-mL fractions in tubes containing 0.1 mL of Tris buffer containing 3% BSA and 0.6% NaN_3 . Pool the major peak (this is eluted with the NaCl), and store at 4°C.

4. Notes

4.1. General Comments

1. The procedure described for the purification of IgG is suitable for rabbit serum. Other sera may require modifications such as an alteration of the pH or ionic strength used for ion exchange. In most cases a compromise between yield and purity must be reached.
2. The efficacy of an antibody–enzyme conjugate may be tested by immobilizing the appropriate antigen in the wells of a microtiter plate, incubating various dilutions of the conjugate for a few hours, washing, adding the substrate, and measuring the amount of product formed. This procedure may also be used to monitor the purification of conjugates by chromatography.

4.2. Alkaline Phosphatase Labeling Using Glutaraldehyde

1. If the alkaline phosphatase is available as a lyophilized powder or as an NaCl suspension, step 3 in section 3.2 should be omitted and step 1 in section 3.2 modified.
2. Dialysis of small volumes can be conveniently done in narrow dialysis tubing by placing a short glass tube, sealed at both ends, in the tubing so that the space available to the sample is reduced. Transfer losses are minimized by carrying out steps 2–5 in the same dialysis bag.

3. We have found that purification of these conjugates by gel filtration does not improve their activity.
4. BSA is routinely added to conjugate solutions to improve stability and minimize adsorption to container walls.

4.3. Horseradish Peroxidase Labeling Using Periodate Oxidation

1. Horseradish peroxidase is often specified in terms of its RZ. This is the ratio of the A_{403} to the A_{280} and provides a measure of the heme content of the preparation. A highly purified sample will have a value of about 3, but this is not a guarantee of high specific activity. Conjugates with an RZ of about 0.4 perform satisfactorily.
2. Preparations of the enzyme vary in their carbohydrate content, and this can affect the oxidation reaction. Free carbohydrate may be removed by gel filtration. Increasing the NaIO_4 concentration to 0.2M can also help, but further increases lead to excess oxidation, which damages the activity of the enzyme.
3. During the course of the NaIO_4 treatment, the mixture changes color from orange to green; this is reversed during the dialysis.
4. NaN_3 should not be used with peroxidase conjugates because it inhibits the enzyme. If an antimicrobial agent is required, 0.02% merthiolate may be used.
5. This method may also be used with the glycoprotein enzymes, glucose oxidase and amyloglucosidase.

4.4. β -Galactosidase Labeling Using MBS

1. The thiol groups of β -galactosidase may become oxidized during purification and storage. Satisfac-

tory conjugates can be made from preparations with about 10 free thiol groups per molecule, and some manufacturers provide information about the thiol content. It is relatively easy to measure the number of thiol groups (6).

2. These conjugates may also be purified by gel filtration in Ultrogel AcA34 (or an equivalent medium) in 10 mM Tris-HCl buffer, pH 7.0, containing 10 mM MgCl₂, 10 mM 2-mercaptoethanol, and 50 mM NaCl.
3. MBS may be used to label IgG with enzymes lacking thiol groups. There are several procedures for introducing these groups into proteins (4,7).

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Chapter 28

Vacuum Blotting

An Inexpensive, Flexible, Qualitative Blotting Technique

Marnix Peferoen

1. Introduction

The first system we used to transfer proteins from polyacrylamide gels onto nitrocellulose was a capillary blotting system, based on the work of Renart et al. (1). Although we were very enthusiastic about the wide range of possible applications created by the protein blotting technique, we were unsatisfied with the slow and inefficient transfer. Electroblothing (2,3) seemed to be much more efficient, but at that time there were serious financial limitations in the department.

Instead of using capillary force to create a flow of buffer through the gel, we decided to speed up the process by using suction force to draw proteins out of the gel (4). Basically, the assembly for vacuum blotting is very similar to the set-up for capillary blotting, but the

entire system is closed and connected to a vacuum pump. Vacuum blotting proved to be significantly faster than capillary blotting.

2. Materials

All buffers are prepared with analytical grade reagents that are dissolved in distilled water. Buffers, when stored at 4°C, are stable for at least 3 mo.

1. Transfer buffer: 3.03 g Tris (25 mM) and 11.26 g glycine (150 mM) are dissolved in 700 mL of water. To this solution, 200 mL of methanol is added, and the final volume is made up to 1000 mL with water. The pH of this buffer is 8.3.
2. Rinsing buffer: 1.14 g Na_2HPO_4 (16 mM), 0.27 g KH_2PO_4 (4 mM), and 3.36 g NaCl (115 mM) are dissolved in 400 mL of water followed by the addition of 1.5 mL of Tween 20. The buffer is made up to 500 mL with water, and the pH should be 7.3 without any adjustment.
3. Transfer membrane: A nitrocellulose membrane of 0.22 μm pore size (Millipore, HAWP 293 25) is used as immobilizing paper.
4. Vacuum blotting equipment. We use a commercial slab gel dryer from Bio-Rad (Model 1125), and vacuum is created with a suction pump (Heraeus, 250W). The pump is protected from buffer by inclusion of a side arm flask and a silica gel column in the vacuum line. The silica gel column is made from a plastic transparent tube (50 x 5 cm) filled with dry silica gel. One also needs a flexible plastic sheet and approximately 20 filter papers (e.g., Whatman No. 2, 16 x 16 cm). A porous polyethylene sheet on the bottom of the slab gel dryer creates an evenly distributed vacuum.

3. Method

3.1. Vacuum Blotting

1. Proteins are separated in SDS-polyacrylamide gels. After electrophoresis the gel is soaked for 20–30 min in the transfer buffer. The gel can be immediately processed or stored frozen in a plastic bag containing some 30 mL of transfer buffer.
2. A sheet of dry filter paper is laid on a porous polyethylene sheet on the bottom of the slab gel dryer. The nitrocellulose membrane is uniformly wet with transfer buffer and is placed on the dry filter paper. A window, some 5 mm smaller than the gel, is cut in the center of a plastic sheet (25 × 25 cm). The plastic sheet is then placed on the dry filter paper, and its window on the nitrocellulose membrane. The gel is put on the nitrocellulose so that its edges are resting on the plastic sheet (Fig. 1A). Air bubbles between the gel and the nitrocellulose membrane should be removed. Some 15 sheets of filter paper are saturated with transfer buffer and placed atop the gel. Again, trapping of air bubbles either between the filter papers and the gel or between the filter papers themselves, should be avoided. The four outer edges of the plastic sheet are folded on top of the saturated filter papers (Fig. 1 B,C).
3. The suction pump is connected via a silica gel column and a side arm flask to the slab gel dryer. The slab gel dryer is then closed with its rubber sheet, and the transfer buffer contained in the filter papers on top of the gel is driven through the gel by vacuum (Fig. 1D).
4. The gel is typically blotted for 1 h.
5. After the transfer, the nitrocellulose membrane is washed in rinsing buffer for 30 min at 37°C, fol-

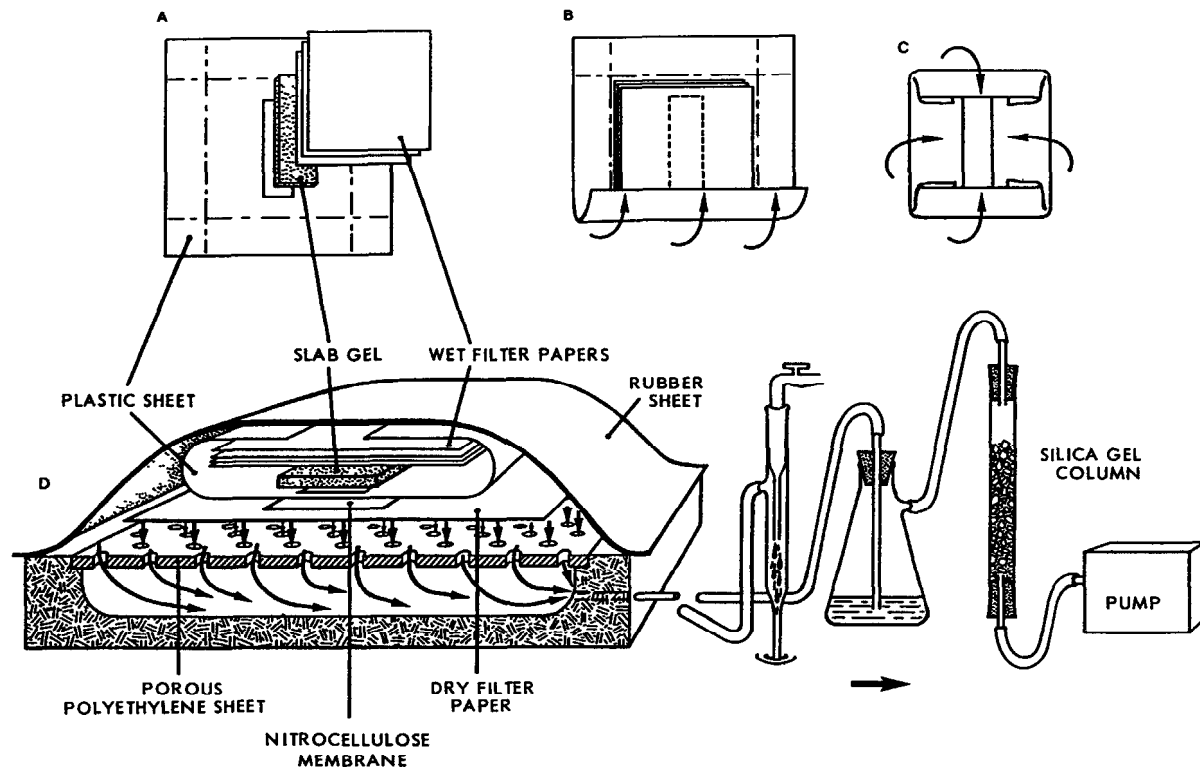


Fig. 1. Vacuum blotting set-up. (A) A nitrocellulose membrane is saturated in transfer buffer and put on a dry filter paper on the bottom of the slab gel dryer. A plastic sheet with a window 5-mm smaller than the gel is placed on the membrane. The gel is put on the nitrocellulose membrane so that its edges rest on the plastic sheet. The gel is covered with approximately 15 sheets of filter paper, saturated with transfer buffer. (B–C). The outer edges of the plastic sheet are folded on top of the filter papers. (D). The slab gel dryer is connected to the suction pump or a water jet pump and is closed with its rubber sheet.

lowed by three times 15 min at room temperature. The membrane is then stained with a general protein stain (Amido black, Indian ink, Aurodye, *see* Chapters 29 and 34 in this volume), or specific proteins are detected by immunostaining (*see* Chapters 31–33 in this volume).

4. Notes

1. In this study we used the transfer buffer originally described by Towbin et al. (2). Since we do not have to worry about high conductivity and high temperature, however, there is no limitation concerning the choice of buffer. One can use the buffer that provides the best conditions for transferring and binding proteins to the nitrocellulose membrane.
2. The rinsing buffer can also be used for immunostaining of membrane bound proteins. The concentration of Tween-20 is high enough to provide efficient blocking of protein binding sites on the nitrocellulose.
3. Nitrocellulose membranes with a pore size of 0.45 μm are less efficient in absorbing proteins of low molecular weight (lower than 20,000), which results in insufficient immobilization, causing leakage or diffusion of the blotted proteins (3,4,5). Vacuum blotting should also be compatible with nylon membranes or the Immobilon PVDF membrane (Millipore). Nylon membranes have higher binding capacities than nitrocellulose, and they are considerably stronger. In order to reduce background staining, nylon membranes require incubations with high concentrations of blocking agents and still give a high background with alkaline phosphatase and glucose oxidase conjugate detec-

tion systems. Nylon membranes are also incompatible with most protein stains (except Ferridye from Janssen Life Science Products and a biotin/avidin detection system). The Immobilon PVDF Transfer Membrane is reported to have high mechanical strength and high protein binding capacity and to be compatible with all commonly used protein and immunostaining techniques.

4. We use a commercial slab gel dryer, but any system that is suitable to dry gels should also be suitable for vacuum blotting. Even a mechanical suction pump is not indispensable, since a good water jet pump does work as well (results not shown). The plastic sheet between the stack of filter papers and the nitrocellulose membrane prevents the transfer buffer from passing around the gel instead of through the gel. It is therefore important to trim the window so that the edges of the gel rest on the plastic sheet. The 15 sheets of filter paper can absorb approximately 300–400 mL of transfer buffer.
5. Since AuroDye is a very sensitive stain, intense background staining can be a problem. A specific staining can substantially be reduced by inserting a sheet of nitrocellulose between the gel and stack of wet filter papers, so that the gel is sandwiched between two nitrocellulose sheets.
6. If the transfer is unsatisfactory after 1 h of blotting, the blotting time can be extended for several hours. In that case, it is necessary to add more transfer buffer to the filter papers. Every hour the blotting is interrupted by lifting the rubber sheet and unfolding the edges of the plastic sheet. Approximately 100 mL of transfer buffer is then poured on the filter papers. After folding the plastic sheet on top of the filter papers, the gel dryer is closed with

the rubber sheet, and the blotting continues. The addition of transfer buffer to the filter papers not only prevents the gel from drying, but also increases the transfer efficiency. The more transfer buffer used, the better the transfer.

7. The quality of the vacuum blotting is illustrated in Fig. 2. Female *Locusta migratoria* hemolymph is collected from a *Locusta* laboratory culture and separated in polyacrylamide-urea tubes by non-equilibrium pH gradient (3.5–10) gel electrophoresis (NEpHGE; 6) for 2000 Vh (volt-hours). The gel tubes are then electrophoresed at right angles on SDS-polyacrylamide gradient (5–15%) gels with the discontinuous buffer system (7). Proteins in the gel are silver stained according to Heukeshoven and Dernick (8). The duplicate gel is blotted for 1 h with 500 mL transfer buffer. Proteins on nitrocellulose are detected with AuroDye (9; Janssen Life Sci. Products). Figure 2a is part of the gel, stained with silver, containing two female specific yolk polypeptides (YP) with molecular weights in the range of 120,000. The blot is stained overnight with AuroDye (Fig. 2b). The vacuum blotting faithfully replicates the high resolution of the two-dimensional electrophoresis on the nitrocellulose membrane.
8. The transfer efficiency of vacuum blotting is determined by blotting [^{14}C]-methylated proteins of different molecular weight (M_r , 14,300–200,000, Amersham). After SDS-polyacrylamide gradient (5–15%) electrophoresis, the gel is cut in four strips, each containing two lanes of separated ^{14}C -methylated polypeptides. Three of these strips are then blotted onto nitrocellulose for 1, 2, or 4 h. Approximately 100 mL of transfer buffer is added every half hour. The proteins in the gel and on the nitrocellu-

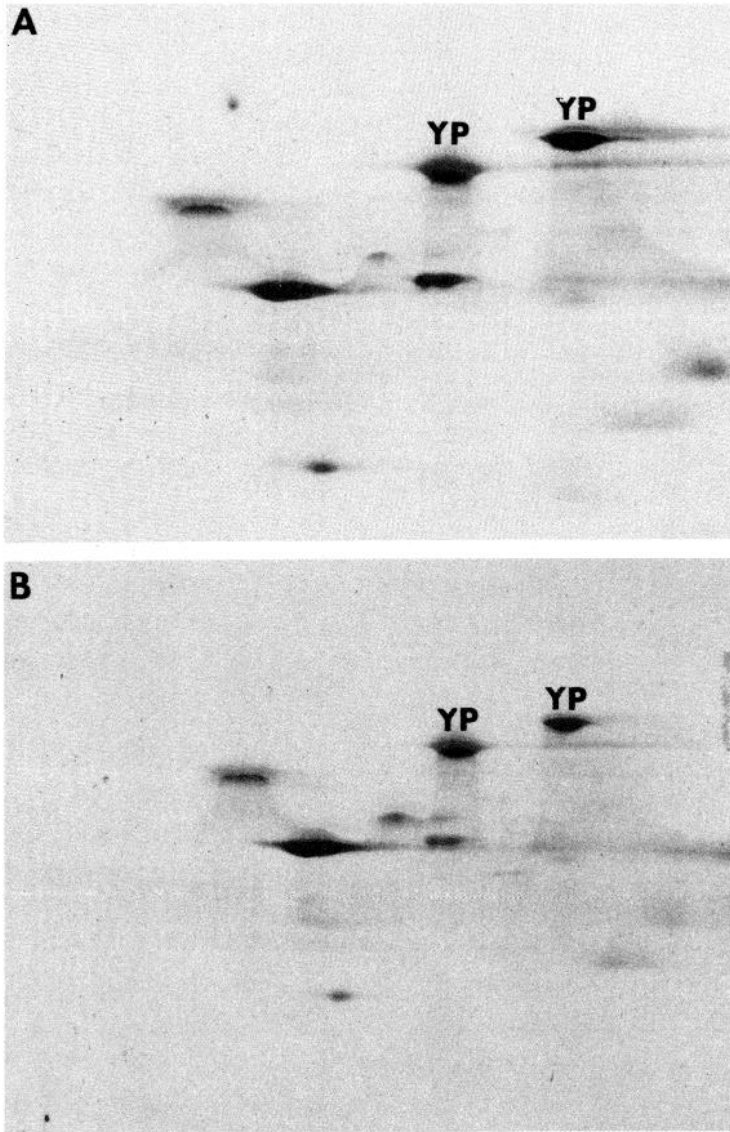


Fig. 2. NEpHGE followed by SDS-PAGE of female *Locusta migratoria* hemolymph. The gel is stained with silver (A) or blotted onto nitrocellulose for 1 h with 500 mL of transfer buffer. Proteins on the nitrocellulose are stained with AuroDye. (B). Only part of the gel and the corresponding piece of nitrocellulose containing two female specific yolk proteins (YP) are shown.

Table 1
Transfer Efficiency of Vacuum Blotting^a

M_r of proteins	Percentage of ¹⁴ C-methylated proteins transferred to nitrocellulose		
	1 h	2 h	4 h
14,300	28	45	64
30,000	23	38	54
46,000	18	31	45
69,000	23	32	44
92,500	18	26	36
200,000	17	21	32

^aAfter SDS-polyacrylamide gradient (5–15%) gel electrophoresis, ¹⁴C-methylated polypeptides are transferred to nitrocellulose by vacuum blotting for 1, 2, and 4 h. Every half hour, approximately 100 mL of transfer buffer is added to the stack of filter papers, on top of the gel. After visualization by radioautography, radioactive polypeptides are cut from the gel and from the nitrocellulose membrane for liquid scintillation counting. Counts in polypeptides that are not blotted are referred to as 100%.

lose are visualized by radioautography, so that proteins can be cut in order to measure the radioactivity by liquid scintillation counting. Counts in electrophoresed polypeptides that are not blotted are referred to as 100%. Counts in transferred polypeptides are expressed as percentages of counts in polypeptides before blotting (Table 1). The transfer efficiency decreases with increasing molecular weight, and after 4 h there is twice as much lysozyme (M_r , 14,300) blotted as there is myosin (M_r , 200,000) (64 and 32%, respectively). These data show that the transfer efficiency of vacuum blotting is comparable to the efficiency of low-powered electroblotting (3,5). This also means that vacuum blotting, like low-powered electroblotting, is not very suitable for quantitative blotting. If, however,

only a qualitative transfer is required, vacuum blotting is a very reasonable alternative to electroblotting. Apart from the fact that vacuum blotting is very inexpensive and easy to use, it may offer some particular advantages. There is for instance absolutely no limitation in the choice of buffers, since there is no increasing conductivity and temperature during transfer. In electroblotting, some proteins are at their isoelectric point and can therefore not be transferred. Vacuum blotting can be used to transfer proteins that otherwise would need buffers with very acidic or basic buffers. It is even possible to transfer proteins selectively from the gel, by adjusting the transfer buffer. Vacuum blotting is used in a wide variety of protein studies (10–14) and is also adapted to transfer DNA (15).

Acknowledgment

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Chapter 29

Blotting with Plate Electrodes

Marnix Peferoen

1. Introduction

Since its introduction by Towbin et al. (1), electroblotting has become one of the most valuable analytical procedures in protein research. Conventionally, the gel and the immobilizing matrix are sandwiched between porous pads and inserted between two electrodes in a buffer tank. Electrodes of platinum wire are mounted in such a way that, with a minimum amount of material, they still generate a uniform electrical field in the center of the buffer chamber, with the minimum amount of material. On the other hand, transfer efficiency is largely dependent of the voltage gradient between the two electrodes. High voltages generate high currents and high temperatures, whereas reducing the distance between wire electrodes seriously disturbs the electrical field (2). Plate electrodes yield essentially homogeneous electrical fields, even when they are moved close together. Vaessen et al. (3) were the first to experiment with plate electrodes. Kyhse-Andersen (4) introduced horizontal

semidry blotting in which the transfer takes place in a stack of buffer-saturated filter papers, inserted between two graphite plates. Svoboda et al. (5) simplified the buffer system from three to two different solutions and used corrosion resistant electrodes. Recently Bjerrum and Schafer-Nielsen (6) described a homogeneous buffer system with transfer efficiency equal to the efficiency of the discontinuous buffer system. The major advantages of the horizontal semidry electroblotting are: Simple assembly, reduced buffer consumption, uniform electrical field, high transfer efficiency, and simultaneous blotting of up to six full-size gels.

2. Materials

Buffers are made with analytical grade reagents dissolved in distilled water. Buffers are stable for at least 3 mo when stored at 4°C.

1. Transfer buffer: 2.91 g of Tris base (48 mM), 1.46 g of glycine (39 mM), and 0.19 g of SDS (1.3 mM) are dissolved in 350 mL of water and adjusted to pH 9.2 with HCl. After addition of 100 mL of methanol, the final volume is made up to 500 mL with water.
2. Rinsing buffer: 1.21 g of Tris base (10 mM) and 8.27 g of sodium chloride (150 mM) are dissolved in 800 mL of water and adjusted to pH 8.0 with hydrochloric acid. The buffer is made up to 1000 mL, and 1 mL of Tween-20 is added.
3. Transfer membrane: Immobilon PVDF transfer membrane (Millipore, IPVH 000 10) is used as an immobilizing matrix.
4. Electroblotting equipment: Proteins are transferred with a horizontal semidry multigel electroblotter, produced by Ancos (Denmark) and distributed in the US by Gelman Sciences Inc. The apparatus consists of a plastic base in which the anodic

carbon plate (17 x 17 cm) is fixed, whereas the cathodic carbon plate of equal dimensions is fixed in the lid of the apparatus. The transfer buffer is soaked in four sheets of absorbent paper.

5. Protein stain: Proteins in the gel are stained with Coomassie brilliant blue R250. Proteins on the Immobilon membrane are stained with Indian ink (0.1% v/v in rinsing buffer) or Amido black 10B (0.1% w/v in methanol and 2% acetic acid).

3. Method

3.1. Electroblotting

1. Proteins are separated in SDS-polyacrylamide slab gels. Four pieces of absorbent filter paper (Bio-Rad, 165-0921) and the transfer membrane, trimmed to the size of the gel, are saturated with transfer buffer.
2. After electrophoresis, two pieces of buffer-saturated absorbent paper are put on the anodic (bottom) plate electrode. The transfer membrane is pre-wet in a small volume of 100% methanol, followed by a brief rinse in transfer buffer and is then layered on the blotting papers. The gel is placed on the membrane and covered with the two other pieces of saturated absorbent paper. During assembly, trapping of air bubbles between any of the layers should be avoided. The semidry electroblotter is then closed with the lid containing the cathode.
3. The blotting apparatus is connected to a power supply and proteins are transferred for 1–2 h at a constant current density of 0.8–1.0 mA/cm² of gel (*see* Note 5 in section 4).
4. Following transfer, blots may be probed using immunodetection methods (*see* Chapters 31–34) or stained for protein as described below.

3.2. Protein Staining

1. After the transfer, the membrane is incubated for 30 min at 37°C in rinsing buffer, followed by three times 15 min at room temperature. The membrane is then stained overnight with Indian ink and rinsed with rinsing buffer.
2. Membranes can also be stained with Amido black for 15 min, followed by destaining for 15 min in 45% methanol, 7% acetic acid. The last traces of background stain are removed by a 1–2 min rinse in 90% methanol. The membrane is then dried for storage.

4. Notes

1. This transfer buffer is one of the many that can be used with the semidry blotter. Originally, a discontinuous buffer system consisting of three different buffers, resulting in isotachophoretic transfer of the proteins, was developed (4). Svoboda et al. (5) use a Tris-glycine buffer (25 mM Tris-192 mM glycine, pH 8.3) with 20% methanol at the anodic side and with 0.1% SDS at the cathodic side. Bjerrum and Schafer-Nielsen (6) describe a continuous buffer system (48 mM Tris, 39 mM glycine, 1.3 mM SDS, 20% methanol, pH 9.2) with transfer efficiency as good as that obtained with the discontinuous buffer system of Khyse-Andersen (4).
2. In this study we use an Immobilon PVDF membrane, which is a polyvinylidene difluoride matrix from Millipore. This membrane is not as brittle as nitrocellulose, is very resistant to high methanol concentrations, and has high protein-binding capacity and a superior retention. The only difference with this membrane is that it has to be pre-wet with

- 100% methanol. (The membrane should be kept wet at all times, and if it does dry out, it should be re-wetted in 100% methanol). The Immobilon PVDF membrane is compatible with all immunochemical stainings and with the AuroDye protein staining. It can also be stained with Coomassie blue and Amido black since the background staining can be removed by rinsing in 90% methanol.
3. Apart from the Ancos apparatus, there are several horizontal semidry electroblotters commercially available (e.g., Kem-En-Tec, Hellerup, Denmark; Sartorius, Gottingen, FRG; Biolyon, Lyon, France). The electroblotter from Idea Scientific Co. (Corvallis, Oregon) also uses plate electrodes, but is a vertical apparatus with a real buffer chamber. Most of the plate blotters use graphite electrodes, which are cheap, but need to be cleaned before each run and have a limited resistance to abrasion. The apparatus of Biolyon presents an elegant solution to that problem, without using the very expensive platinum plates. The anode is a surface conductive glass plate (Hortiplus from Glaverbel, Brussels, Belgium), whereas the cathode is a stainless steel plate (stainless steel does not resist anodic oxidation).
 4. Most horizontal semidry blotters allow transfer of proteins from several (up to six) gels, stacked on top of each other. The transfer efficiency gradually decreases toward the cathode, however (6). Dialysis membranes must be interleaved to prevent contamination of the gels and membranes with substances being transferred from other gels in the same stack. When small gels (less than half the size of the plate electrode) are to be blotted, short circuiting should be prevented by inserting isolating material (e.g., slab gel spacers) between the electrodes.

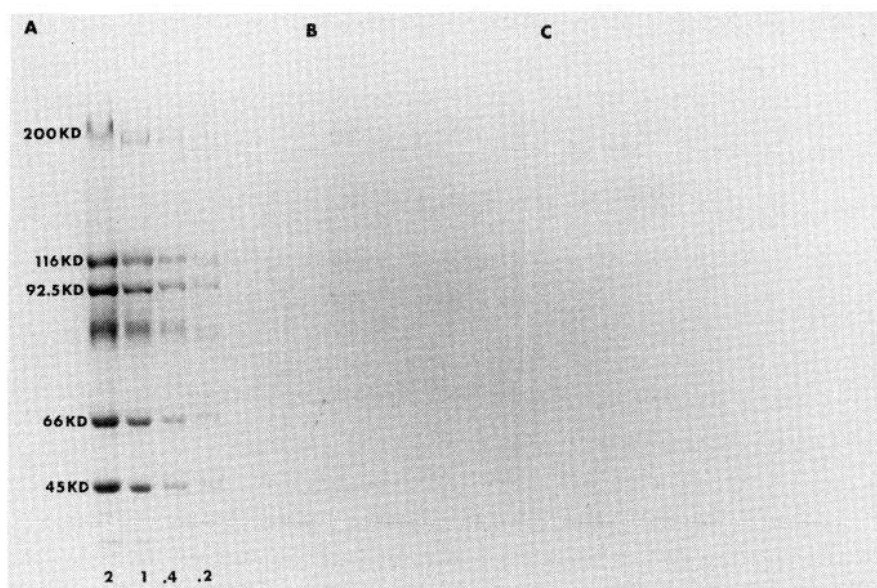


Fig. 1. SDS-polyacrylamide gradient (5–15%) gel electrophoresis of a high molecular weight mixture (45,000–200,000). The protein mixture is applied at 2, 1, 0.4, and 0.2 μg of individual protein per lane, stained with Coomassie brilliant blue (A) or first blotted for 60 (B) and 90 (C) min, and then stained.

5. The transfer can be conducted either at constant voltage or at constant current. If the transfer is performed at constant current, the size of the gel should be considered so that the current density ranges between 0.8 and 1.0 mA/cm^2 . Under these conditions, the voltage will increase during the transfer, from about 10 V at the beginning to about 40 V after 2 h. Using constant voltage, the distance between the electrodes should be considered. As a rule of thumb, 20 V $\times$ h should result in an almost complete transfer. The power limit depends on the heat dissipation capacity of the apparatus. Bjerrum and Schafer-Nielsen (6) measured a core tempera-

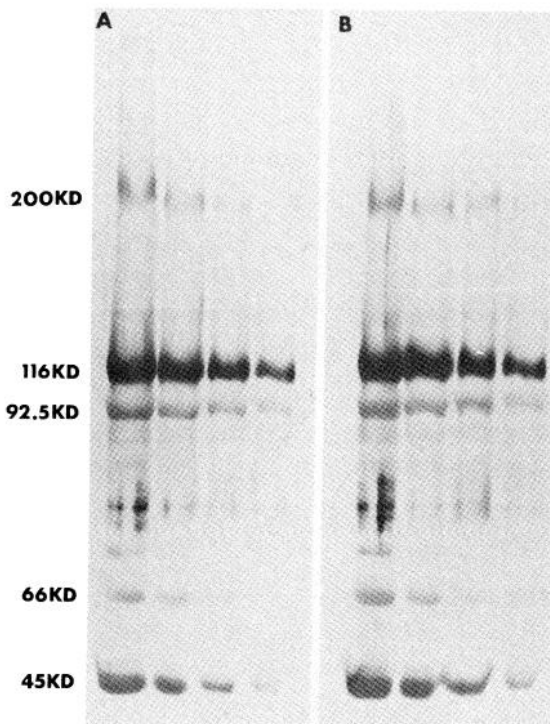


Fig. 2. Proteins transferred onto the Immobilon PVDF membrane from gels after 60 (A) and 90 (B) min or blotting and stained with Indian ink and Amido black. Blots (A) and (B) corresponds to gels 1 (B) and 1(C) respectively.

ture of 40°C after 2 h of blotting at 18 W. At a 1 mA/cm² of gel (16 x 16 cm) and 40 V, the power is 10 W, well within the limit.

6. The transfer quality and efficiency is illustrated by blotting a protein mixture with molecular weights varying between 45,000 and 200,000 (Fig. 1). This protein mixture is electrophoresed at different concentrations (2, 1, 0.4, and 0.2 µg of individual protein) in a SDS polyacrylamide gradient (5–15%) gel. Part of the gel is stained with Coomassie brilliant blue (Fig. 1A), and two identical sets of gel lanes are

transferred at a current density of 1 mA/cm² for 60 and 90 min. The blotted gels (Figs. 1B,C) are then stained with Coomassie blue, and corresponding membranes are stained with Indian ink, followed by staining with Amido black (Fig. 2A,B). A rough estimate shows that after 60 min, more than 90% of the proteins smaller than mol. wt. 116,000 are transferred, whereas some 20% of the mol. wt. 200,000 protein is still in the gel. A 90-min transfer reduces the amount of the mol. wt. 200,000 protein in the gel to about 10%, and the lower molecular weight proteins are almost completely removed from the gel.

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Chapter 30

Use of Dried Milk for Immunoblotting

Rosemary Jagus and Jeffrey W. Pollard

1. Introduction

Immunoprobng for specific peptides on solid supports such as nitrocellulose requires prior blocking of nonspecific protein binding sites. Many proteins, or protein mixtures, have been used for this, including bovine serum albumin, gelatin, and calf serum, but the cheapest and most efficient method makes use of dried milk protein.

The method was first described, and compared with other methods, by Johnson et al. (1), who refer to the blocking cocktail as BLOTTO (bovine lacto transfer technique optimizer). BLOTTO consists of a 5% solution of dried milk in Tris-buffered saline, along with antifoam and antimicrobial agents. This cocktail is used for blocking, as an incubation medium, and for subse-

quent washing to remove unreacted reagents. The initial description of the technique includes blocking protocols for immunoblots (Western blots), and the probing of RNA and DNA blots with radiolabeled DNA (Northern and Southern blots). However, the main usefulness of the procedure has been for immunoblotting, along with the probing of protein blots with radiolabeled RNA and DNA (North- and South-Western blots). The reagent is effective and easy to prepare, as well as being considerably cheaper than any available alternatives.

2. Materials

1. 10 x Tris-buffered saline (10 x TBS): 1.5M NaCl, 250mM Tris-HCl, pH 7.5. Solution should be autoclaved and stored at 4°C.
2. 1x TBS: 1:10 dilution of 10x TBS in sterile distilled water. Store at 4°C.
3. BLOTTO: 5% (w/v) nonfat dried milk, 0.01% (v/v) Antifoam A (Sigma), 0.01% (v/v) sodium azide, in TBS. Store at -4°C.
4. [¹²⁵I]-protein A, specific activity, 2–10 µCi/µg. Store at -4°C.
5. Antibody directed against protein of interest. This may be a monoclonal antibody or antiserum. Immunoglobulins should be separated from whole serum by ion exchange chromatography (2). Antibodies should be stored at -70°C in TBS.

3. Methods

The following procedure is for processing two 13 x 15 cm blots.

1. Rinse the nitrocellulose blots, to which proteins have been transferred, in 50 mL of TBS. Discard the TBS.

2. Add 50 mL of BLOTTO and incubate at room temperature (15–20°C), with shaking, for 1 h.
3. Discard the BLOTTO. Add 50 mL of fresh BLOTTO containing the first antibody.
4. Incubate at room temperature for 2–15 h, with shaking.
5. Rinse briefly in 50 mL of TBS. Discard the TBS.
6. Rinse three times in 50 mL of BLOTTO for 5 min each rinse, at room temperature, with shaking.
7. Add fresh BLOTTO containing the second antibody, if used (*see* section 4).
8. Incubate at room temperature for 2 h, with shaking.
9. Rinse briefly in 50 mL of TBS, then discard the TBS.
10. Rinse three times in 50 mL of BLOTTO for 5 min each rinse, at room temperature, with shaking.
11. Add fresh BLOTTO containing 20 μ Ci of [125 I] protein A, 0.1% (v/v) Triton X-100 and 0.1% (w/v) sodium dodecyl sulfate.
12. Incubate at room temperature for 2 h, with shaking.
13. Rinse briefly in 50 mL of TBS. Discard the TBS.
14. Rinse three times in 50 mL of BLOTTO containing 0.1% (v/v) Triton X-100 and 0.1% (w/v) sodium dodecyl sulfate, for 10 min each rinse, at room temperature, with shaking.
15. Rinse three times, briefly, in TBS.
16. Dry, mount blots on card, cover with plastic wrap, and expose to Kodak XAR-5 film, sandwiched between intensifying screens, at –70°C for 5–24 h.

4. Notes

1. Protein A, from *Staphylococcus aureus*, recognizes the Fc region of many, but not all, IgGs. For instance, rabbit and human IgGs are recognized by protein A, but mouse, sheep, and chicken IgGs are

not. If using a first antibody not recognized by protein A, it is necessary to use a second, "sandwich," antibody. For instance, if the available antibody is a mouse monoclonal antibody, then it is necessary to use, for instance, rabbit anti-mouse IgG as the second antibody. An additional advantage of using a sandwich antibody is that, since most antibodies are divalent, amplification of the signal occurs, increasing the sensitivity of the method.

2. The sensitivity of the method varies according to the avidity of the first antibody for the protein of interest, but can be sensitive down to 0.5 ng. Figure 1 shows a BLOTTO-processed immunoblot of 0.5–2 ng of the beta-subunit of eukaryotic initiation factor 2 (eIF-2), using affinity-purified sheep anti-beta-subunit as the first antibody. Details of the methodology can be found in Austin et al. (3).
3. Other methods of detection can be used. For instance, instead of being ^{125}I -labeled, protein A can be conjugated to horseradish peroxidase (4), alkaline phosphatase (5), or colloidal gold (4). Alternatively, second antibodies radiolabeled with ^{125}I or conjugated to the aforementioned components can be used (*see* Chapters 31–34 in this volume).
4. Even with the use of an antimicrobial agent, BLOTTO can turn sour. The use of putrid BLOTTO gives rise to high backgrounds, and the "smell" test is recommended. The nitrocellulose should be handled with gloved hands; finger "extracts" contain proteins and oils and can prevent wetting. Sodium azide inhibits horseradish peroxidase and therefore, if this method of detection is employed, the azide should be left out of the BLOTTO mix. Such a mix should be used for only a few days before it is discarded.

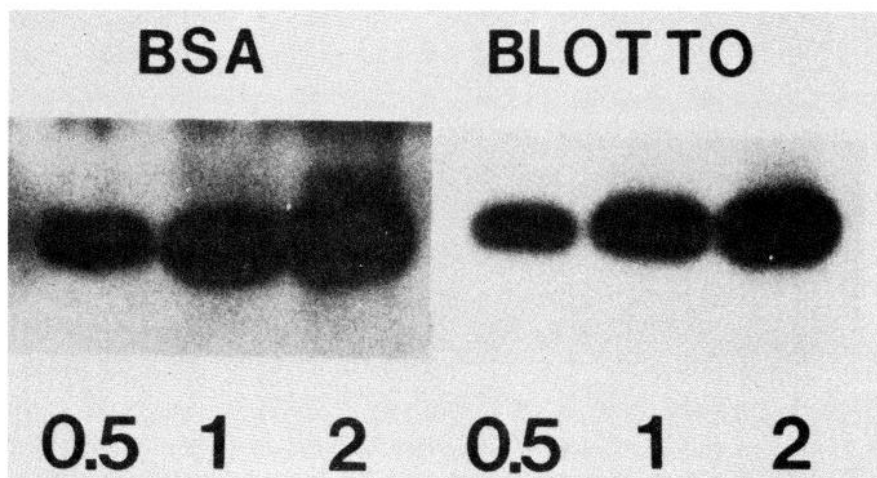


Fig. 1. Comparison of BLOTTO with BSA for blocking nonspecific protein binding sites on immunoblots. Purified eIF-2 (0.5–2 pmol) was subjected to electrophoresis by SDS-PAGE (15% polyacrylamide), and the separated subunits were transferred electrophoretically to nitrocellulose. Nonspecific protein binding sites were blocked with either BLOTTO or 3% BSA in TBS for 1 h. The blots were probed with affinity-purified antibody to the beta-subunit of eIF-2 and processed as described in this chapter. Autoradiographs are shown here.

5. Although BLOTTO has been described for use in the processing of Northern and Southern blots, such a technique gives rise to higher backgrounds than when using the conventional Denhardt's reagent. Nucleic acid binding to membrane-bound proteins (North- and South-Western blotting) appears to be significantly improved by the use of BLOTTO, however, and may require its use to guarantee specificity (6).

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Chapter 31

Immunodetection of Proteins on "Western" Blots Using ^{125}I -Labeled Protein A

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1. Introduction

Immunoblotting provides a simple and effective method for identifying specific antigens in a complex mixture of proteins. Initially, the constituent polypeptides are separated using SDS-polyacrylamide gel electrophoresis, or a similar technique, and then are transferred either electrophoretically or by diffusion onto a nitrocellulose filter. The transfer of proteins from gels onto membranes is discussed in vol. 1 of this series and Chapters 28 and 29, this volume. Once immobilized on a sheet of nitrocellulose, specific polypeptides can be identified using antibodies that bind to antigens retained on the filter. In this chapter we describe conditions appropriate for binding antibodies to immobilized proteins and a method for locating these antibody-antigen complexes using ^{125}I -labeled protein A. These methods are based on those of Burnett (1) and Towbin et al. (2).

Although there are several different techniques for visualizing antibodies bound to nitrocellulose, radiolabeled protein A is both convenient and sensitive. Protein A specifically binds the Fc region of immunoglobulin G (IgG) from many mammals (*see* Chapter 26). Thus this compound provides a general reagent for detecting antibodies from several sources. Using this method, as little as 0.1 ng of protein may be detected, though the precise amount will vary with the specific antibody titer (3). The principal disadvantage of protein A is that it fails to bind effectively to major IgG subclasses from several experimentally important sources, such as rat, goat, and sheep. For antibodies raised in such animals, a similar method using iodinated protein G may be suitable (*see* section 4).

Both antigens and antisera can be screened efficiently using immunoblotting. Probing of a crude extract after fractionation by SDS-polyacrylamide gel electrophoresis indicates the specificity of an antiserum. The identity of the antigen can be confirmed using a complementary technique, such as immunoprecipitation of enzyme activity. This information is essential if the antibodies are to be used reliably. Once characterized, an antiserum may be used to identify antigenically related proteins in other extracts using the same technique. Examples of the potential of immunoblotting have been described by Towbin and Gordon (4).

2. Materials

1. Electrophoretic blotting system, such as Trans-Blot, supplied by Bio-Rad.
2. Nitrocellulose paper: 0.45 μm pore size.
3. ^{125}I -labeled protein A, specific activity 30 mCi/mg. Affinity-purified protein A, suitable for blotting, is available commercially (*see* Note 1 in section 4). ^{125}I

emits γ -radiation. Check the procedures for safe handling and disposal of this radioisotope.

4. Washing solutions: Phosphate buffered saline (PBS): Make 2 L containing 10 mM NaH_2PO_4 , 150 mM NaCl adjusted to pH 7.2 using NaOH. This solution is stable and may be stored at 4°C . It is susceptible to microbial contamination, however, and is usually made as required.

The other washing solutions used are made by dissolving the appropriate weight of bovine serum albumin or Triton X-100 in PBS. Dissolve bovine serum albumin by rocking the mixture gently in a large, sealed bottle to avoid excessive foaming. The "blocking" and "antibody" solutions containing 8% albumin may be stored at -20°C and reused several times. Microbial contamination can be limited by filter-sterilizing these solutions after use or by adding 0.05% (w/v) NaN_3 . Other solutions are made as required and discarded after use.

5. Staining solutions: These are stable at room temperature for several weeks and *may be reused*.

Amido black stain (100 mL): 0.1% (w/v) Amido black in 25% (v/v) propan-2-ol, 10% (v/v) acetic acid.

Amido black destain (400 mL): 25% (v/v) propan-2-ol, 10% (v/v) acetic acid.

Ponceau S stain (100 mL): 0.2% (w/v) Ponceau S, 3% (w/v) trichloroacetic acid.

Ponceau S destain (400 mL): distilled water.

3. Method

3.1. Immunodetection of Polypeptides

1. Following SDS-polyacrylamide gel electrophoresis, electroblot the polypeptides from the gel onto

nitrocellulose at 50 V for 3 h using a Bio-Rad Trans-Blot apparatus. This technique and other suitable transfer methods have been described in detail in vol. 1 of this series and Chapters 28 and 29 of this volume.

2. After blotting, transfer the nitrocellulose filters individually to plastic trays for the subsequent incubations. Ensure that the nitrocellulose surface that was closest to the gel is uppermost. Do not allow the filter to dry out, since this often increases non-specific binding and results in heavy, uneven backgrounds. The nitrocellulose filter should be handled sparingly to prevent contamination by grease or foreign proteins. Always wear disposable plastic gloves, and only touch the sides of the filter.
3. Rinse the nitrocellulose briefly with 100 mL of PBS. Then incubate the blot at room temperature with the following solutions, shaking gently (*see* Note 2 in section 4).
 - (a) 50 mL of PBS/8% bovine serum albumin for 30 min. This blocks the remaining protein-binding sites on the nitrocellulose.
 - (b) 50 mL of PBS/8% bovine serum albumin containing 50–500 μ L of antiserum for 16 h (*see* Note 3 in section 4).
 - (c) Wash the nitrocellulose at least five times, each time using 100 mL of PBS for 15 min, to remove unbound antibodies.
 - (d) 50 mL of PBS/4% bovine serum albumin containing 1 μ Ci 125 I-labeled protein A for 2 h.
 - (e) Wash the nitrocellulose at least five times, each time using 100 mL PBS/1% Triton X-100 for 5 min, to remove unbound protein A.

To ensure effective washing of the filter, pour off each solution from the same corner of the tray and replace the lid in the same orientation.

4. If desired, stain the blot for total protein as described below.
5. Allow the filter to dry. Do not use excessive heat—nitrocellulose is potentially explosive when dry.
6. Mark the nitrocellulose with radioactive ink to allow alignment with exposed and developed X-ray film.
7. Fluorograph the blot using suitable X-ray film and intensifying screens (*see* vol. 1 of this series). Expose the film at -70°C for 6–72 h, depending on the intensity of the signal.

3.2. Staining of Total Protein

Either of the following stains is suitable for visualizing polypeptides after transfer onto nitrocellulose. Each can detect bands containing about 1 μg of protein. Coomassie blue is unsuitable, since generally it produces heavy background staining. An alternative staining method using colloidal gold is given in chapter 34.

3.2.1. Amido Black

Incubate the filter for 2–5 s in 100 mL of stain solution. Transfer immediately to 100 mL destain solution, and wash with several changes to remove excess dye. Unacceptably dark backgrounds are produced by longer incubation times in the stain solution.

3.2.3. Ponceau S

Incubate the filter with 100 mL of Ponceau S stain solution for 30 min. Wash excess dye off the filter by rinsing in several changes of distilled water. The proteins may be destained by washing the filter in PBS (*see* note 4 in section 4).

4. Notes

1. Iodination of protein A using Bolton and Hunter reagent labels the ϵ -NH₂ group of lysine, which apparently is not involved directly in the binding of protein A to the Fc region of IgG. This method is preferable to others, such as those using chloramine T or iodogen, which label tyrosine. The only tyrosine residues in protein A are associated with Fc binding sites, and their iodination may reduce the affinity of protein A for IgG (5).
2. Nonspecific binding is a common problem in immunoblotting. Several factors are important in reducing the resulting background.

First, the filter is washed in the presence of an "inert" protein to block the unoccupied binding sites. Bovine serum albumin is the most commonly used protein, but others, such as fetal calf serum, hemoglobin, gelatin, and nonfat dried milk (*see* Chapter 30, this volume), have been used successfully. Economically, the latter two alternatives are particularly attractive.

The quality of protein used for blocking is important, since minor contaminants may interfere with either antigen-antibody interactions or the binding of protein A to IgG. These contaminants may vary between preparations and can be sufficient to inhibit completely the detection of specific polypeptides. Routinely we use bovine serum albumin (fraction V) from Sigma Chemical Co. (product number A 4503), but no doubt albumin from other sources is equally effective. The suitability of individual batches of protein should be checked using antisera known to react well on immunoblots.

Second, the background may be reduced further by including nonionic detergents in the appropri-

ate solutions. These presumably decrease the hydrophobic interactions between antibodies and the nitrocellulose filter. Tween 20, Triton X-100, and Nonidet P-40 at concentrations of 0.1–1.0% have been used. In our experience, such detergents may supplement the blocking agents described above, but cannot substitute for these proteins.

Third, the nitrocellulose must be washed effectively to limit nonspecific binding. For this, the volumes of the washing solutions should be sufficient to flow gently over the surface of the filter during shaking. The method described in this chapter is suitable for 12 x 7 cm filters incubated in 14 x 9 cm trays. If substantially larger filters are used, the volumes of the washing solutions should be increased.

3. The exact amount of antibody to use will depend largely on its titer. Generally it is better to begin by using a small amount of antiserum. Excessive quantities of serum tend to increase the background rather than improve the sensitivity of the technique. Nonspecific binding can often be reduced by decreasing the amount of antibody used to probe the filter.
4. If desired, the nitrocellulose filter may be stained with Ponceau S immediately after electroblotting. This staining apparently does not affect the subsequent immunodetection of polypeptides, provided that the filter is thoroughly destained using PBS before incubation with the antiserum. Initial staining allows tracks from gels to be separated precisely and probed individually. This is useful when screening several antisera.
5. Protein G, a cell wall component of group G streptococci, binds to the Fc region of IgG from a wider

range of species than does protein A (6). Therefore antibodies that react poorly with protein A, particularly those from sheep, cow, and horse, may be detected by a similar method using ^{125}I -labeled protein G. In principle, the latter is more versatile. At present, however, the limited availability of protein G means that, when appropriate, labeled protein A is still preferable.

6. Particular care should be taken when attempting to detect antigens on nitrocellulose using monoclonal antibodies. Certain cell lines may produce antibodies that recognize epitopes that are denatured by detergent. Such "conformational" antibodies are unlikely to bind to the antigen after SDS-polyacrylamide gel electrophoresis.
7. Even prior to immunization, serum may contain antibodies, particularly against bacterial proteins. These antibodies may recognize proteins in an extract bound to the nitrocellulose filter. Therefore, when characterizing an antiserum, control filters should be incubated with an equal amount of pre-immune serum to check whether such preexisting antibodies interfere in the immunodetection of specific proteins.
8. Quantitation of specific antigens using this technique is difficult and must be accompanied by adequate evidence that the radioactivity bound to the filter is directly related to the amount of antigen in the initial extract. This is important, since polypeptides may vary in the extent to which they are eluted from the gel and retained by the nitrocellulose. Additionally, in some tissues proteins may interfere with the binding of antigen to the filter. Therefore the reliability of the technique should be checked for each extract.

Perhaps the best evidence is provided by determining the recovery of a known amount of pure antigen. For this, duplicate samples are prepared, and to one is added a known amount of antigen comparable to that already present in the extract. Ideally the pure antigen should be identical to that in the extract. The recovery is calculated by comparing the antigen measured in the original and supplemented samples. Such evidence is preferable to that obtained from only measuring known amounts of pure antigen. The latter indicates the detection limits of the assay, but does not test for possible interference by other components in the extract.

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Chapter 32

Detection of Protein Blots Using the Avidin–Biotin System

Michael J. Dunn and Ketan Patel

1. Introduction

The high resolution capacity of polyacrylamide gel electrophoresis for complex protein mixtures can only be fully exploited if suitable procedures exist for the characterization of the separated components. One approach to this problem is to investigate the interactions of the separated proteins with specific antibodies or other ligands such as lectins. This can be achieved by applying antisera directly to polyacrylamide gels after electrophoresis, but this technique of "immunofixation" is inefficient and time-consuming because of the slow rate of diffusion of antibodies into the gel matrix. This has resulted in the development of methods for transferring the pattern of separated proteins out of the gels onto thin substrates. This technique is known as "Western" blotting. The most popular substrate for proteins is current-

ly nitrocellulose. Positively charged nylon membranes have a greater protein-binding capacity than nitrocellulose and can be used successfully with antibodies, but they have not proved popular because of the lack of an easy-to-use general staining procedure. Protein transfer can be achieved by capillary, contact-diffusion, or vacuum techniques, but more rapid and efficient transfer is achieved if the proteins are electrophoretically removed from the gel to the support by application of an electric field perpendicular to the plane of the gel. This technique, based on the method of Towbin et al. (1), is known as electroblotting. The methodology was described in vol. 1 of this series (2) and is reviewed in refs. 3-5.

After proteins have been transferred to nitrocellulose, all the unoccupied potential binding sites on the blot must be blocked before probing with a specific ligand. This is often achieved by soaking the blot in a solution of bovine serum albumin (1) or gelatin (6). These proteins do not always prove to be nonreactive during subsequent probing, however, particularly if lectins are used as probes, because of the presence of contaminating glycoproteins in the blocking reagent. An alternative strategy is to use a dilute solution of nonionic detergent for the blocking step, and polyethylene sorbitan monolaurate (Tween 20) has been found to be particularly suitable (7). It should be remembered, however, that certain proteins can be displaced from nitrocellulose by nonionic detergents.

Once the blot has been blocked, it can then be reacted with the specific antibody or ligand. Although it is feasible to label the primary antibody to give a direct visualization of the blot, indirect sandwich methods using a second antibody are generally employed because of the increased sensitivity that can be obtained.

The second antibody can be fluorescently labeled (e.g., fluorescein isothiocyanate, FITC), radiolabeled (usually with ^{125}I), or conjugated to an enzyme such as horseradish peroxidase or alkaline phosphatase. Fluorescent methods are not popular because of the requirement for UV illumination to visualize the result. Methods employing radiolabels are very sensitive, but since an autoradiographic step is required, there is a considerable time delay before the result is obtained. This has made methods dependent on enzyme activity very popular, as in the case of enzyme-linked immunoassays (ELISA), since they are very sensitive and yield rapid results. The secondary antibody can be replaced in some cases with *Staphylococcus aureus* protein A, which specifically binds to the Fc regions of IgG. A recent development is the use of gold-labeled secondary antibodies (8), which are very sensitive and have the added advantage that they are directly visible (red color) without further development. In addition, sensitivity can be increased by a silver enhancement procedure (see also Chapter 34).

A technique that has been found to increase the sensitivity of detection of blots exploits the specificity of the interaction between the small molecular weight vitamin biotin and the protein avidin. Many biomolecules (antibodies, lectins) can be readily conjugated with biotin and used as the second reagent in blot probing. The blots are then visualized using in a third step, avidin conjugated with an enzyme (e.g., peroxidase, alkaline phosphatase, β -galactosidase, glucose oxidase). Even greater sensitivity can be achieved at this stage by using preformed complexes of a biotinylated enzyme with avidin since many enzyme molecules are present in these complexes producing an enhanced signal. Egg white avidin (mol. wt. 68,000) is often used in these

procedures, but has two distinct disadvantages: (1) it is highly charged at neutral pH so that it can bind to proteins nonspecifically, and (2) it is a glycoprotein that can interact with other biomolecules such as lectins via the carbohydrate moiety. It is, therefore, advantageous to use streptavidin (mol. wt. 60,000), isolated from *Streptomyces avidinii*, which has a pI close to neutrality and is not glycosylated.

2. Materials

1. Phosphate buffered saline (PBS) containing 0.05% (w/v) Tween 20 (PBS/Tween). For 1 L of solution, use:
0.2 g KH_2PO_4
1.15 g Na_2HPO_4
0.2 g KCl
8.0 g NaCl
0.5 g Tween 20
2. Biotinylated affinity-isolated rabbit immunoglobulins to mouse immunoglobulins (DAKO) diluted 1:200 with PBS/Tween (biotinylated second antibody).
3. Peroxidase-conjugated avidin (DAKO), diluted 1:10,000 with PBS/Tween.
4. Diaminobenzidine (DAB) solution: 0.05% (w/v) Diaminobenzidine and 0.01% (v/v) Hydrogen peroxide (from 30% stock solution) in PBS/Tween. Prepare fresh when required.

3. Method

Note: All steps are carried out with gentle shaking.

1. The blot is blocked by incubation in PBS/Tween for 90 min at room temperature.
2. The blot is then incubated for 60 min at room temperature with the primary antibody diluted at

an appropriate concentration in PBS/Tween. In the example illustrated in Fig. 1 (left panel), mouse monoclonal antibody to desmin (Amersham) was diluted 1:250 (*see* Note 4 in section 4).

3. The blot is washed for 3 x 10 min with PBS/Tween at room temperature.
4. The blot is incubated for 60 min at room temperature with biotinylated secondary antibody diluted in PBS/Tween.
5. The washing step 3 is repeated.
6. The blot is incubated for 60 min at room temperature with the streptavidin-peroxidase diluted in PBS/Tween.
7. The washing step 3 is repeated.
8. The blot is visualized by the addition of DAB solution for 10 min at room temperature. Brown bands of peroxidase activity are detected (*see* Fig. 1, left panel). Other suitable substrates for peroxidase-linked antibodies are given in Chapter 33.

4. Notes

1. If lectins are used to detect glycoproteins, a similar method to that described here can be used if an antibody against the lectin is available. Alternatively, if the lectin is conjugated with biotin (many conjugates are available commercially), it can be used directly so that the secondary antibody procedure (step 4 and 5 in section 3) can be omitted. An example of this technique using biotinylated concanavalin A and alkaline phosphatase-conjugated avidin visualized with Fast Red is shown in Fig. 1 (right panel).
2. Other enzymes conjugated with avidin can be used with appropriate substrates at step 6 in section 3

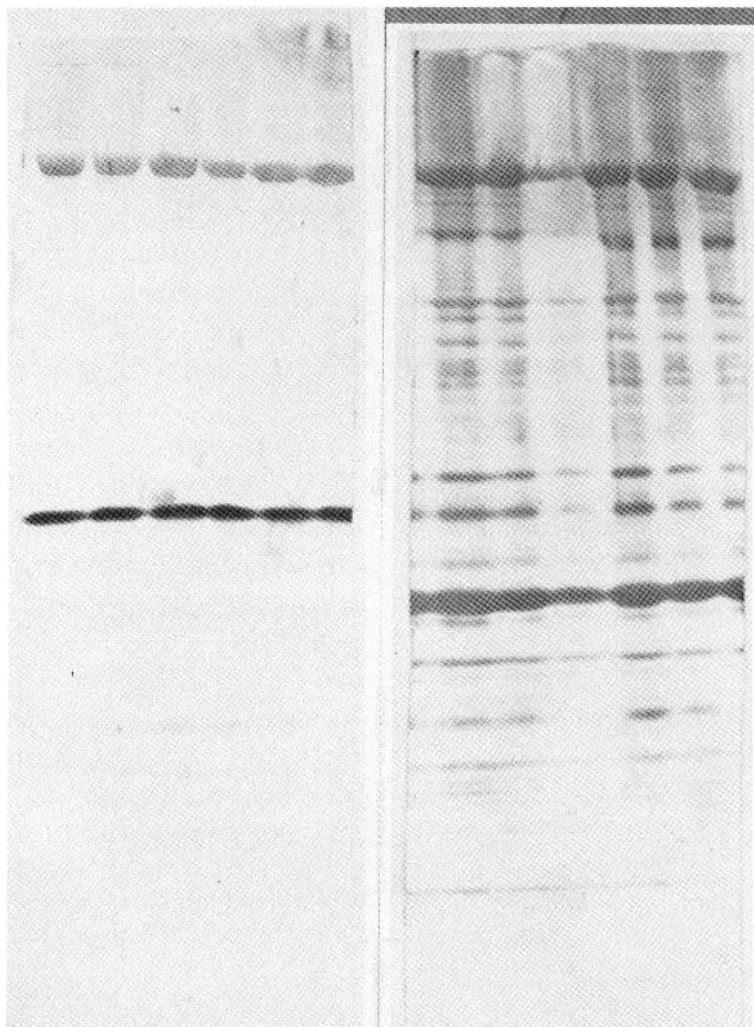


Fig. 1. Human skeletal muscle tissue proteins were separated by SDS-PAGE and transferred to nitrocellulose by electroblotting. The blots were blocked using PBS/Tween. (left) The blot was reacted sequentially with (1) mouse monoclonal antibody to desmin, (2) biotinylated antibody to mouse immunoglobulin, and (3) peroxidase-conjugated avidin, and (4) visualized using DAB. (right) The blot was reacted sequentially with (1) biotinylated Concanavalin A and (2) alkaline phosphatase-conjugated avidin, and (3) visualized with Fast Red.

- (e.g., β -galactosidase, alkaline phosphatase (*see also* Chapter 33).
3. If an avidin-biotinylated enzyme complex (ABC) is required to enhance sensitivity, it is used at step 6 (section 3) suitably diluted in PBS/Tween. Reagents for these complexes are available from Vector Laboratories (avidin with biotinylated peroxidase, alkaline phosphatase, or glucose oxidase), DAKO (avidin with biotinylated peroxidase or alkaline phosphatase), and Amersham (streptavidin with biotinylated peroxidase or β -galactosidase).
 4. No general guide can be given to the appropriate dilution for any particular primary antibody. This must be established for each antibody using serial dilutions, so that a strong specific signal is obtained, with minimal nonspecific and background staining.

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Chapter 33

Detection of Protein Blots Using Enzyme-Linked Second Antibodies or Protein A

J. M. Walker and W. Gaastra

1. Introduction

The immunodetection of a specific protein on a protein blot relies initially on the ability of a primary antibody to bind specifically to the protein of interest. This is simply achieved by washing the nitrocellulose paper in a dilute solution of the antibody. Before this step is carried out, however, unblocked protein binding sites on the nitrocellulose paper must be masked by washing with a general blocking solution (e.g., bovine serum albumin solution, dilute bovine serum, gelatin solution, hemoglobin solution, and so on). In this way, nonspecific binding of the antibody (which is itself a protein) to nitrocellulose is prevented, and any antibody that does bind to the paper will be doing so by a

genuine antigen–antibody reaction alone. This antigen–antibody reaction must now be visualized. This can be achieved by washing the paper with an appropriately labeled ligand that binds specifically to the primary antibody. This ligand is generally of two possible types:

1. Protein A (which binds specifically to IgG molecules) either radiolabeled or linked to a marker enzyme (see below).
2. A second antibody (anti-IgG) that binds to the primary antibody. This second antibody is used linked to a marker enzyme. Alternatively this second antibody is linked to biotin. This biotinylated antibody is then treated with avidin linked to a marker enzyme (avidin binds with high specificity to biotin).

The general principles involved in detecting proteins on blots is shown in Fig. 1. The use of protein A is described in detail in Chapter 31, and the use of biotinylated antibodies is described in Chapter 32. With the exception of the use of radiolabeled protein A, however, all these approaches use the same principle for the ultimate detection of the required product—the use of a marker enzyme. Marker enzymes most commonly used are alkaline phosphatase or horseradish peroxidase. Both can be linked with high efficiency (while maintaining activity) to other proteins (such as antibodies, protein A, or avidin) without interfering with that protein's function. Such enzyme-linked second antibodies (or protein A or avidin) are commercially available, but can also be prepared quite easily in the laboratory (*see, for example, Chapter 27*). By washing the blot in a suitable substrate for the enzyme, the enzyme converts the colorless substrate into a colored product that

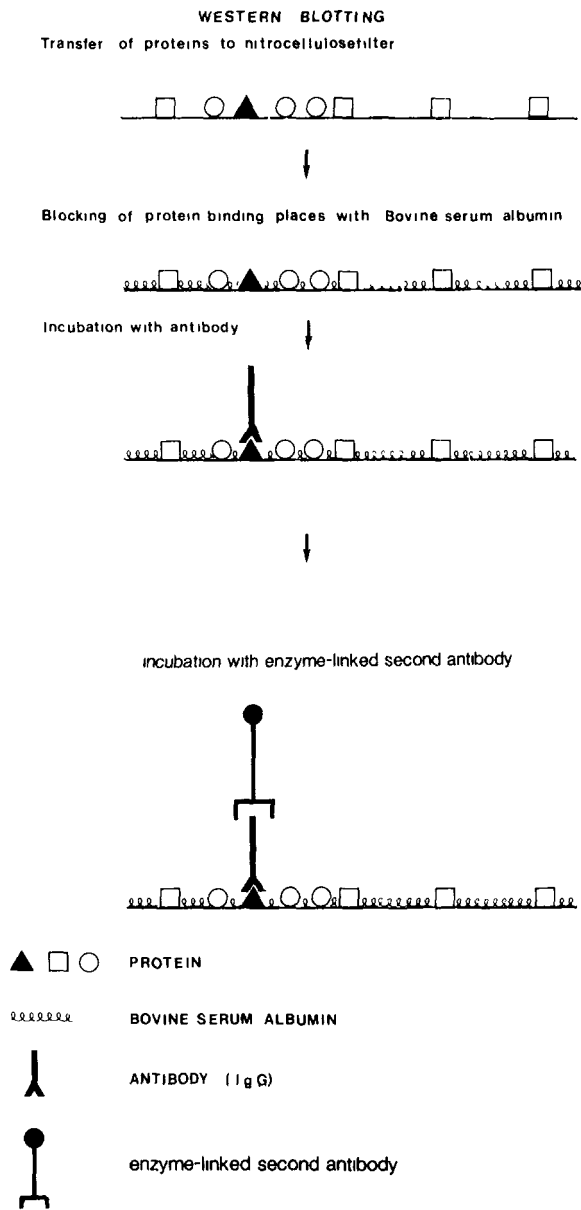


Fig. 1. Diagram showing the general principles involved in the immunodetection of protein blots.

precipitates in the paper. In this way the position of the original antigen-antibody reaction is determined. Even if only very low levels of enzyme-linked antibody (or protein A) are bound to the paper, given enough time, sufficient substrate is converted into product to allow for visualization of the reaction. The linked enzyme therefore greatly amplifies the primary antibody-second antibody interaction.

Although the basic steps involved in the immunodetection of protein blots are very straightforward, each protein blot experiment has its own problems. One of the main problems in the immunodetection of protein blots is minimizing high backgrounds. Basically, one is trying to maximize color development (i.e., maximum sensitivity), while retaining a clear background. This is especially important when low levels of antigens are being detected. Any problems can only be overcome by trial and error and can involve modifying any of the following:

1. The dilution of antisera used.
2. The protein solutions used for blocking nonspecific binding sites.
3. The composition of washing buffers and the solutions used to dilute the primary and secondary antibody.
4. The chromogenic substrate used to detect the linked enzyme.

This chapter will therefore describe the steps involved in carrying out the immunodetection of a protein blot using a primary antibody and an enzyme-linked second antibody, but in particular will describe the different blocking and washing solutions that can be used, and will examine the choice of substrates available

for detecting linked enzymes (alkaline phosphatase and horseradish peroxidase).

2. Materials

1. Protein stain: 0.15% Amido black in 62:25:10, water:methanol:acetic acid. Destain: methanol:acetic acid:water; 90:2:100. Coomassie brilliant blue is not used, since it gives high background staining with nitrocellulose.
2. Washing buffer (PBS-Tween): NaCl (8.0 g), KCl (0.2 g), Na_2HPO_4 (1.15 g), KH_2PO_4 (0.2 g), Tween 80 (0.5 mL). Make to 1 L with distilled water (*see* note 6 in section 4).
3. Blocking solutions: The use of a range of different solutions has been reported in the literature. There is no suggestion that one is better than another. For any given experiment, the most suitable solution (i.e., that which gives lowest backgrounds) is found by trial and error. The following is an example of some of the solutions that have been used and could be tried.
 - a. 4% Bovine serum albumin in 0.05M Tris-HCl, 0.15M NaCl, 0.5% Tween 20, pH 10.3.
 - b. 2% Tween in PBS.
 - c. Non-fat dried milk (*see* Chapter 30).
4. Primary and secondary antibody solutions: These are usually used at dilutions of between 1:500 and 1:2000. The optimum dilution must be determined by trial and error, however. Antibodies are diluted in solutions containing protein and detergent to minimize nonspecific adsorption to the paper. Some workers recommend the use of relatively high pH buffers to minimize nonspecific binding, but this correspondingly decreases the binding of

antibody to antigen. This can be a problem when one is using low concentrations of antibodies (e.g., culture supernatants containing monoclonal antibodies), in which case more neutral buffers should be used. The following are some typical solutions used to dilute antisera:

- a. 0.1% BSA, in 0.05M Tris-HCl, 0.15M NaCl, 0.5% Tween 20, pH 10.3 (or adjust to pH 8.5 if necessary).
- b. 0.25% Gelatin, 0.05% Nonidet P40, 50 mM Tris-HCl, 5mM Na₂ EDTA, 0.15M NaCl, pH 7.5.
- c. PBS containing 2% polyethylene glycol 6000 and 0.05% Nonidet NP-40.
5. Enzyme substrates:
 - a. BCIP substrate.
 1. 5-Bromo-4-chloro-3-indolyl phosphate (BCIP).
 2. Either 0.1M Tris-HCl, 1 mM MgCl₂, pH 8.6, or 1M ethanolamine, pH 9.8.
 - b. BCIP plus NBT substrate.
 1. Stock solutions: 5-bromo-4-chloro-3-indolyl phosphate (BCIP), 5 mg/mL in dimethylformamide. Store at -20°C.
 2. Nitro blue tetrazolium (NBT). 1 mg/mL in Veronal acetate buffer. Prepare fresh as required.
 3. 2M MgCl₂.
 4. 0.15M Veronal acetate, pH 9.6.
 - c. 4-methylumbelliferyl phosphate.
 1. 4-methylumbelliferyl phosphate.
 2. 1M Diethanolamine, 1 mM MgCl₂, pH 9.8.
 - d. Naphthol ASMX phosphate plus fast red TR salt.
 1. 3-Hydroxy-2-naphthoic acid 2,4-dimethylanilide phosphate (naphthol ASMX phosphate).
 2. Fast red TR salt. Fast red TR salt can be replaced by a range of other diazo dyes including fast blue BB salt, fast blue RR salt, fast garnet GBC, and so on.
6. 0.2M Tris-HCl, pH 8.2.

- e. 3-Amino-9-ethylcarbazole substrate.
 - 1. Stock solution: 3-amino-9-ethylcarbazole (1 mg/mL) in acetone. Store at 4°C.
 - 2. 0.05M sodium acetate buffer, pH 5.0.
 - 3. 30% (v/v) hydrogen peroxide.
- f. Diaminobenzidine.
 - 1. Stock solution: diaminobenzidine (1 mg/mL) in acetone. Store at -20°C.
 - 2. 0.05M Sodium acetate buffer, pH 5.0.
 - 3. 30% (v/v) Hydrogen peroxide.
- g. 4-Chloro-1-naphthol.
 - 1. Stock solution: 4-chloro-1-naphthol (2%) in methanol. Store at -20°C.
 - 2. 0.05M Tris-HCl, pH 7.4.
 - 3. 30% (v/v) Hydrogen peroxide.

3. Method

All incubation and washing steps are carried out with gentle shaking.

3.1. Immunodetection of Protein Blots

- 1. Gently remove the nitrocellulose filter from the gel after blotting.
- 2. Be sure to indicate on which side of the filter the proteins are bound, and if not done so before, also indicate right, left, top, and bottom of the gel.
- 3. Remove, if included, the lanes containing the molecular weight markers from the filter with a pair of scissors. The use of a scalpel on wet filters is not recommended, since this tends to damage the filter.
- 4. Stain the strip with molecular weight markers in amido black solution for 5 min. Destain. This shows the efficiency of the electrophoretic transfer and gives the necessary molecular weight markers

if needed. Alternatively the filter strip can be stained with Indian ink (1) or aurodye (see Chapter 34). Both these stains are considerably more sensitive than amido black.

5. If the transfer is satisfactory, place the rest of the nitrocellulose filter in a small plastic dish and wash it, with gentle shaking, in blocking solution (20 mL) for 30 min or longer (overnight is often used). This will block remaining binding sites on the nitrocellulose sheet. In all washing steps, take care that the nitrocellulose sheet is mobile in the washing solution and not stuck to the bottom of the plastic dish.
6. Pour off the blocking solution and wash the nitrocellulose sheet in a solution of first antibody (20 mL) for 30 min (again, this incubation can be increased if necessary: some workers use an overnight incubation).
7. Remove the first antibody solution, and wash the nitrocellulose sheet (4 x 5 min) in PBS-Tween (4 x 20 mL). This should remove all traces of any unbound first antibody.
8. Wash the nitrocellulose sheet in the appropriate second antibody solution (20 mL) for 30 min, then pour off this solution and wash the sheet (4 x 5 min) in PBS-Tween (4 x 20 mL). Finally, wash the sheet in the buffer used for the enzyme substrate (20 mL) for 5 min.
9. Pour off the buffer solution and wash the sheet in *freshly prepared* substrate solution (see below). A colored band (blue or red, depending on the enzyme and substrate used) should appear on the nitrocellulose sheet within 5–30 min. When sufficient color has developed, wash the sheet in distilled water and blot dry between two sheets of filter paper.

3.2. Preparation of Substrates

All papers should be washed in the appropriate buffer for a few minutes before adding substrate solution to bring the paper to the optimum pH for enzyme reaction.

3.2.1. Substrates for Alkaline Phosphatase (AP)

3.2.1.1. 5-Bromo-4-chloro-3-indolyl Phosphate (BCIP)

In this method (2), the enzyme converts the BCIP to the corresponding indoxyl compound, which precipitates, tautomerizes to a ketone, oxidizes, and dimerizes to form a blue indigo, which is deposited in the nitrocellulose. This substrate has been used by workers at concentrations between 0.5 and 2.0 mg/mL in various buffers (see note 1 in section 4). Two typical solutions are described below. Dissolve the BCIP in a small volume of DMSO, then add to the appropriate buffer.

1. 2.0 mg/mL in 1M ethanolamine, pH 9.8.
2. 0.5 mg/mL in 0.1M Tris-HCl, 1 mM MgCl₂, pH 8.6.

3.2.1.2. BCIP with Nitro Blue Tetrazolium (NBT)

In this method (3), the hydrogen ion released from dimerization (see above) reduces the NBT salt to the corresponding diformazan, which is intensely purple. This compound is therefore deposited along with the indigo giving an enhanced signal. To produce the substrate solution, mix the following:

1. 9 mL of Veronal-acetate buffer.
2. 0.1 mL of BCIP solution.
3. 1 mL of NBT solution.
4. 20 μ L of MgCl₂ solution.

3.2.1.3. 4-Methylumbelliferyl Phosphate

This substrate is converted to 4-methylumbelliferone, which is detected as a blueish-white fluorescent band using UV (360 nm) light. The substrate solution is dissolved at 0.1 mg/mL in 1M diethanolamine, 1 mM MgCl_2 , pH 9.8.

3.2.1.4. Naphthol ASMX-Phosphate Plus Fast Red TR Salt

Mix equal volumes of naphthol ASMX phosphate (0.4 mg/mL in distilled water) and Fast Red TR salt (6 mg/mL in 0.2M Tris-HCl, pH 8.2). Both compounds are poorly soluble, and each should be stirred for 20 min before mixing. When mixed, filter the solution, then use directly. The reaction is terminated by washing the nitrocellulose sheet in distilled water (5).

3.2.2. Substrates for Horseradish Peroxidase (HP)

All HP substrates are susceptible to oxidation and should be stored in dark bottles. When preparing solutions, rigorous stirring should be avoided to prevent unwanted oxidation.

3.2.2.1. 3-Amino-9-ethylcarbazole

Mix the following:

1. Stock solution (0.8 mL).
2. Sodium acetate buffer, pH 5.0 (20 mL).
3. 30% Hydrogen peroxide (30 μL).

This substrate produces a red-brown insoluble deposit.

3.2.2.2. Diaminobenzidine

Mix the following:

1. Stock solution (2 mL).
2. Sodium acetate buffer, pH 5.0 (50 mL).

3. 30% H_2O_2 (30 μL).

This substrate produces a red-brown insoluble deposit.

3.2.2.3. 4-Chloro-1-naphthol

Mix the following:

1. Stock solution (1.5 mL).

2. Tris-HCl buffer, pH 7.4 (50 mL).

3. 30% H_2O_2 (30 μL).

This substrate produces a blue deposit.

4. Notes

1. Knecht and Diamond (2) reported that for alkaline phosphatase, BCIP substrate gave higher backgrounds, with no increase in sensitivity, when diethanolamine buffer was used in place of Tris.
2. Although sensitivity limits using HRP or AP have been reported, they vary considerably (in the ng–pg range), and comparison of different methods is not possible since each study was carried out using different antigens, primary antibodies, and protocols. *In general*, however, AP appears to be more sensitive than HRP when direct comparisons have been made (*see*, for example, ref. 4). This may be caused in part by the fact that with AP, color development can continue for many hours with extended incubation time, whereas HRP is substrate-inactivated, resulting in an effective reaction time of the order of 30 min. Also AP-developed blots are less susceptible to fading or photobleaching and thus maintain their color for longer periods of time. HP-developed blots should be kept in the dark and preferably photographed as soon as possible after development of the color.

3. Prolonged reaction times are possible with AP using the BCIP substrates and methylumbelliferyl phosphate since when prepared, the reagents are stable and do not precipitate. When using ASMX-phosphate, however, nonspecific precipitation is a problem when using prolonged reaction times.
4. A minor disadvantage of the methylumbelliferyl substrate for AP is the need to immediately UV photograph the blot to preserve the result.
5. For all staining methods, the reaction is terminated by washing the blot in distilled water, blotting the nitrocellulose between two sheets of filter paper, and air-drying.
6. If nonspecific adsorption, leading to high backgrounds, continues to be a problem, increase the protein and detergent concentrations in the blocking and antibody solutions, and include protein in the washing buffer. The use of protein to concentrations as high as 10% is not uncommon. In addition to gelatin and BSA, many other blocking proteins have been used, e.g., hemoglobin, fetal calf serum. The only requirement of a blocking protein is that it does not cross-react with either the primary or secondary antibody. Increasing the numbers of washing steps and the volume of washing solution used can sometimes reduce background staining.
7. When using horseradish peroxidase, do not use buffers containing azide [many workers include this in low concentrations (0.05%) in buffers as an antimicrobial agent]. Azide inhibits HRP.
8. When using horseradish peroxidase, 4-chloro-1-naphthol is often the substrate of choice since both diaminobenzidine and 3-amino-9-ethylcarbazole are reported to be mutagens.
9. At no stage allow your nitrocellulose paper to dry

out. This will lead to heavy, uneven background staining.

10. Nonionic detergents are used in all solutions to minimize hydrophobic interactions between antibodies and nitrocellulose that would lead to high backgrounds. Tween 20, Triton X-100, and Nonidet NP-40 can all be used at concentrations between 0.1 and 2%.
11. In our own work we have found 2% Tween in PBS to be an excellent blocking agent, but others have reported that this is not the case. This just stresses the point that optimal conditions must be found by trial and error for each experiment.
12. Although this is stating the obvious, it is of course essential that the enzyme-linked second antibody is against IgG of the *species* in which the primary antibody was raised.
13. High background can be a particular problem when studying proteins from *Escherichia coli*. This is because the animal in which the primary antibody is raised has, at a particular time in its life, been through an infection with *E. coli*. This gives rise to anti-*E. coli* antibodies that can give substantial background staining when screening for the production of a particular protein in *E. coli*. In those cases the antiserum should be absorbed with an *E. coli* extract before use.

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Chapter 34

Colloidal Gold for the Detection of Proteins on Blots and Immunoblots

Alan Jones and Marc Moeremans

1. Introduction

Immunoblotting techniques, in which electrophoretically separated proteins are transferred to the surface of nitrocellulose or nylon membranes ("Western" blots), are becoming increasingly popular in the analysis of protein systems. Transfer techniques in protein blotting have been covered in this series (Chapter 20 in vol. 1, and Chapters 28 and 29 in this volume). In this chapter we will describe the use of the colloidal gold marking system for the detection of proteins on blots.

Immunodetection of specific bands within a separated "field" of proteins on a blot can be done in a number of ways. Protein A-¹²⁵I or peroxidase are common labels used for the visualization of these immunoblotted proteins (*see* Chapters 31 and 33). These labels require fur-

ther processing by autoradiography or chromogenic development, respectively, in order to localize the antigen to the naked eye. Using immunogold staining (IGS), a red signal is produced without further development, thus simplifying the process (1). In contrast to autoradiography, the technique is faster and operator safety is improved since the handling of shorter shelf-life radioactive substances is avoided (2–4). Nevertheless, autoradiography has the advantage that major bands and minor bands can be detected on the same blot by adapting the exposure time, without the necessity of repeating the whole experiment.

The sensitivity of the red color produced by indirect immunogold staining yields a sensitivity comparable to that of indirect peroxidase. We would, however, now strongly recommend the additional and more sensitive silver intensification step. Amplification of the signal by silver enhancement, in which the red bands of the gold-marked antibodies on the antigen become intensely black, increases the sensitivity by at least one order of magnitude, making the immunogold-silver staining (IGSS) system even more sensitive than the PAP and ABC immunoperoxidase systems (1). IGSS also produces a highly contrasted and stable signal with practically no background. Enzymatic methods also usually require possibly carcinogenic developers, and the signal tends to fade with time. The possibility of evaluating the result first with immunogold alone and then after with silver enhancement is yet another advantage of immunogold over the enzymatic methods. Figures 1 and 2 illustrate diagrammatically the IGS and particularly the IGSS techniques, and Fig. 3 compares these various detection methods.

Because of their convenience and reliability, the IGS and IGSS techniques have proved very useful for

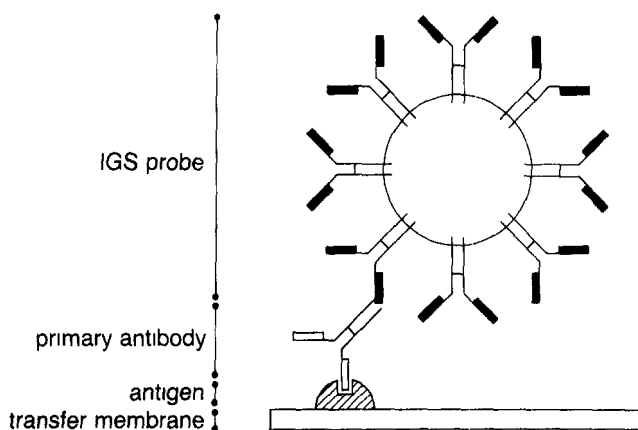


Fig. 1. Principle of the indirect visualization of antigens with primary antibodies recognizing the immobilized antigen and gold-labeled secondary antibodies (or IGS probes) directed against the primary antibody. Colloidal gold particles of about 20 nm linked to the affinity-purified secondary antibodies give the best results. The actual pattern of binding of the antibodies to the gold particle is unknown, and the illustration above is simply pictorial.

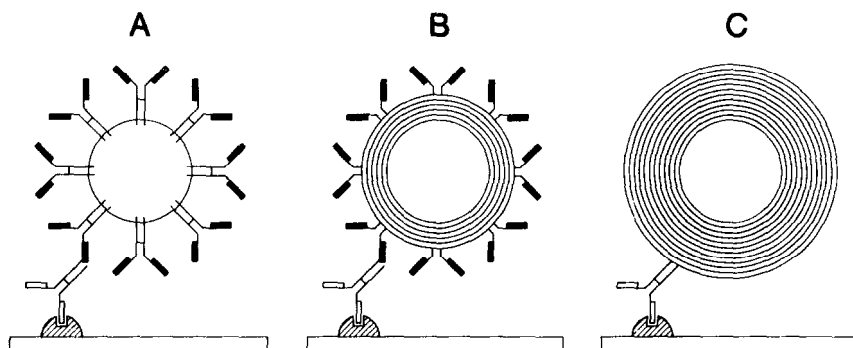


Fig. 2. Schematic representation of the silver-enhancement process. In the initial phase (A) the gold-probe piggybacks on the primary antibody, which in turn sticks to the antigen. During the silver enhancement step layers of silver selectively precipitate on the colloidal gold surface (B). The result is a significantly larger particle (C), a silver surface that generates a much more intense macroscopic signal.

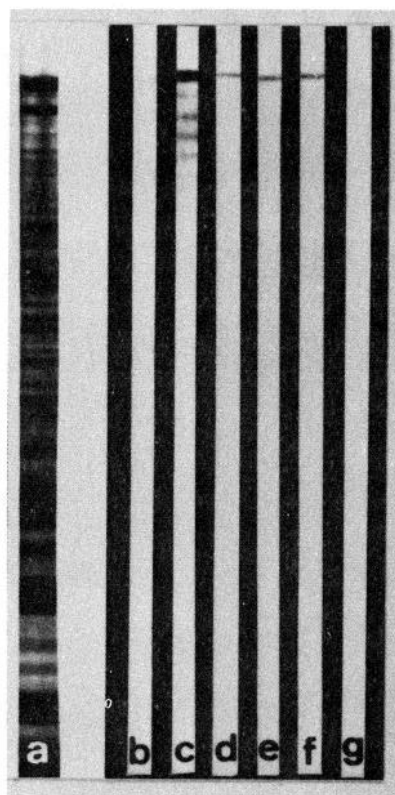


Fig. 3. Comparison of various detection methods on strips of identical nitrocellulose transfers. Strip (a) shows overall protein pattern of chicken lung epithelial cell proteins as stained by colloidal gold, i.e., AuroDye™. Strips (b)–(f) were incubated with an affinity-purified rabbit anti-filamin antibody [directed against chicken gizzard filamin (1)] at a concentration of 1 $\mu\text{g}/\text{mL}$. Strip (g) is a control of the immunogold/silver staining in which the primary antibody was omitted. Visualization method: (b) and (c) immunogold and immunogold/silver staining using AuroProbe BL GAR; (d) indirect peroxidase using GAR-HRP; (e) avidin—biotinylated horseradish peroxidase complex method (according to the instructions of the manufacturer); and (f) peroxidase anti-peroxidase. For the enzymatic methods the strips were developed with 4-chloro-1-naphthol.

screening monoclonal antibodies on strips of Western-blotted protein mixtures. It can also be expected that these techniques should become useful for serological testing, e.g., in tests of HTLV-III virus proteins. The IGS and IGSS techniques are also applicable to the specific detection of translation products from cloned c-DNA fragments in overlay assays of phage plaques or lysed bacterial colonies.

The sensitivity of immunoblotting techniques can be very high (as low as pg levels). This has created the need for an ultrasensitive staining method on a duplicate or even on the same protein blot, to make possible direct correlation between the total electrophoretogram (1-D or 2-D) and detected bands or spots in the immunoblot. Such a procedure would also be very useful for assessing the quality of the transfer from the separating gel to the blotting membrane. Conventional stains on nitrocellulose membranes are known to be very insensitive, and they do not work at all on nylon membranes.

An India ink staining method has been described (5) and in our hands has given good results. It is, however, not always possible to find a good batch of India ink. Alternative methods are based on the immunochemical detection of protein bands after derivation with 2,4-dinitrofluorobenzene, pyridoxal 5'-phosphate, or TNP (6-8).

We (9) have found that colloidal gold particles, appropriately buffered at a pH of 3.0 [and stabilized with Tween 20 (9)], can be used as a highly sensitive total protein stain on nitrocellulose membranes (Fig. 4). A similar application of colloidal gold has also been developed independently by others (10). We have called this gold stain AuroDye™, and it is available from Janssen Life Sciences Products. Gold staining of proteins transferred to nitrocellulose has subsequently become very

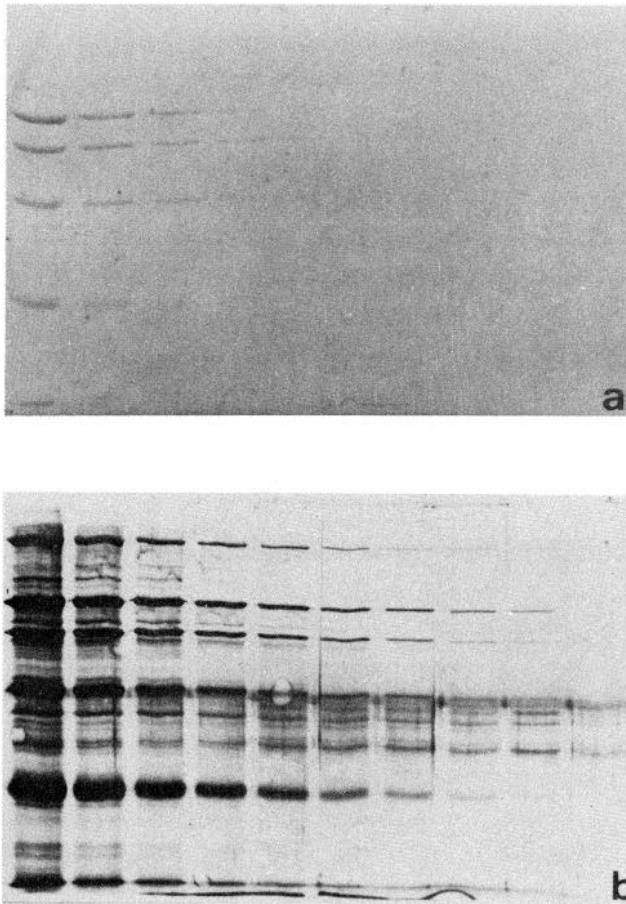


Fig. 4. Comparison of Amido Black and colloidal gold staining on nitrocellulose (b) with silver staining of SDS-polyacrylamide gel (e) and the same for FerriDye™ transferred to either nitrocellulose (c) or Zetaprobe (d). Nine dilutions (twofold) of Bio-Rad high molecular weight standards separated on a 7.5% gel. Amount per protein: (Lane 1) 1000 ng; (Lane 2) 500 ng; (Lane 3) 250 ng; (Lane 4) 125 ng; (Lane 5) 62.5 ng; (Lane 6) 31.3 ng; (Lane 7) 15.6 ng; (Lane 8) 7.8 ng; (Lane 9) 3.9 ng. The last well (10) was loaded with sample buffer alone.

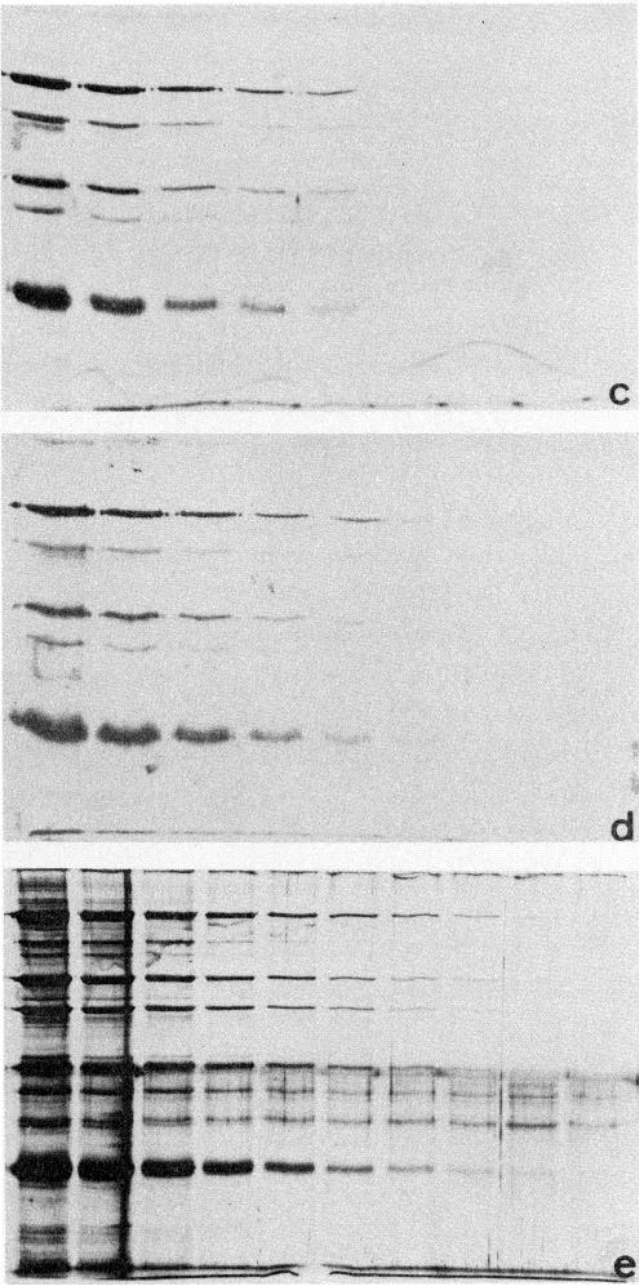


Fig. 4 (continued)

useful in our laboratories, especially since it can detect more spots in a blotted 2-D protein pattern than silver staining can detect on a polyacrylamide gel (11) (Fig. 5).

The IGSS technique can also be combined with total protein staining using colloidal gold (or iron) on the same membrane (12). This double-staining technique, especially in 2-D blots, makes the exact location of an immunodetected protein within a complex protein pattern possible (Fig. 6). A new technique recently published, describes the use of colloidal gold for negative staining of nucleic acids on nylon based filters (13). One outstanding advantage of the colloidal gold marker system lies in the fact that these same reagents can be used in immunoelectron microscopy (14, 15) and immunocytochemistry (16–18) techniques that can be used along side that of immunoblotting.

Apart from antibody–gold conjugates, colloidal gold can also be linked to a wide range of other affinity probes—enzymes, hormones, lectins, and so on. Gold linked to streptavidin is the latest in these gold conjugated probes (19).

Streptavidin–gold can either be used with biotinylated antibodies, as in "normal" immunoblotting (*see* Chap. 32), or indeed in nonantibody techniques (e.g., biotinylated hormones), and there is currently interest (20) in the clinical application of these streptavidin labels for use with biotinylated cDNA and cRNA probes for hybridization experiments on Southern and Northern blots.

Although this chapter will only describe the use of colloidal gold reagents, it is worth mentioning that, in the colloidal sphere(!), a cationic iron stain for general protein staining on nylon membranes has just been described (21) (Fig. 7).

FerriDye™ is supplied by Janssen Life Sciences Products and contains a colloidal iron sol supplemented

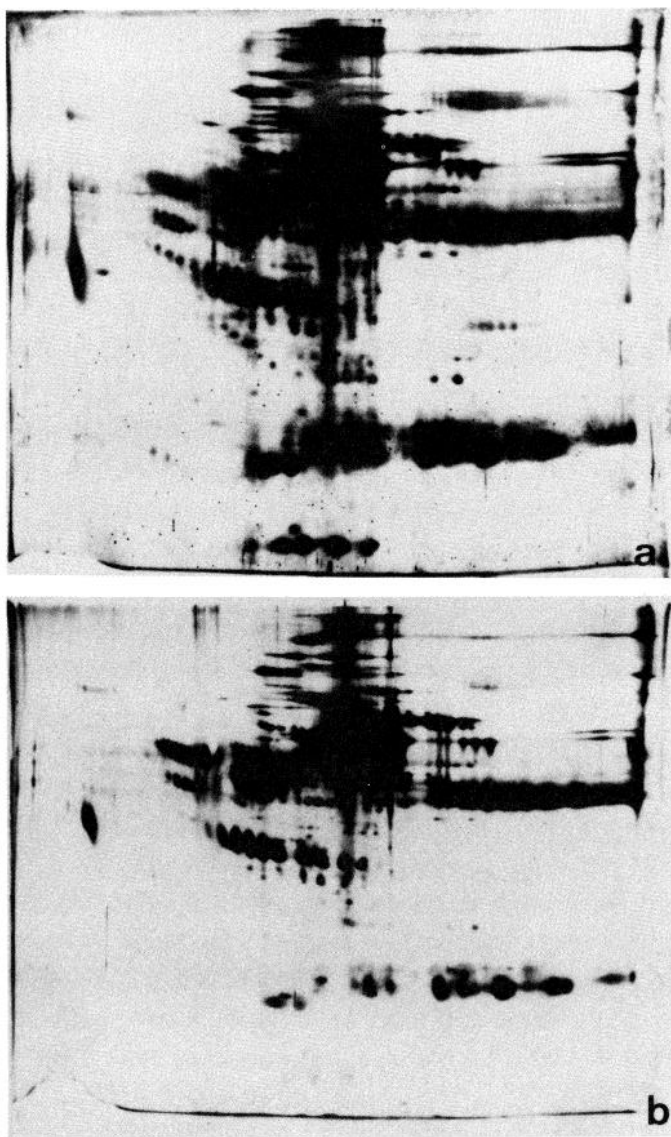


Fig. 5. Comparison of AuroDye™ on nitrocellulose (a) with silver staining of polyacrylamide gel (b). Human pleural fluid (30 mg) was run on a 2-D gel. Proteins separated in the first dimension with IEF using 2% LKB ampholines, pH 3.5–10 and 5–8, in a ratio 3:1 and in the second dimension with SDS-electrophoresis using a 10% T 2.6% C polyacrylamide gel. A duplicate gel was electrotransferred to nitrocellulose. Courtesy of Prof. Rabaey and Dr. Segers (State University of Gent, Belgium).

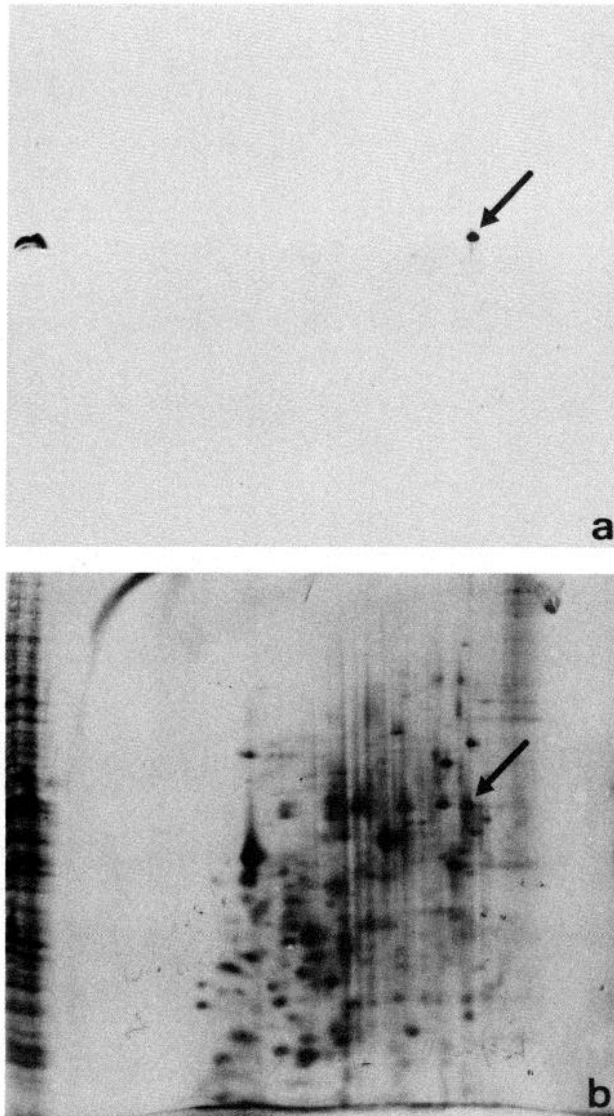


Fig. 6. A whole cell extract of rat hepatoma cells was prepared for 2-D electrophoresis according to Garrels (24). After NEPGHE in the first dimension, proteins were separated according to their molecular weight with SDS-electrophoresis on a 12% gel and electrotransferred to nitrocellulose. Immunodetection (indirect immunogold-silver staining) of tubulin (arrow) with rabbit anti-tubulin and AuroProbe™ BL GAR (a). Sequential overall protein staining of the same blot with AuroDye (b).

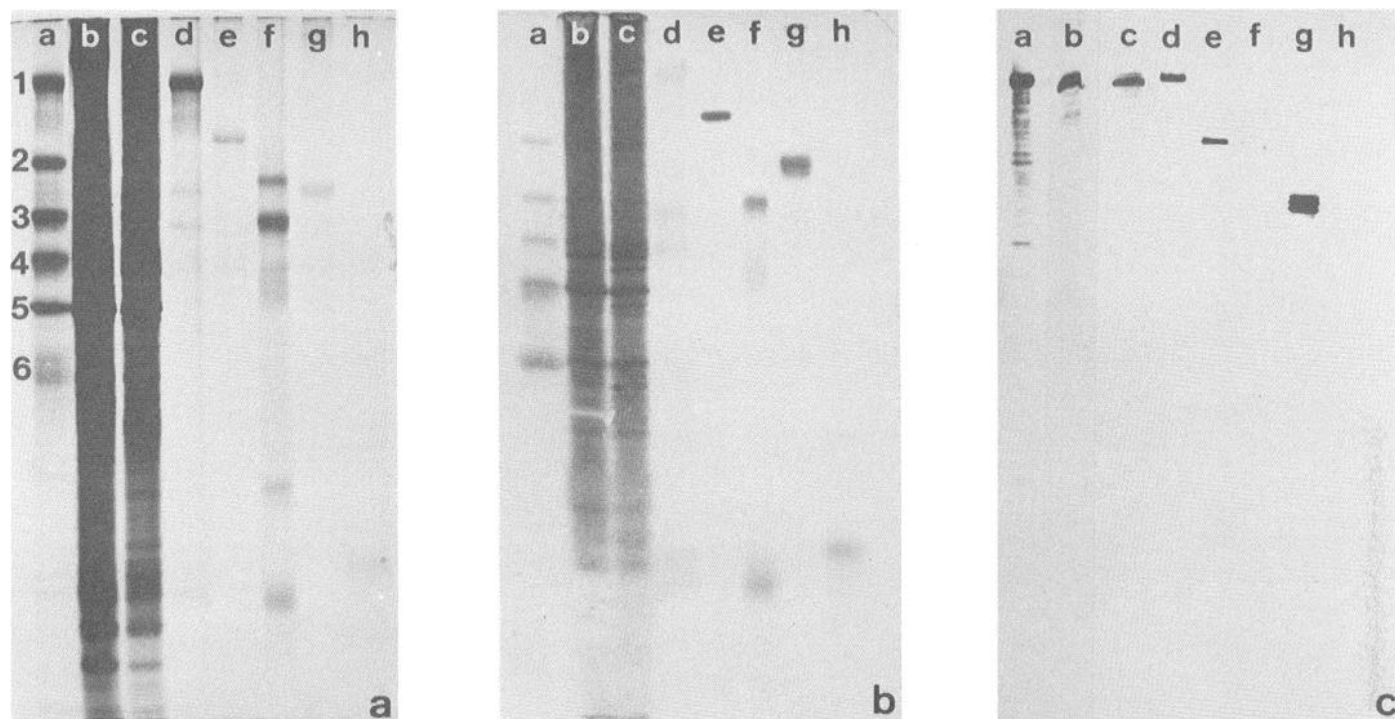


Fig. 7. Characterization of an antibody to chicken gizzard light meromyosin (25). (Panel a) Silver staining of SDS-polyacrylamide gel. (Panel b) Visualization of total protein pattern on ZetaProbe by colloidal iron, i.e., FerriDye™. (Panel c) Immunodetection of antigen after incubation with antibody: (a) purified reference proteins at 0.5 mg per band: (1) Chicken gizzard myosin heavy chain; (2) Chicken gizzard alpha-actinin; (3) bovine serum albumin; (4) rat brain tubulin; (5) chicken gizzard actin; (6) pig stomach tropomyosin. (b) Chicken heart fibroblast. (c) Chicken lung epithelial cells. (d) Chicken gizzard myosin heavy chain (0.25 mg). (e) Chicken gizzard myosin rods (0.25 mg). (f) Chicken gizzard myosin, Subfragment 1 (g) Chicken gizzard light meromyosin. (0.25 mg). (h) Chicken gizzard L20 (0.25 mg).

with additives that promote the selective binding of iron particles to sodium dodecyl sulfate (SDS)-denatured protein bands. FerriDye™ is not only markedly more sensitive than Amido black or Coomassie blue, it is also the first stain to work on nylon-based membranes (e.g., ZetaProbe, GeneScreen, Nytran, Biotodyne, and so on). Because of its high sensitivity (about 1 ng/mm²), FerriDye™ can also be successfully used with nitrocellulose blotting membranes. FerriDye™ can also be used for positive staining of nucleic acids (13). For applications in which proteins need to be stained on nitrocellulose with the highest sensitivity, the colloidal gold reagent AuroDye™, should be used. AuroDye™ is about eight times more sensitive than FerriDye™ and is compatible with proteins in both their denatured and native state. AuroDye™, however, requires more care while using, takes longer to develop, and cannot be used on positively charged membranes.

2. Materials

2.1. General

Nitrocellulose (Bio-Rad, Hoefer, Schleicher and Schull, Amersham, and so on), or nylon-based membranes such as ZetaProbe (Bio-Rad), GeneScreen (New England Nuclear), Nytran (Schleicher and Schull), Hybond-N (Amersham), or Nylon 66 (Hoefer) can be used as an immobilizing matrix. A newly introduced hydrophobic membrane, Immobilon (Millipore), is also fully compatible with the techniques described in this chapter. For good results we would recommend normal goat serum (NGS, heat-inactivated, 5 mL lyophilized), bovine serum albumin (BSA, 50 g), and high-quality gelatin (5 g) supplied as components of AuroProbe™ BL

Kits (Janssen Life Sciences Products). Recommended total protein-staining reagents are Ponceau red (reversible) or Amido black (low-sensitivity nitrocellulose), FerriDye™ (intermediate-sensitivity nylon/nitrocellulose), or AuroDye™ (high-sensitivity nitrocellulose). The latter two are available from Janssen Life Sciences Products. Reagents for the silver enhancement step are now in a kit form (IntenSE™, Janssen Life Sciences Products).

Analytical grade chemicals should be used throughout, and water should be deionized and glass-distilled. If silver enhancement is used, the glassware used in all the steps of the procedure from immunoincubation to silver enhancement should preferably be machine-washed, including an acid wash step. It should be rinsed with the same quality of water used in the silver enhancement. If washed manually, the use of brushes with metallic parts should be avoided since they give off metallic ions (such as zinc) that interfere with the enhancement process.

2.2. Immunogold/Silver Staining Procedure

The standard procedure refers to BSA-Tris. This is a buffer composed of 20 mM Tris-base, 150 mM NaCl, adjusted to pH 8.2 with 1N HCl, supplemented with 20 mM NaN_3 , and containing a (w/v)% BSA indicated by the preceding figure, e.g., 0.1% BSA-Tris.

1. Tris buffered saline (TBS): 20 mM Tris-HCl, 150 mM NaCl, 20 mM NaN_3 , pH 8.2.
2. Blocking solution (nitrocellulose): 5% (w/v) BSA in TBS.
3. Blocking solution 2 (for charge-modified nylon): 10% (w/v) BSA in TBS.

4. Washing buffer: 0.1% (w/v) BSA in TBS.
5. Gelatin buffer: (Washing buffer supplemented with gelatin.) TBS containing 0.4% (w/v) gelatin and 0.1% (w/v) BSA. Heat to 50°C to dissolve the gelatin.
6. AuroProbe™ BL: AuroProbe™ BL reagent 1% (v/v) in gelatin buffer.
7. IntenSE™ kit (enhancer, initiator, and fixing solution)
 - a. Enhancer
 - Enhancer powder, 2.35 g
 - Bidistilled water, to 50 mL
 - b. Initiator
 - Initiator powder, 3.9 g
 - Bidistilled water, to 50 mL
 - c. Fixing solution
 - Fixing solution, 10 mL
 - Bidistilled water, to 100 mL

2.3. Alternative Silver Enhancement Method *(Light Sensitive)*

The principal solutions required for this step are a citrate buffer, the silver enhancer, and the fixing solution.

1. A 2M stock solution of citrate buffer is prepared by dissolving 23.5 g of trisodium citrate • 2H₂O and 25.5 g of citric acid • H₂O in 100 mL of water. Dilute half of it ten times.
2. Next prepare the silver enhancement developing solution (77 mM hydroquinone and 5.5 mM silver lactate in 200 mM citrate buffer, pH 3.85). This solution is light-sensitive. It should be prepared in a container wrapped in aluminum foil immediately prior to use. Mix 10 mL of the citrate buffer stock solution with 60 mL double distilled water (solu-

tion A). Dissolve 0.85 g hydroquinone in 15 mL water (solution C). The solutions B and C are also light sensitive, and they should be protected by aluminum foil. Add Solution B to solution A. Add solution C and mix well. The developer should always be freshly prepared immediately before use.

2.4. Staining Blotted Proteins with Gold Particles

1. Phosphate buffered saline, pH 7.2 (PBS):
 - a. Potassium dihydrogen phosphate, 0.2 g.
 - b. Disodium hydrogen phosphate, 1.15 g.
 - c. Sodium chloride, 8.00 g.
 - d. Potassium chloride, 0.2 g.
 - e. Distilled water, to 1000 mL.
2. Washing and blocking buffer: 0.3% (v/v) Tween 20 in PBS.
3. Tween 20 stock solution: 2% (v/v) Tween solution in bidistilled water.

2.5. Double Staining Using IGSS and AuroDye™ Total Protein Staining

1. Phosphate buffered saline, pH 7.2.
2. Blocking solution: 0.3% (v/v) Tween 20 in PBS.
3. Washing buffer: 0.05% (v/v) Tween 20 in PBS.
4. Enhancer, initiator, fixing solution (for the light-insensitive method, using an IntenSE™ kit).
5. Citrate buffer, silver lactate, hydroquinone (for the light sensitive method, *see above*).

3. Methods

3.1. Immunogold/Silver Staining Procedure

3.1.1. Immunogold Staining

1. Cut a narrow strip off the separating gel to silver stain the protein pattern after electrophoresis. We

recommend the use of 0.75-mm thick gels. Fix this strip with fixing solution for 30 min, then silver stain (see vol. 1 of this series, and Chapter 13 of this volume).

The silver staining method matches the sensitivities of the IGSS technique. Alternatively the gel strip can be stained with Coomassie Brilliant Blue R-250 using the standard procedure. This is not particularly recommended, however, because of its nonmatching relative low sensitivity.

2. Transfer the proteins from the remainder of the gel to nitrocellulose or nylon-based membranes, preferably using electroblotting. One can either use conventional methods (e.g., Hoefer's Transphor) or the recently introduced semidry blotter (22; see Chapter 29 in this volume). Recommendations for obtaining optimal transfer and low background are given in section 4. *To avoid skin contact with the transfer membrane, handle blots by their edges with clean plastic tweezers.*
3. Silver stain the gel afterward to evaluate the efficiency of transfer.
4. When nitrocellulose is used, a strip of the transfer membrane can also be stained with Amido black or Ponceau Red. A dramatically increased sensitivity can be obtained, however, by using the newly developed colloidal gold stain, AuroDye™. Its sensitivity matches that of silver staining of the gel and immunodetection with IGSS on the blot membrane. [When nylon-based membranes are used, FerriDye™ (Janssen Life Sciences Products) is the only direct protein stain available for this purpose.]
5. Saturate the nonoccupied protein-binding places by incubating the transfer membrane in blocking solution 1 at 37°C for 45 min. Nylon-based mem-

branes, with their higher binding capacities, should be incubated overnight at 37°C in blocking solution 2. Do not throw away the blocking solution afterwards; it can be reused several times. After quenching excess binding sites, the blot should then be washed in 100 mL of washing buffer in a Petri dish for 2 × 5 min.

6. Incubate the saturated transfer membrane, or strips of it, under constant agitation in washing buffer supplemented with 1% normal serum and containing 1–2 µg of the primary antibody per mL (if purified). If unpurified serum is used, a dilution of preferably higher than 1/500 should be used. Use about 1 mL for every 10 cm² of transfer membrane surface. To reduce the amount of antibody, one can prepare tailored plastic bags that, after addition of the incubation fluid, are heat sealed. In this case a volume of 15–20 mL is sufficient for a 10 × 15 cm membrane. A typical incubation time, at room temperature, is 2 h. To ensure that the whole membrane is flooded by the antibody solution, the plastic bag must be placed horizontally on a tilting apparatus. A negative control consists of replacing the primary antibody solution by washing buffer plus 1% normal serum (prepared from the same species as the secondary antibody).
7. Wash the transfer membrane in washing buffer three times for 5 min. All washing steps are performed in Petri dishes with 100 mL of washing fluid.
8. Dilute the Immunogold staining (AuroProbe™ BL) reagent 1:100 in gelatin buffer. Gelatin comes with AuroProbe™ BL kits or can be purchased from Difco (Bacto gelatin 0143-01); it has been shown to inhibit nonspecific binding of gold probes to cer-

tain proteins, especially tropomyosin, without affecting the immunoreactivity.

9. Incubate the transfer membrane overnight in the diluted AuroProbe™ BL reagent under constant agitation. When a shorter secondary incubation (2 h) is required, use a more concentrated solution (1:25 dilution). To reduce the amount of AuroProbe™ BL reagent used, the incubation can again be performed in tailored plastic bags. About 15–20 mL gold reagent is sufficient for a 10 x 15 cm blot.
10. Do not throw away the incubation fluid afterward; it can be reused several times. Store at 4°C.
11. Wash the transfer membrane for 2 x 10 min in washing buffer.
12. Evaluate the result. If no signal amplification is needed, the membrane can be air-dried. More often, however, it is best to stain the transfer membrane by the silver enhancement procedure as described below. This will give the highest possible sensitivity.

3.2. Silver Enhancement (Light Insensitive) Using Janssen Life Sciences Products IntenSE™ Kit

Before proceeding with this step, it is highly recommended to carefully read the additional information in section 4.

1. Perform the IGS method as outlined above.
2. Wash the gold-stained transfer membrane for 2 x 5 min in excess of double-distilled water to remove chloride ions.
3. Just before use, prepare the solutions required for silver enhancement. These can be made from scratch or from IntenSE™ kits, produced by Jans-

sen Life Sciences Products. Make an appropriate volume (about 100 mL for a 10 x 15 cm blot) of developer solution by mixing 50 mL of enhancer solution with 50 mL of initiator solution.

4. Pour the mixed enhancer and initiator solution into a glass Petri dish. Put the gold-stained transfer membrane in this silver enhancer *immediately* afterward.
5. Enhance the gold signal on the transfer membrane. The enhancement should be monitored visually. Typical development times range from 5 to 15 min. For a further description, please read the additional information in section 4.
6. Put the transfer membrane in a Petri dish with the diluted fixing solution for 5 min (watch the signal intensity).
7. Wash the membrane for 3 x 5 min in excess water, after which it can be air-dried.

3.3. Alternative Silver Enhancement Method

The above method has been given for silver enhancement using the light-insensitive kit, IntenSE™, from Janssen Life Sciences Products. Alternatively, the silver enhancement step can be performed by a light-sensitive procedure. The principal solution required for this step are a citrate buffer, the silver enhancer, and the fixing solution.

1. The immunogold-labeled blotting membranes are further washed in an excess of distilled and deionized water for 5 min twice. It is essential that all chloride ions be removed, since chloride impurities will induce a precipitate. Then the strips or sheets are washed again, but only for 2 min, in 100 mL of the diluted citrate buffer.

2. The blotting membranes are then incubated in a glass Petri dish or the like, filled with developer, for 5–15 min. Here, too, aluminum foil should be used to protect the reaction from daylight. Strips can be developed in a dish of 15-cm diameter and sheets in a dish of 25-cm diameter. The silver enhancement can be visually checked by regularly taking a look. Avoid, however, exposing the developer to strong daylight. Possible turbidity of the developing fluid could indicate contamination with chloride ions.
3. After the reaction has been completed, the strips or sheets are transferred into the fixing solution for 5 min and then washed three times in an excess of water. Use a photographic fixer (Agfa-Gevaert), diluted 1:4, for 2 min, or alternatively, a 5% aqueous solution of sodium thiosulfate.

The treatment with fixing solution should not take too much time because the enhancement obtained then diminishes. If this seems to occur before the 5 min have passed, this period of time can be shortened. To conclude the procedure, the immunogold-silver stained transfer membranes are air-dried on a piece of filter paper.

3.4. Staining Blotted Proteins with Gold Particles

Because of the very high sensitivity obtained with AuroDye™, any protein impurities will also appear, and it is necessary to work under very clean conditions during the electrophoretic run as well as during electrotransfer. When using conventional electrotransfer in a buffer tank, putting an additional nitrocellulose membrane at the cathodic side of the gel is very helpful in filtering and adsorbing impurities out of the system that

would otherwise adsorb on the protein blot. This is not necessary when using "semidry" electrotransfer (see Chapter 29).

1. After transfer, that part of the blot to be stained with gold particles (AuroDye™) is cut off the transfer membrane. It usually contains a duplicate protein pattern that is to be probed with an (immune) overlay assay. The membrane is washed extensively on a gyratory shaker with excess washing and blocking buffer at 37°C for 30 min and three times for 15 min at room temperature. Use 0.4 mL of buffer per cm² of blotting membrane. About 100 mL (four times) is sufficient for a blot of 15 × 10 cm. This extensive washing is necessary to remove adherent polyacrylamide gel particles and residual SDS. The blot is then rinsed in excess distilled water (200 mL) for 3 min.
2. The washed blot is drained off and incubated in AuroDye™, undiluted, at a minimum 0.2 mL AuroDye™/cm² blotting membrane. With high protein loads, 0.5 mL/cm² may be needed. Sheets are incubated in sealed plastic bags and strips in stoppered tubes under constant agitation on a tilting apparatus (e.g., Hoefer's Red Rocker). If a tilting apparatus is not available, best results are obtained by using excess AuroDye™ in a clear glass or polycarbonate tray on a reciprocal or rotary shaker. Incubation should continue until optimum color formation has been obtained. In general, incubation time is in the order of 4 h to overnight. Overstaining does not occur. Background only forms on proteinaceous impurities.

The colloidal gold stain may turn blue, probably by reaction with proteins released from the blot.

Table 1
Staining Procedure with Colloid Gold

Wash extensively in excess PBS with 0.3% Tween 20 30 min at 37°C
↓
Wash extensively in excess PBS with 0.3% Tween 20 15 min at room temperature
↓
Wash extensively in excess PBS with 0.3% Tween 20 15 min at room temperature
↓
Wash extensively in excess PBS with 0.3% Tween 20 15 min at room temperature
↓
Incubate in AuroDye™ 4 h to overnight
↓
Wash blot in excess H ₂ O for 15 min and air dry

This happens when the proteins are present in high concentrations and can be relieved by diminishing the protein load, if possible. The capacity of nitrocellulose is limited to about 0.8 $\mu\text{g}/\text{mm}^2$. Otherwise, it is recommended to replace the gold sol immediately after a brief wash with excess bidistilled water.

3. After staining, the blot is washed several times with excess water and air dried. The stain turns darker upon drying and does not fade. These steps are summarized in Table 1.

3.5. Immunogold Staining of Protein Dots

Materials and procedures are identical to the above, except for the spotting procedure, which is given below.

1. Soak the transfer membrane in water.
2. Dry it in air for 15 min.
3. Dilute the antigen in an appropriate buffer containing 50 μg of BSA per mL.
4. Apply 1 μL of the diluted antigen on the transfer membrane with a pipet or a microsyringe (for example the Drummond Microdispenser).
5. Dry it in air for 15 min.

Recently several companies (e.g., BRL, Bio-Rad, and so on) have introduced small manifolds that enable the user to apply 96 dots on a sheet of transfer membrane. Using these manifolds generally results in smooth and regular dots. Immunochemical manipulations such as incubation and wash steps are also greatly facilitated.

3.6. Combined Immunogold-Silver Staining and AuroDye™ Total Protein Staining of Electrophoretically Separated Proteins Transferred to Nitrocellulose

This double staining method results in a combination of immunodetection with the staining of the complete electrophoretogram on the same membrane. Such a technique makes an exact localization of an immunodetected band within a complex protein pattern possible. It is based on the use of Tween 20 (polyoxyethylene sorbitan monolaureate) as a blocking agent and is particularly interesting for 2-D blots. The procedure requires silver enhancement materials.

1. Transfer the electrophoretically separated proteins to a nitrocellulose membrane. For optimum results, follow the recommendations regarding clean conditions as explained above. One of these recom-

mendations is the use of an additional blotting membrane at the cathodic site in the sandwich to absorb impurities from the blotting system.

2. After transfer, saturate the blotting membrane in the blocking solution for 30 min at 37°C.
3. Wash for 3 x 5 min in the blocking solution at room temperature under constant agitation. About 100 mL is sufficient for a blot of 10 x 15 cm. The extensive washing is necessary to remove adherent polyacrylamide gel particles and residual SDS. All subsequent steps are also to be carried out at room temperature. After quenching of excess binding sites with blocking buffer, the blot is washed again with washing buffer in a Petri dish for 2 x 5 min.
4. Incubate the saturated and washed transfer membrane under constant agitation in washing buffer containing 1–2 µg/mL of the primary antibody (or a suitable dilution of the primary antiserum) for 2 h. Use about 1 mL for every 10 cm² of transfer membrane. Normal serum is not used in this instance in order to reduce the protein loading and to keep any background staining to a minimum.
5. Wash for 3 x 10 min in washing buffer.
6. Dilute the AuroProbe™ BL reagent (up to 1:100) in washing buffer (solution 3). Gelatin cannot be used in this AuroProbe™ dilution because gelatin is not compatible with the AuroDye™ step that follows later. Since gelatin blocks the nonspecific binding of gold probes with certain proteins like tropomyosin, its obligatory omission may result in some nonspecifically immunostained bands. This can be controlled by omitting the primary antibody.
7. Incubate the transfer membrane for 2 h (or overnight) in the diluted AuroProbe™ BL gold reagent under constant agitation.

8. Wash for 3 x 5 min in washing buffer.
9. Wash for 2 x 5 min in double-distilled water.
10. Perform a silver enhancement step on the immunogold-stained transfer membrane as described previously.
11. Wash for 3 x 5 min in a large excess of distilled water.
12. Wash for 5 min in blocking solution.
13. Take a photograph of the result.
14. Incubate the immunogold-silver transfer membrane overnight in AuroDye™.
15. Rinse the double-stained membrane in excess water and dry it in air before storing.
16. Take another photograph of the result.

4. Notes

4.1. Use of Colloidal Gold for General Protein Staining

1. The stabilized colloidal gold sol AuroDye™, containing gold particles of 20 nm in size, has been adjusted to a pH of about 3, at which the negatively charged particles will now bind very selectively to proteins by hydrophobic and ionic interactions. The latter may result in different staining intensities of proteins differing in isoelectric point. This is to our knowledge a common feature of all protein staining methods (e.g., silver staining, Coomassie Blue, and so on). However, AuroDye™ stains more proteins on a 2-D blot that can be shown in a gel with silver staining. Indeed, some proteins are not detected with silver staining, but become clearly visible with AuroDye™. It cannot be used on nylon-based membranes since, because of these interactions, the whole membrane will be stained. The

method works well on blots of both denatured and nondenatured proteins.

2. Use of filter papers in the sandwich set-up. Whatman 3MM (No. 3030917) or Schleicher and Schull (No. 604A-24343) are recommended. Other brands should be checked for the presence of substances that are released during the transfer and produce a very disturbing speckled background.
3. The high sensitivity of AuroDye™ used in the protein staining procedures requires special care and handling to avoid background staining. High-quality chemicals should be used throughout. Many researchers routinely work with far less sensitive stains such as Amido black or with relatively insensitive overlay detection techniques. As a result, they are used to rather heavy protein loads.

Heavy protein loads on gels, when transferred to nitrocellulose, will not only be heavily stained, but will leak protein off the membrane from saturated sites. Excessive protein leakage will cause an aggregation of the gold particles and destroy the AuroDye™ reagent. The problem will be more severe with 1-D gels than with 2-D gels, in which the proteins are distributed as spots over the entire blotting membrane. In general, for 1-D gels it is recommended to use protein loads that, in the case of a complex mixture, give resolvable banding patterns after silver staining. Single bands in 1-D gels should not exceed 1000 ng/band. The ideal load for molecular weight standards is 200 ng/band. Lower loads lead to better separations and help save precious samples. For immunovisualization, AuroDye™ is best used in conjunction with a detection system such as the AuroProbe™ BL IGSS kit, which matches its high sensitivity.

Because it is practically impossible to avoid proteinaceous impurities in the transfer apparatus, it is advisable to sandwich the gel between two nitrocellulose membranes. One will give you the exact replica of the gel; the other will adsorb these impurities. As mentioned previously, however, this is unnecessary when using "semidry" blotting.

4. It is important to wear gloves when handling gels and blots. For extra care, handle blots by their edges with clean plastic (not metallic!) forceps, since gloves can leave smears. All current laboratory electrophoresis protocols may be used. 2-D gels should be thoroughly washed in several changes of excess transfer buffer, in order to remove the ampholytes, unless the first-dimension gels have been fixed selectively for proteins before use in the second dimension.
5. In a 1-D gel system, blots contain one to several lanes. Identical blots can react in parallel with various probing agents (e.g., antibodies). Usually one blot with an additional lane containing molecular weight standards is stained with AuroDye™.

For 2-D gels, the combination of an immunogold-silver overlay assay with subsequent AuroDye™ staining is particularly effective, because it facilitates correlation of detected polypeptides with the total electrophoretogram. Note that Tween 20, not BSA, should be used when double staining.

6. It is advisable not to reuse the AuroDye™ solution. The concentration of the particles will probably be too low to give optimal results.
7. Low ionic strength transfer buffers are recommended for blotting in conventional apparatus (i.e., 25 mM Tris, 192 mM glycine, 20% methanol, pH 8.3). When SDS is used in the transfer buffer to

SANDWICH SET-UP

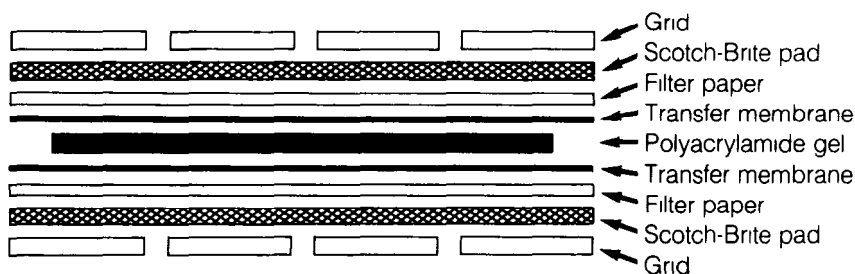


Fig. 8. Schematic representation of the protein blotting sandwich setup.

increase transfer efficiency, the following procedure is recommended. An initial transfer is done overnight at 4 V/cm with initial transfer buffer (25 mM Tris, 192 mM glycine, 20% methanol, 0.05% SDS, pH 8.3). The initial transfer buffer is then removed (can be reused once), and the apparatus, cassettes, and cooling coils are rinsed with distilled water. The apparatus is filled with secondary transfer buffer (25 mM Tris, 192 mM glycine, 20% methanol, pH 8.3), and the transfer continued at 4 V/cm for 4 h to remove SDS from nitrocellulose.

8. For obtaining optimal AuroDye™ protein staining with minimal background, it is recommended to use the following sandwich set-up for conventional electrotransfer: The cathode-oriented grid—a Scotch-Brite pad—two sheets of filter paper—a nitrocellulose sheet—the gel—a nitrocellulose sheet—two sheets of filter paper—a Scotch-Brite pad—the anode-oriented grid (Fig. 8).

We have found that the use of an extra nitrocellulose sheet at the cathodic side of the gel and of a high-quality filter paper leads to a sharp, high-contrast staining with negligible background. Wash the Scotch-Brite pads extensively with distilled water after each run to remove protein contaminants. In special cases, the AuroDye™ signal can be silver enhanced with Janssen's IntenSE™ silver-enhancement kit. This is not, however, to be considered a standard procedure, since the use of the stain by itself already gives very sensitive detection.

4.2. Double-Staining Method

1. By using Tween 20 as blocking agent, instead of BSA, it is possible that some background staining may occur because of higher nonspecific interactions. It is, therefore, essential to include a negative control. In addition, because of the subsequent protein staining, it is not possible to include gelatin in the immunogold reagent, which may result in nonspecific binding of the gold probe. Normally this is no problem because nonspecific interactions (if they occur) are shown by the negative control and can be easily distinguished from the positive signal.
2. It is sometimes difficult to recognize an immunodetected band or spot after overall protein staining. It is advisable then to take a photograph after immunodetection or to mark the detected spot before performing the overall protein staining.

4.3. Summary of the Use of Colloidal Gold General Staining

1. Colloidal gold may be used for staining the proteins on duplicate blots of overlay assays to make corre-

lation of detected bands with the total protein transfer accurate and simple. For quick reference, blots may also include an additional lane with molecular weight markers.

2. Colloidal gold may be used for staining proteins on blots that have previously been subjected to immunogold-silver staining. This is particularly useful for Western blots of 2-D gels.
3. Colloidal gold may be used for high-sensitivity staining of Western blots of 2-D gels to maximize the number of stained spots. In cases in which an extremely low background occurs, the AuroDyeTM signal can be further amplified with IntenSETM. This is not, however, to be considered as a standard procedure.
4. Colloidal gold may be used to negatively stain transfers of nucleic acids on nylon-based membranes prior to hybridization. This facilitates correlation of hybridized bands with the total electrophoretogram (13).

Should problems arise during the use of colloidal gold in general protein staining, the trouble-shooting guide in Table 2 may be useful.

4.4. Use of Colloidal Gold for Immunoblotting

1. Immunoblotting combined with immunogold staining (IGS) or immunogold-silver staining (IGSS) can be applied in a variety of experiments. The most common field of application is the study of antigens and antibodies. The technique can be used for antigen detection as well as for antibody characterization in both a qualitative and a quantitative way. Blots from electrophoretically fraction-

Table 2
Trouble-Shooting

Observations	Probable cause	Remedy
AuroDye™ turns purplish during staining	Agglutination of gold particles by proteins released from the blotting membrane	Wash blot briefly with excess water. Replace AuroDye™. If possible, use lower protein loads
AuroDye™ turns colorless during staining	Adsorption of all the gold particles by excess protein on blotting membrane	Replace AuroDye™ and double the volume/cm ² . If possible, use lower protein loads
Spotty background	Impurities released from filter paper adsorbed onto the blotting membrane during blotting	Use high-quality filter paper in electrotransfer apparatus during blotting
High background	Interference with contaminants of chemicals	Contact your distributor for a list of high quality chemicals
	Interference with proteinaceous contaminants	Use sandwich set-up as outlined above. If possible, use lower protein loads. Replace Scotch-Brite pads and follow recommendations as outlined above.
Smears on background	Incorrect handling	Handle blot with plastic forceps and avoid contact with gloves

ated antigens, for example, are most helpful in the evaluation of monoclonal antibodies. The dot blot technique on the other hand is very convenient for large-scale screening assays or for measuring the activity of an antibody. In the latter case a serial dilution series of the antigen is applied as spots (e.g., 250–0.1 ng/spot) on transfer membrane strips. The least detectable amount of antigen is a measure of the activity of the primary antibody. Similarly, protein dot blots are used to assess cross-reactions of antigens or relative sensitivities of detection systems. Cross-reaction can be expressed as the proportion between the least detectable amount of homologous antigen over the least detectable amount of heterologous antigen multiplied by 100. Another useful way is to spot, e.g., in a mini-manifold, constant amounts of antigen and to test a dilution series of the immunogold reagent.

2. The incubation time in the immunogold-reagent is influenced by the concentration of the reagent. A dilution of 1:100 ($OD_{520} = 0.05$) or even 1:200 ($OD_{520} = 0.025$) can be used overnight. When a dilution of 1:25 ($OD_{520} = 0.20$) is used, however, optimal results are obtained within 2 h.

The immunogold staining reagent can be used several times, certainly when it is diluted 1:25. Of course by reusing the reagent, longer incubation times may become necessary. The number of times the reagent can be used will be dependent on the amount consumed during the assay. This can be reduced by diminishing protein loads and relying on the stronger possible silver enhancement for obtaining satisfactory signals.

3. Controls are important. It is advisable to incubate one strip without primary antibody or if possible

with a nonrelated primary antibody. A positive control, i.e., the use of an antibody of known reactivity, is helpful to judge the experimental conditions. When nonpurified antigens give multiple bands, absorption with excess pure antigen will indicate which detected band corresponds to antibodies against it.

4. It is essential to add gelatin to the immunogold reagent. In our hands this prevented the nonspecific binding between the immunogold-reagent and tropomyosin, as demonstrated by the negative control. The source of the gelatin is equally important. If other brands of gelatin are used than the one supplied with the kit, it may be necessary to include a negative control.

4.5. Silver Enhancement

1. The enhancement reagents are extremely sensitive to the purity of the water used. Low-quality water results in the formation of precipitates that lower the reactivity of the enhancement reagent and in most cases produce a background. We therefore advise using double-distilled water throughout the procedure.

As well as the high purity of the water used, the glassware used with the silver enhancement reagents should be very clean. Contact with metallic objects and the silver enhancer or its components should be avoided.

2. Using the Janssen Life Sciences IntenSE™ kit, the enhancer and initiator solutions have to be prepared on the same day the actual silver enhancement will take place. Using solutions that are kept longer than a day will result in a deteriorated silver

Table 3
Variations in Enhancement Time Depending
on Ambient Temperature

Temperature, 0°C	Typical enhancement time, min	Typical silver enhancer stability, min
16	13.0	18.0
18	11.0	16.0
20	10.0	13.5
22	9.0	11.5
24	8.0	9.5

enhancer. The enhancer and initiator solutions do not have to be shielded from normal (day) light. We advise that the user, however, not expose them to extreme heat or light. Use care: The silver enhancer or its components can stain the skin. The actual silver enhancer is prepared by mixing equal volumes of enhancer and initiator solution. Since this silver enhancer is only usable for about 15 min (see below), its components should be mixed immediately prior to the actual silver enhancement. The silver enhancer does not have to be shielded from normal (day) light. Note: At the time of press, an advanced IntenSE™-II kit was being prepared that obviates the need for such stringency regarding water purity and will be ready to use in liquid form.

Both the time required for enhancement and the time the silver enhancer can be used are considerably sensitive to the temperature of the environment. Table 3 illustrates this.

Both the typical enhancement time and typical silver enhancer stability are somewhat arbitrary in that for some applications it may be necessary to enhance longer than suggested, and for others,

shorter. Since the enhancement can be monitored visually, the researcher can judge this personally. A comfortable margin between enhancement time and silver enhancer stability time exists around room temperature. For certain very demanding applications, in which a very strong signal amplification is required, it is possible to repeat the silver enhancement. Enhance the gold signal for 5 min or less, rinse with distilled water, and put in fresh silver enhancer for another 5 min before fixing.

After enhancement, put the sample in the fixing solution. Fixing the sample too long may cause bleaching of the signal obtained. It is also necessary, therefore, to remove any remaining fixing solution afterward by thoroughly washing the sample in distilled water.

3. Because both metallic gold and silver catalyze the reduction of silver ions, the reaction, which is initially catalytic, soon becomes autocatalytic. Anti-gen spot tests show that high concentrations of antigen (100 and 50 ng/mm²) result in the same signal intensity as judged with the naked eye. We would, therefore, advise monitoring the enhancement from time to time. Long incubations can result in the visualization of bands that are of minor importance compared to the main signal.

As already explained in the procedure, the incubation in fixing solutions must be monitored. If the signal intensity diminishes before the 5 min have passed, the fixation time can be shortened. This is especially true for weak signals. As also indicated in the procedure, it is necessary to remove the fixative by thoroughly washing with distilled water.

The light-insensitive developer makes it possible to monitor the development continuously in nor-

mal daylight. The IntenSE™ kit can be stored at room temperature.

4. Bovine serum albumin is expensive. Because of the addition of 20 mM NaN_3 to the blocking solutions, these solutions can be reused several times.
5. Recently we have formulated a new buffer that increases the speed of the immunogold color development as well as the sensitivity. Further, after silver enhancement a sharper contrast is obtained. This is a BSA-phosphate-Tween 20 buffer composed of 50 mM phosphate buffer, pH 7.5, 0.5% Tween 20 (v/v), 0.5% BSA (w/v), and 0.05% sodium azide (w/v). This buffer can be used through the whole procedure, i.e., for blocking, washing, and incubating with immunoreagents. In addition no normal serum is needed to dilute the antibody, and no gelatin is used to dilute the AuroProbe™ BL reagent.

4.6. Preparation of Colloidal Metals and Immunoprobos

For the sake of convenience in this review, the use of commercially available gold probes and colloidal metals has been referred to throughout. Please refer to ref. 23 for the preparation of gold sols and probes, ref. 9 for the preparation of colloidal gold for protein staining, and ref. 21 for the preparation of colloidal iron for general protein staining.

Acknowledgments

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Daneels, Marc De Raeymaeker, Bart De Wever, and Jan De Mey are acknowledged for their contribution to the methods described in this chapter.

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Note. A range of publications concerning the colloidal metal marker system in immunochemistry, immunocytochemistry, and immunoelectron microscopy is available free on request from the authors of this chapter. These include:

Volumes No. 1 and 2 of the colloidal gold marker bibliography

A booklet, *Colloidal Gold Sols for Macromolecule Labelling*

A booklet, *A Guide to the Immunogold-Silver Staining (IGSS) of Transferred Proteins*

A booklet, *Colloidal Gold Reagents for the Staining of Protein Blots on Nitrocellulose Membranes*

Chapter 35

Enzyme Immobilization by Adsorption

M. D. Trevan

1. Introduction

Much has already been written on the subject of biocatalyst (cell/enzyme) immobilization, and it is not the function of a text such as this to go into all the various aspects of the field. For general reviews on the area, the reader is referred to refs. 1–3. The introduction to this chapter is also meant, however, to serve as an introduction to the succeeding three chapters, and is thus more general than is usual in this series.

Biocatalyst immobilization has been defined as the imprisonment of an enzyme (or cell) in a phase (the biocatalyst phase) that is distinct from, but communicates with, the phase (the bulk phase) in which substrate product or effector molecules are dispersed and monitored. Thus the usual form of the biocatalyst phase is

some type of "polymer" matrix made to retain the biocatalyst by interacting with it either physically or chemically; the "polymer" may be soluble or insoluble in the bulk phase.

Immobilization may alter the behavior of a biocatalyst. The effect of immobilization on enzymes has been studied in depth and extensively reviewed (1,4,5). The effect of immobilization on cells has hardly been studied. Immobilization has been shown to affect all aspects of an enzyme's behavior: stability may be affected; K_m , V_{max} , and pH optimum may go up or down; sensitivity to inhibitors or activators may be altered; normal kinetic patterns may be totally disrupted. The changes may be attributed to a direct real effect of immobilization on the enzyme's intrinsic characteristics *per se* and/or to apparent changes in the enzyme's characteristics. This latter is usually caused by disturbances in solute concentrations in the immediate vicinity of the enzyme molecule, brought about by the imposition on the enzyme of a microenvironment determined by the physicochemical nature of the immobilizing "polymer" matrix.

Methods for biocatalyst immobilization probably number several hundred and present a bewildering choice to one intent upon a first attempt at immobilization. Broadly, however, immobilization methods can be divided into three major categories; adsorption, covalent bonding, and entrapment (Fig. 1). By far the greatest number of methods fall into the covalent bonding category, and it is for that reason that several protocols are given in Chapter 37.

Immobilization of biocatalysts is carried out for a number of reasons: to facilitate reuse of the biocatalyst; to aid retention of the biocatalyst in a continuous reactor; to enhance the biocatalyst's useful characteristics, for example stability or productivity; to facilitate higher

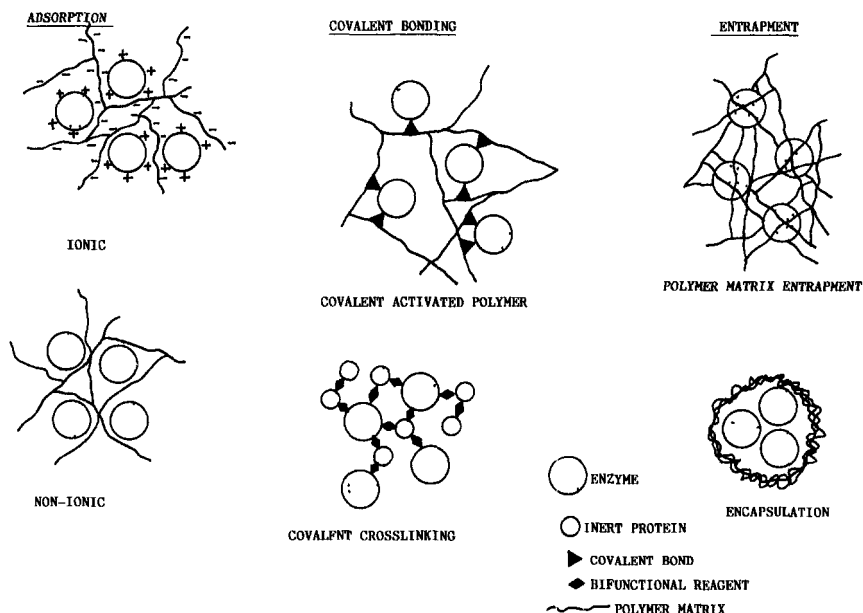


Fig. 1. Forms of biocatalyst immobilization.

concentrations of biocatalysts (in particular, viable cells) within bioreactors than would be possible with the unimmobilized biocatalyst; to protect large cells from high shear forces often found in bioreactors; or to study enzyme reactions or cellular physiology in defined heterogeneous systems. No one immobilization method is therefore of universal use, and most of the art of biocatalyst immobilization lies in selecting the least inappropriate method. A variety of factors may be involved in the choice of an appropriate method of immobilization. For example, does the method of immobilization inactivate the biocatalyst or detrimentally alter its characteristics? Is the "bond" between polymer matrix and biocatalyst strong enough to retain the molecules/cells under the conditions of their use? Does the polymer material exist in the desired physical form; for example,

as a membrane? Does the polymer material have appropriate mechanical characteristics, for example, incompressibility (important in a packed bed reactor), density (important in a fluidized bed reactor), or nonbrittleness (important in a stirred tank reactor)? How simple/cheap is the immobilization procedure; is it economically viable on a large scale? How safe is the method? Will the chosen immobilization method alter the real or apparent characteristics of the biocatalyst in a beneficial way; for example, increasing stability? It is hardly surprising that for one of the first major industrial uses of immobilized enzymes, the amino acylase process for the production of L-amino acids, some forty immobilization procedures were tested, of which just three were selected for further investigation (6).

Not only must the appropriate immobilization method be chosen, but also the form of the biocatalyst. At one extreme, biocatalysts are highly purified enzymes; at the other end of the spectrum, they are viable, growing cells. Somewhere in between there is a gray area in which enzyme becomes dead cell, becomes nongrowing viable cell. The selection of the appropriate form of biocatalyst will invariably depend upon individual circumstance. Thus in certain applications, e.g., biosensors, purified enzyme may have to be used in order to maintain either very high activity in the immobilized form or precise specificity. The expense of enzyme purification may be avoided by the use of "dead" cells or crude cell extracts, providing that the immobilized biocatalyst does not display any unwanted side reactions. Thus in most industrial-scale immobilized enzyme processes involving a single hydrolytic or isomeric reaction with a pure substrate stream, e.g., the conversion of glucose to fructose by xylose isomerase (E.C. 5.3.1.5), dead cell preparations rather than purified

enzyme are usually used, since significant side reactions are unlikely to occur. When the desired conversion is a complex series of reactions or involves the use/regeneration of coenzymes, then whole cells (nongrowing) may well provide a more useful system than purified enzymes. Thus, the form of the biocatalyst will also in part dictate the type of immobilization method chosen. In general, covalent binding is usually reserved for purified enzymes, with the notable exception of glutaraldehyde cross-linking used to immobilize flocculated "dead" cells containing xylose isomerase activity. Adsorption techniques have been applied to both enzymes and cells, but work more effectively with the former. Entrapment techniques are usually reserved for cell immobilization, for the reasons given in Chapter 38.

The rest of this introductory section is concerned specifically with immobilization by adsorption, probably the oldest and simplest form of enzyme immobilization. The adsorption forces involved may be nonspecific, hydrophobic interactions or ionic interactions (the most common). Typically adsorbent materials used have included clay, charcoal, hydroxyapatite, and, most frequently, ion-exchange materials. Of the latter, ion-exchange celluloses and like materials have proved more suitable than traditional ion-exchange resins, since the high degree of ionic substitution of the resins too often results in protein denaturation. A number of characteristics of the absorbent material must be considered. For the highest possible enzyme loadings, the material must have a high available surface area-to-volume ratio. This can be achieved in one of two ways: finely dividing the absorbent material or providing a macroporous structure so that the inner core of the material, not just its external surface, is available to the enzyme. Neither approach is without its drawbacks. Very small

particles are likely to be mechanically fragile, thus rendering them vulnerable to attrition by the stirrer mechanism in a stirred tank reactor, and are likely to present practical problems in maintaining true plug-flow conditions in a packed-bed reactor (because of overpacking and channeling of the bed material). Conversely, although larger particles are likely to be mechanically tougher and exhibit better flow characteristics in packed bed reactors, the distribution of the majority of the enzyme throughout the core of the adsorbant particle enhances the likelihood that significant mass transfer (diffusion) limitations will occur. This in turn will reduce the overall catalytic efficiency of the immobilized enzyme (*see, however, ref. 7*). Another significant, potential disadvantage of using physical adsorption to immobilize enzymes is that the binding forces between enzyme and polymer are relatively weak, and hence the loss of enzyme, by desorption, is highly likely. Thus, pH, ionic strength, and temperature must all be closely controlled. Even so the addition of substrate to the immobilized enzyme occasionally produces sufficient conformational change in the enzyme to cause its desorption! Assuming that all these factors can be controlled, however, the potential desorption of the enzyme can be put to good use. Thus, in their evaluation of immobilization methods for the aminoacylase process, Chibata and Tosa (6) chose DEAE-Sephadex-bonded enzyme, because it allowed them to regenerate the packed-bed enzyme reactor's activity every so often (without unloading the reactor), by flushing the system with high salt concentrations and low pH to remove the denatured enzyme, subsequently adding fresh enzyme directly to the reactor to allow immobilization to take place *in situ*. The original DEAE-Sephadex was left in their reactors for at least 10 yr. This case of immobilized enzyme regeneration, coupled to the almost perpetual

reuse of the otherwise expensive polymer support material, added significantly to the overall economy of the process. The major advantage of this type of method is, of course, the ease with which it can be performed and the mild conditions employed during the immobilization procedure. The only other limitation to this technique is the necessity that the enzyme must be active within the narrow range of environmental conditions (particularly pH) required to keep the enzyme bound to the polymer matrix. The alteration of an enzyme's apparent pH optimum as a consequence of immobilization onto a polyelectrolyte matrix does not matter in this context, because the apparent *pI* value of the enzyme will, in all probability, also be altered!

The method given below is for the binding of amyloglucosidase (E.C. 3.2.1.3) to DEAE-Sephadex. Within the constraints mentioned above, almost any enzyme can be used.

2. Materials

1. Amyloglucosidase solution (5 g in 100 mL of 0.02M acetate buffer, pH 4.2).
2. DEAE-Sephadex Type A 25 (10 g).
3. 0.02M Acetate buffer, pH 4.2.
4. 0.5M HCl.
5. 0.5M NaOH.

3. Method

1. Suspend 10 g of DEAE-Sephadex in 50 mL of 0.5M HCl and stir for 20 min.
2. Filter, resuspend in 50 mL of 0.5M NaOH, and stir for 20 min.
3. Filter, wash with distilled water until washings are neutral, suspend in 100 mL of acetate buffer, pH 4.2,

and stir for 10 min.

4. Filter, resuspend in 100 mL of acetate buffer, and leave to equilibrate overnight. Filter dry.
5. Add the prepared DEAE-Sephadex to 100 mL of the amyloglucosidase solution, stir at room temperature for 2 h, and then filter.
6. Reslurry the enzyme-Sephadex complex in 200 mL of acetate buffer pH 4.2, stir for 15 min, and filter.
7. Repeat step 6 until the filtrate is free of protein as judged by its absorbance at 280 nm.
8. Resuspend the amyloglucosidase-Sephadex complex in 50 mL of acetate buffer and store at 4°C.

4. Notes

1. This method is virtually foolproof — it is on occasion run by the author as a second-year undergraduate class practical, and it invariably works.
2. When using other enzymes, care should be taken to check that the pH used for adsorption and also for assay is at least 1.5 units higher than the pK_a values for DEAE-Sephadex.
3. Maximum loadings obtainable will depend in part on the molecular weight of the enzyme and the net charge density on the enzyme molecule. In practice with this (and any other methods of immobilization), the law of diminishing returns applies to enzyme loading; after a given point the polymer becomes effectively saturated with enzyme, and up to that point the relationship between the quantity of enzyme used and the amount bound is nonlinear.
4. In general, it is also better to use a small volume of concentrated enzyme solution (rather than a large volume of dilute enzyme), since this enhances the

degree of substitution of enzyme molecules onto the polymer.

5. The addition of substrate (or inhibitor) during the immobilization process may increase the final activity of the immobilized enzyme.

Acknowledgment

The materials and methods used in this chapter are copyrighted by John Wiley & Sons (1980), and are reproduced, with permission, from ref. 1.

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Chapter 36

Enzyme Immobilization by Entrapment

M. D. Trevan

1. Introduction

Entrapment methods of immobilization are mostly used in cell immobilization procedures, but have found some application with enzymes, particularly when the enzyme is essentially a dead cell or crude homogenate(1). In theory the entrapped enzyme is not attached to the polymer; its free diffusion is merely restrained. In practice, however, some or all of the enzyme may be an integral part of the polymer matrix; for example, chain termination in polyacrylamide gel formation may occur onto the enzyme (*see also ref. 2*).

The great virtue of entrapment methods for immobilizing enzymes is that they are fast, cheap, very easy, and usually involve fairly mild conditions. The most used method seems to be that based upon entrapment in

polyacrylamide gels, although the concentrations of acrylamide and quantity of crosslinking reagent used vary from method to method. The major drawback with this form of immobilization for pure soluble enzymes is the tendency of the enzyme to leak from the gel. This may be overcome by increasing the degree of cross-linking of the gel matrix, thus reducing pore size. This will in turn, however, increase the likelihood of mass transfer/diffusion limitation occurring, a particular problem with entrapment of biocatalysts. The effect of mass transfer/diffusion limitation may be ameliorated to a large extent by reducing the size of the entrapping polymer particle. The method given here easily produces a very finely divided, highly cross-linked polyacrylamide gel. It should be noted that mass transfer/diffusion limitation will be increased with increasing enzyme loading, such that the apparent specific activity of the entrapped enzyme falls as the amount of enzyme entrapped is increased (3).

When the enzyme is already insoluble or is part of a dead cell, leakage of the entrapped enzyme from the gel matrix is seldom a problem (*see* section 4). Clearly, enzymes immobilized by entrapment are likely to be inactive with high molecular weight substrates. The method given below is for the entrapment of urease in an acrylamide gel, although almost any enzyme may be used.

2. Materials

1. Urease (EC 3.5.1.5).
2. Tris buffer, 0.1M, pH 7.0, containing 10^{-4} M EDTA.
3. Tris buffer, 0.1M, pH 7.0, containing 10^{-4} M EDTA and 0.5M NaCl.
4. Methylene-*bis*-acrylamide (MBA).

5. Acrylamide (ACR).
6. Dimethylamino-propionitrile.
7. Potassium persulfate solution 10 mg/mL.

3. Method

1. Make up a solution of 110 mg MBA and 10 mg ACR dissolved in 10 mL of Tris buffer, pH 7.0, and place in a stoppered flask.
2. Dissolve 15 mg of urease in the MBA:ACR solution and gas with nitrogen for 20 min.
3. Add 0.2 mL of dimethylamino-propionitrile and mix gently.
4. Add 0.5 mL of potassium persulfate solution, mix gently, stopper the flask, and allow to stand at room temperature for 20 min. An opaque gel will form.
5. Disrupt the gel by shaking the flask and passing the contents through a fine-needle syringe.
6. Filter on a Buchner funnel and wash the gel with 50 mL of 0.1M Tris buffer, pH 7.0, containing 0.5M NaCl.
7. Repeat step 6 three times.
8. Resuspend the enzyme gel in 15 mL of 0.1M Tris buffer, pH 7.0.

4. Notes

1. The method given here is routinely used as an undergraduate student laboratory exercise, largely because of the speed of preparation.
2. In practice, little leakage of enzyme occurs from the gel. The gel particles are not particularly robust, however, and thus vigorous stirring may break them down and release enzyme.
3. At the suggested level of addition, about 20% of the enzyme is actually retained in the gel. Using less

enzyme will result in a higher percentage being entrapped.

4. This method can be easily adapted for use with dead cells or nonpurified enzymes. In those cases in which diffusion of the "enzyme" out of the gel is likely to be less of a problem, however, it would be better to increase the quantity of acrylamide in the gel in order to increase the gel strength. The exact proportions and concentrations of acrylamide to methylene-bis-acrylamide are best determined by experimentation.
5. Soaking the formed immobilized enzyme in a 2.5% solution of glutaraldehyde for 30 min at pH 7.5 will reduce any enzyme leakage to virtually zero, but may inactivate the enzyme. Inactivation may be overcome by adding a high concentration of competitive inhibitor (or substrate).

Acknowledgment

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Chapter 37

Enzyme Immobilization by Covalent Bonding

M. D. Trevan

1. Introduction

Enzyme immobilization by covalent bonding to an insoluble polymer has the supposed advantage of irreversible binding of the enzyme to the support matrix. Many of the bonds commonly employed are not 100% stable in use, however. Unless some care is taken to ensure that the enzyme can withstand the reaction conditions, substantial loss of enzyme activity will occur. Even under favorable conditions, 60% loss of activity is not uncommon.

Covalently bound immobilized enzymes are not only suitable for use in large-scale industrial processes [although their use in this field is in practice limited (1)] and in the construction of biosensors, but also as the immobile phase in affinity chromatography for the purification of enzyme inhibitors, activators, or substrates.

A wide variety of binding chemistries has been studied, and it is possible to bind an enzyme to almost any polymer or solid material (e.g., cellulose, agarose,

polyacryl-type compounds, glass, nylon, and so on). The choice of chemistry and support is still largely an empirical matter, and few studies have been performed to compare the efficacy of the various methods. In a recent study, McManus and Randerson (2) compared the effect of chemistry and support on the uptake, stability, and immunoreactivity of a covalently immobilized antibody. They found that both the support and chemistry affected the uptake, the former largely because of variations in surface area, the latter in an unpredictable manner. For example, aldehyde groups were effective in binding the antibody when attached directly to the polymer support or via support-bound glutaraldehyde, but not in the form of support-phenylaldehyde. The nature of the support was shown not to affect the stability of the antibody-polymer bond, whereas the chemistry involved in producing that bond had a marked effect on stability. In general they concluded that the order of decreasing stability was binding via glutaraldehyde to amino groups, *N*-hydroxysuccinide esters, tresyl bonds, and, least stable, isourea bonds formed in cyanogen bromide reactions. None of the methods affected the K_s for binding of antibody to antigen.

Many commercially prepared activated supports for enzyme immobilization are available with a wide range of support materials and chemistries. There is obviously a premium to be paid for this however; cyanogen bromide-activated Sepharose 4B costs some 10 times more than the Sepharose 4B, and with an enzyme ready immobilized to Sepharose 4B, the cost soars by another 100 times!

There is obviously room for the do-it-yourself enthusiast. The four different chemistries that are detailed here are among the most commonly employed and are: coupling to cyanogen bromide-activated agarose; cou-

pling to tresylchloride-activated agarose; coupling to 2-amino-6-chloro-s-triazyl-cellulose, and coupling by glutaraldehyde.

The covalent coupling of enzymes, proteins, and other ligands to polymer-support matrices activated by cyanogen bromide (3) is probably the most widely used immobilization method. The coupling relies on the formation of an isourea bond between a hydroxyl group on the polymer and an amino group on the protein. The immobilization method is essentially a two-stage process. The first stage, the activation of the polymer with cyanogen bromide, is potentially hazardous, and it is possible to buy (at some cost) ready-activated polymers from many suppliers of chromatography materials. The second stage, the coupling of protein to activated polymer, is carried out under very mild conditions (neutral pH, low temperature), and even sensitive proteins can be coupled with very little loss of activity. Any hydroxyl-containing polymer can be used, but among the most useful, from the view point of enzyme loading, high surface area-to-volume ratio, mechanical stability, and flow characteristics, are the beaded cross-linked agaroses such as Sepharose 4B. The major problem with this technique is the relative instability of the isourea bond causing loss of enzyme.

Activation of hydroxyl-containing polymers by tresyl chloride (2,2,2-trifluoroethyl sulfonate chloride) has been proposed as an alternative to bromide activation (4). Although tresylchloride is not exactly innocuous (a corrosive, lachrymator), it is certainly less potentially hazardous than cyanogen bromide. Coupling of the activated polymer to enzyme occurs under very mild conditions, and, in addition, the tresylchloride-activated polymer is stable for several months at pH 3 in aqueous suspension, whereas cyanogen bromide-acti-

vated polymers rapidly lose their enzyme-binding ability in aqueous suspension. Activation by tresylchloride is thought to occur by tresylation of a hydroxyl group on the polymer (with elimination of HCl). At neutral pH, this then is available for nucleophilic attack by an NH_2 group on the enzyme. At pH values above 8.5, it is possible that substitution of individual fluorines occurs. The resulting bond between the polymer and enzyme seems in practice to be 100% stable. The degree of activation of the polymer must be carefully controlled, because there appears to be an optimum level above which percentage yields of enzyme activity markedly decrease. This optimum level probably varies with both support material and enzyme.

Coupling of enzymes to cyanuric chloride (trichloro-*s*-triazine)-activated polymers again relies on the presence of hydroxyl groups on the polymer and amino groups on the enzyme. The original method (5) first reacted the trichloro-*s*-triazine to cellulose and then coupled the enzyme to the dichloro-*s*-triazinyl cellulose compound, quenching the third chloro-group with ammonium salts. The highly reactive nature, however, of both the tri- and dichloro compounds means that reaction times for the activation of the cellulose are measured in seconds, and for coupling the enzyme, in minutes. The apocryphal story of the exact reaction time for the cellulose-activation step being achieved by dropping a flask containing the reaction mixture down the well of a four-story staircase remains one of the more curious of scientific anecdotes—that the story originated at all testifies to the "art" involved in many immobilization methods. It is difficult to prevent the cellulose from becoming crosslinked by the activation step, and loss of 90% of the enzyme activity is quite usual. A more common version of this method, therefore, is first to synthe-

size dichloro-s-triazinyl ethanolamine, then monochloro-s-triazinylethanolamine-cellulose, and use this to couple the enzyme (5). Since the monochloro-compound is relatively unreactive, the process is much easier to control. This method has not been widely used, probably because it was covered by a fairly comprehensive patent up to the late 1970s. It has a number of inherent advantages over the cyanogen bromide, however. First the enzyme-polymer bond is quite stable. Second, cyanuric chloride is a trifunctional reagent; any suitable ligand may also be bound to the polymer along with the enzyme. Thus the polymer may be made polyanionic, polycationic, or even hydrophobic in character. Third, although it is not exactly harmless, cyanuric chloride is not as difficult to handle, nor are its reactions as potentially lethal as cyanogen bromide.

Glutaraldehyde, strictly a dialdehyde, will react with amino groups under very mild reaction conditions. In the presence of ammonium ions, it will undergo polymerization. It is thus a potentially useful bifunctional molecule for enzyme immobilization. It has been used in one of two different ways; either to crosslink enzyme molecules to each other or to "activate" amino group-containing polymers. When used as a crosslinking reagent, it has a tendency to react with the enzyme's active site, unless this is protected by the presence of a competitive inhibitor. Glutaraldehyde is used to good effect to crosslink already flocculated (dead) cells in the industrial-scale preparation of immobilized enzymes (e.g., xylose isomerase for high-fructose corn syrup production). The method given here is a very simple process for the formation of immobilized enzyme membranes. It can be adapted for any sheet or membranous material to which enzymes can be adsorbed.

2. Materials

2.1. Coupling via Cyanogen Bromide

1. Sepharose 4B.
2. 2M Phosphate buffer, pH 12.1.
3. 5M Phosphate buffer, pH 12.1.
4. Cyanogen bromide solution: 100 mg in 1 mL of 5M phosphate buffer, pH 12.1.
5. α -Chymotrypsin (EC 3.4.21.2): 100 mg dissolved in 4 mL of 0.3M KHCO_3 .
6. Glycine.
7. 0.1M NaHCO_3 .
8. 10^{-3}M HCl.
9. 0.5M NaCl.

2.2. Coupling via Tresyl Chloride

2.2.1. For Activation

1. Sepharose 4B: 10 mL of swollen gel (about 3.2 g dry weight).
2. Tresylchloride (2,2,2-trifluoro-ethyl sulfonate chloride). Caution: corrosive.
3. 30:70, 60:40, and 80:20 mixtures of acetone:water (100 mL of each).
4. Dried acetone (200 mL), dried over molecular sieve 30 g/L acetone.
5. Pyridine.
6. 30:70, 50:50, 70:30, and 85:15 mixtures of 1 mM HCl:acetone (100 mL of each).
7. 1 mM HCl (250 mL).

2.2.2. For Coupling

1. Coupling buffer: 0.2M Sodium phosphate plus 0.5M sodium chloride, pH 8.0 (500 mL).

2. Trypsin (EC 3.4.21.4): 100 mg dissolved in 10 mL of coupling buffer at 4°C.
3. 0.1M Dithiothreitol (10 mL).
4. 0.2M Sodium acetate buffer, pH 3, containing 0.5M sodium chloride (100 mL).
5. 0.5M Sodium chloride (100 mL).

2.3. Coupling via Cyanuric Chloride

1. Ammonia solution (sp. gr, 0.88).
2. Cyanuric chloride.
3. Dioxane.
4. Toluene.
5. Acetone.
6. Cellulose powder.
7. Na_2CO_3 ; 15% solution.
8. 1M HCl.
9. HCl concentrated.
10. 0.05M Phosphate buffer, pH 7.0.
11. 0.05M Borate buffer, pH 9.0.
12. 0.05M Borate buffer, pH 9.0, containing 1.0M NaCl.
13. Papain (EC 3.4.22.2).

2.4. Formation of Immobilized Enzyme Membrane with Glutaraldehyde

1. Cellulose acetate sheet, 30 μm thick.
2. Alkaline phosphatase (EC3.1.3.1) solution : 10 mg/mL in 0.02M phosphate buffer, pH 6.8.
3. 0.1M Glycine solution adjusted to pH 9.0.
4. Glutaraldehyde solution: 25% solution diluted to 2.5% with 0.02M phosphate buffer, pH 6.8.

3. Methods

3.1. Coupling via Cyanogen Bromide

3.1.1. Activation of Sepharose 4B

1. Wash 10 g of Sepharose 4B with 2.0M phosphate buffer, pH 12.1, and filter dry on a Buchner funnel.
2. Slurry the Sepharose in 10 mL of cold 5M phosphate buffer, pH 12.1, and add 20 mL of cold distilled water.
3. To the Sepharose slurry, at 5°C, add the cyanogen bromide solution dropwise, such that the whole volume is added within 5 min.
4. Stir the mixture for a further 10 min.
5. Transfer the gel to a glass filter and wash with 100 mL of cold distilled water.

3.1.2. Coupling of α -Chymotrypsin to Cyanogen Bromide-Activated Sepharose 4B

1. Suspend the activated Sepharose in NaHCO_3 (50 mL) of and add to the solution of α -chymotrypsin.
2. Stir gently for 24 h at 4°C.
3. Add 1.5 g of glycine to the slurry and stir for a further 24 h.
4. Filter the α -chymotrypsin-Sepharose complex and wash in turn with 200 mL aliquots of water, 0.1M NaHCO_3 , 10^{-3} M HCl, 0.5M NaCl, and finally distilled water.

3.2. Coupling via *Tresylchloride*

3.2.1. Activation of Sepharose 4B

1. Place 10 mL of swollen Sepharose 4B in a glass filter funnel.
2. Wash with 100 mL of water.
3. Wash with 100 mL each of 30:70, 60:40, then 80:20 acetone:water mixtures.
4. Wash twice with 100-mL aliquots of acetone.

5. Wash three times with 50-mL aliquots of *dry* acetone.
6. Place the gel in a small beaker containing 2 mL of *dry* acetone and 1 mL of pyridine and stir vigorously.
7. Add 500 mg of tresyl chloride (3 mmol), dropwise over 1 min.
8. Continue stirring for 10 min.
9. Transfer the gel to a glass filter and wash twice with 100-mL aliquots of acetone.
10. Wash with 100 mL each of 30:70, 50:50, 70:30, and 85:15 solutions of 1 mM HCl:acetone.
11. Wash twice with 100-mL aliquots of 1 mM HCl and store at 4°C in a minimum volume of 1 mM HCl.

3.2.2. Coupling of Trypsin to Tresylchloride-Activated Sepharose 4B

1. Wash the gel briefly with coupling buffer (10 mL) on a glass filter.
2. Add the washed gel to 10 mL of enzyme solution at 4°C.
3. Stir or shake gently for 18 h at 4°C (or 2 h at 25°C).
4. Filter off excess liquid.
5. Add the gel to 10 mL of 0.1M dithiothreitol.
6. Stir at 4°C for 2 h.
7. Wash three times with 100-mL aliquots of coupling buffer.
8. Wash successively with 100-mL aliquots of acetate buffer, sodium chloride solution, distilled water, and coupling buffer.

3.3. Coupling via Cyanuric Chloride

3.3.1. Activation of Cellulose

1. Slurry 184 g of cyanuric chloride in 1.2 L of di-oxane:toluene solvent (5:1 v/v). Cool to 5°C.

2. Gas the slurry with a dried stream of gas produced by passing nitrogen through the ammonia solution. Continue the gassing until a thick suspension is formed.
3. Filter the reaction mixture and wash the precipitate with 0.5 L of dioxane. Discard the precipitate and retain both filtrate and washings.
4. Evaporate the filtrate and washings to dryness on a rotary evaporator.
5. Redissolve the 2-amino-4,6-dichloro-s-triazine in the minimum volume of 1:1 (v/v) acetone:water.
6. Evaporate the acetone under reduced pressure.
7. Redissolve the precipitate formed in boiling water.
8. Cool rapidly to room temperature.
9. Dissolve 1 g of the 2-amino-4,6-dichloro-s-triazine in 50 mL of 1:1 acetone:water at 50°C.
10. Add 10 g (dry weight) of cellulose and stir for 5 min.
11. Mix 12 mL of 1.0M HCl with 8 mL of the Na₂CO₃ solution, add to the cellulose slurry, and stir for a further 5 min.
12. Reduce the pH of the slurry to below pH 7.0 with conc. HCl.
13. Filter the reaction mixture and wash three times with 100-mL aliquots of the acetone:water solvent.
14. Wash the amino-chloro-s-triazyl-cellulose copiously with distilled water.
15. Wash with 100 mL of 0.05M phosphate buffer, pH 7.0, and store at 2°C.

3.3.2. *Coupling of Papain to Cyanuric Chloride-Activated Cellulose*

1. Suspend 5 g of the activated cellulose in 50 mL of borate buffer, pH 9.0.
2. Dissolve 0.5 g unactivated papain in 50 mL of borate buffer, pH 9.0.

3. Mix the papain solution and cellulose suspension and stir at 4°C for 24 h.
4. Filter the enzyme–cellulose complex and wash with 100 mL of borate buffer, pH 9.0, followed by 100 mL of borate buffer, pH 9.0, containing 1.0M NaCl and, finally, with 100 mL of borate buffer, pH 9.0.

3.4. Formation of Alkaline Phosphatase Membrane with Glutaraldehyde

1. Spread 5 mL of alkaline phosphatase solution over a 10-cm² piece of cellulose acetate in a Petri dish.
2. Leave at 4°C until desiccated.
3. Spread 5 mL of glutaraldehyde solution over the membrane and leave at 4°C until desiccated.
4. Soak the membrane in glycine solution for 10 h at 4°C.
5. Wash the membrane in distilled water until the washings show no absorbance at 280 nm.
6. Store wet at 4°C.

4. Notes

4.1. Cyanogen Bromide Method

1. *Cyanogen bromide is extremely hazardous; use of a fume cupboard is essential, and cyanide antidote should be at hand.*
2. It is essential during the reaction of the Sepharose with the cyanogen bromide that the pH does not fall below 11.5, hence the use of 5M buffer. It is prudent to monitor the pH with a meter and to have 10M sodium hydroxide available.
3. Any hydroxy-containing polymer can be activated by this method, and it can easily be adapted to the

activation of soluble polymers, for example, polyethylene glycol or dextran.

4. Like most enzyme-immobilization methods, the amount of enzyme coupled expressed as a fraction of the amount added falls as the latter increases. Thus, typically if 5 mg of enzyme is added to 1 g of activated Sepharose, most of it will become bound, whereas adding 20 mg may result in only 10 mg being bound. For Sepharose 4B, the usual practical limit to the binding of enzyme occurs at around 15–20 mg/g. At these levels of substitution, however, the specific activity of the enzyme declines markedly, probably as a result of mass transfer/diffusion limitation or steric hindrance of the active sites caused by the high local enzyme concentration (in the region of 5 mg/mL). Thus there is an optimum level of enzyme loading beyond which it may be uneconomic to stray (*see, however, ref. 7*).
5. In general, coupling occurs most efficiently between pH 8 and 10, although it can be carried out at as low as pH 5 and is usually complete after 2 h at 25°C. For sensitive enzymes, coupling is best carried out at 4°C for at least 18 h.
6. Where possible, the enzyme–Sepharose coupling should be carried out at relatively high ionic strength (approximately 0.5) in order to minimize protein interactions.
7. CNBr-activated polymers are not stable in aqueous suspensions and neutral pH values; therefore they should be used immediately or lyophilized and stored cold.

4.2. *Tresylchloride Method*

1. The important feature of this method is the ratios of the various volumes or reagents.

2. The ratio of pyridine:tresylchloride should be kept at 2/1 (v/v).
3. Varying degrees of substitution may be obtained by altering the tresylchloride volume in the range of 0.05–0.5 mL.
4. With other types of support material, more acetone may have to be added in step 6 of the activation procedure in order to achieve adequate stirring.
5. Coupling may be carried out at pH values from 7.0 to 8.5.
6. The wash in acetate buffer (step 8, coupling procedure) may be omitted or the pH modified if the enzyme to be coupled is unstable in these conditions.
7. Similarly the concentration of sodium chloride used may be reduced if necessary, although at the expense of total desorption of physically bound enzyme.
8. Enzymes immobilized by this method generally show high stability and are best stored at 4°C, pH 7–8.
9. The coupling buffer can be modified (no amino groups) as necessary, depending on the enzyme to be coupled.

4.3. Cyanuric Chloride Method

1. Though not as hazardous as cyanogen bromide, cyanuric chloride should be handled with care. Certain individuals may be particularly sensitive, and even brief exposure may cause an irritant reaction.
2. This method has been used to bind enzymes to a variety of materials (filter paper, filter cloth, dialysis tubing) and invariably works, although fairly

large (e.g., 80%) losses of enzyme activity can occur as a result of coupling.

3. The use of this method for activation of soluble polymers is not recommended because of the difficulty of preventing cross-linking of the polymer.

4.4. Glutaraldehyde Method

1. This method either works well or not at all! A lot depends on the enzyme chosen and its specific activity.
2. Inclusion of a competitive inhibitor or substrate of the enzyme will help protect the active site.
3. Better yields (in terms of activity coupled to activity added) can often be achieved by adding bovine serum albumin to the enzyme solution at a ratio of up to 10:1, albumin:enzyme.
4. Despite, or perhaps because of, the impurities, technical-grade glutaraldehyde seems to work better than EM grade.
5. Sometimes drying down a particular enzyme solution causes it to become denatured. In such cases allowing the membrane to soak in a solution of the enzyme (at about 1–5 mg/mL) for a couple of days and then transferring this to the glutaraldehyde solution may solve the problem.
6. If for some reason a thick sheet of immobilized enzyme is required, the above method may be easily adapted thus. Cast a sheet of cellulose nitrate by dissolving cellulose nitrate in a 49:49:2 ether:ethanol:water solvent (at 4% concentration). Pour onto a flat glass surface to the desired thickness, allow the solvents to evaporate, and soak the sheet off the glass with water (the glass must be scrupulously clean). Keep the sheet moist at all times.

Soak the sheet for a number of days (the thicker the sheet, the longer the time—allow about 5 d/1 mm) in a 0.1% solution of the desired enzyme with or without added albumin. Allow 100 mL of enzyme solution per cm^3 of membrane, and proceed as above. If desired, the cellulose nitrate may subsequently be dissolved in methanol, leaving a pure protein sheet.

7. Glutaraldehyde may be used to activate amino group carrying polymers (e.g., amino ethyl cellulose or amino silane-treated control pore glass). The following guidelines may aid in method development. React the polymer with a 2.5% glutaraldehyde in phosphate buffer, pH 7–8, at a ratio of about 10 mL of glutaraldehyde per g of polymer. The reaction is fairly rapid and should be complete in 1 h at 25°C. Wash off any unbound glutaraldehyde. Add enzyme solution at concentrations of the order of 1–10 mg/mL at 10 mL/g polymer. The enzyme should be in a neutral, nonamino group-containing buffer. React for 1–2 h at 25°C or overnight at 4°C. Quench any remaining aldehyde groups on the polymer with 0.1M glycine or ethanolamine, depending on the ionic nature required of the polymer. Wash appropriately with high ionic strength buffer.

Acknowledgment

The materials and methods used in this chapter for coupling cyanogen bromide, cyanuric chloride, and glutaraldehyde are copyrighted by John Wiley & Sons (1980), and are reproduced, with permission, from ref. 8.

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Chapter 38

Cell Immobilization

M. D. Trevan

1. Introduction

We will be concerned in this chapter with methods that may be used to immobilize live rather than dead cells. The use of immobilized viable cells is seen as an alternative approach to extracellular metabolite production by traditional fermentation technology. There are a number of reasons why this may be advantageous. First, immobilized cells are easily utilized in continuous flow reactors without, theoretically, the contamination of the product stream by cellular material. Second, cell densities in the reactor of at least one order of magnitude greater than with traditional fermentation techniques are possible. Third, immobilization has been reported to enhance the longevity of cells, in particular plant cells, which is therefore useful in extending the productive life of slow-growing cells. Fourth, it has been suggested that immobilization may enhance the release of secondary metabolites from cells, particularly when (as in

plant cells) these metabolites are normally stored intracellularly.

The European Federation of Biotechnology (1) has published guidelines on the classification of cellular biocatalysts, but these have not yet in practice been universally adopted; some confusion therefore remains in the literature. As we have seen (Chapters 35 and 37), nonviable cells (that is, effectively crude enzyme preparations) may be immobilized by many of the methods used for pure enzymes. With viable (resting) or growing (reproducing) cells, something more subtle is called for. When growing (i.e., reproducing) cells are to be immobilized, one particular problem is that caused by cell reproduction; that is, how to retain the daughter cells. Covalent immobilization would clearly be impractical, as would most absorptive methods. With the exception of anchorage-dependent cells (e.g., many animal cells) that may be immobilized onto polymer microcarriers using their natural adherent properties, the immobilization methods usually employed therefore fall into the general category of entrapment. Clearly the methods used must not debilitate the cells, at least not beyond the point of recovery/regeneration, and must therefore be relatively mild—methods based on polyacrylamide gel entrapment are liable not to work, and there are a number of other suitable entrapment methods.

In practice, however, cell "leakage" from immobilized cell preparations made by an entrapment method can still be a significant problem. For the most part, "leakage" seems to be related to cell growth, and thus stationary phase cells are much less prone to this problem (2), as well as being advantageous for their predicated superior secondary metabolite production.

A common observation of immobilized viable cells is that, after a period of time, the cells appear to be located only around the periphery of the immobilization ma-

trix. This is probably a result of cell death in the center of the matrix coupled to cell growth at the periphery. For the alga *Chlorella emersonii* immobilized in calcium alginate beads and grown phototrophically, this has been shown to be caused by poor penetration of CO₂ into the alginate beads (3); the cells at the periphery of the beads effectively scavenge all the available CO₂.

Supplementing the medium with CO₂ in this study not only encouraged cell growth throughout the bead, but also resulted in a more stable bead structure, suggesting that the cells, in this case at least, form an integral part of the immobilized cell bead structure. It is likely that the limiting factor on cell growth and immobilization matrix colonization for any viable cell is poor mass transfer of necessary gases, not so much because the concentration is usually low, but because cellular demand is usually high, and thus significant concentration gradients of dissolved gas will be found in the immobilized cell particle. Recognizing this problem, Alder-cruz et al. (4) coimmobilized *Chlorella pyrenoidosa* with *Gluconobacter oxydans* in calcium alginate, on the grounds that the oxygen evolved by *C. pyrenoidosa* would stimulate the microorganism's growth and productivity. Interestingly, *C. emersonii* immobilized in calcium alginate and grown heterotrophically (i.e., illuminated and with a carbon source, glucose, in the medium) colonized the whole alginate matrix to a very high density (5). Clearly, for maximum productivity, maximum cell density in the immobilization matrix is required. If dissolved gas mass transfer cannot be improved in a given system, the problem can in part be overcome by producing smaller immobilized cell particles (approximately 0.2 mm in diameter) (see section 4).

Of the methods given here, neither is perfect, and many variants exist in the literature. Both methods involve mixing a cell suspension with a solution of the

entrapping polymer and then causing the polymer to gel. In the case of κ -carrageenan, the solution-gel transition is effected by lowering the temperature, and in the case of alginate, by metal ion substitution. Thus the former gel is sensitive to disruption by increased temperature, and the latter is sensitive to disruption by the presence of metal-ion chelating agents (e. g., phosphate or citrate).

There is another form of immobilization by "entrapment" not given here, that is, "passive" entrapment. In this technique, cells are allowed to invade preformed polymer matrices (e.g., polyurethane foams). This immobilization method, however, results in a preparation that does not generally retain cells well, and it is difficult to prepare particles for use in packed-bed types of reactors.

Since the methods recorded here have almost universal applicability, no specific cell cultures are suggested. Hence the medium used in the methods must be suitable for the cells chosen. Experience shows that almost any microbial or plant cell can be immobilized successfully by these methods, with the possible exception of normally highly motile cells, which do not appear to favor being fixed.

2. Materials

All reagents should be autoclaved when necessary.

2.1. *Calcium Alginate Entrapment*

1. 5% Solution of sodium alginate in water (5 mL).
2. 0.1M Calcium chloride (500 mL).
3. Appropriate growth medium.

4. Cell suspension between 2 and 32% (w/v) (5 mL).
5. Laboratory detergent (nonionic).

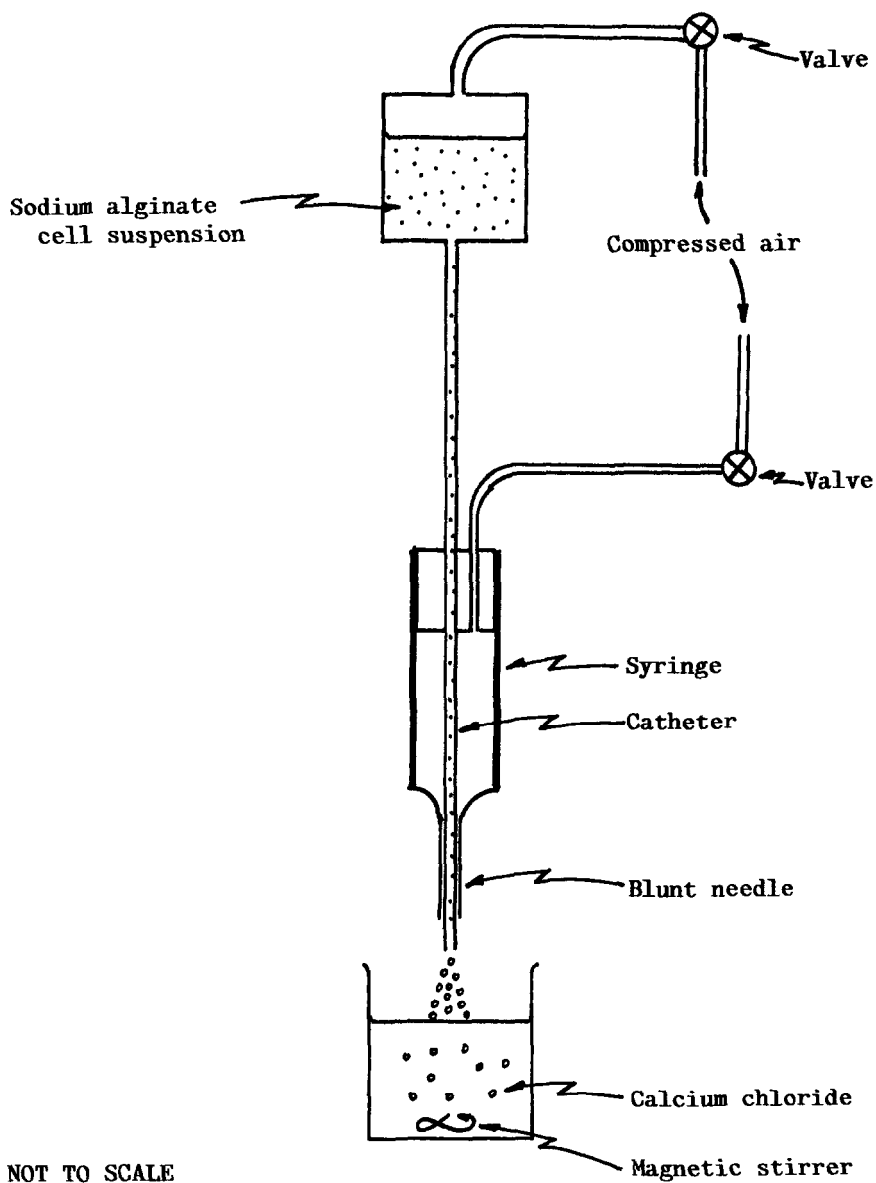
2.2. κ -Carrageenan Entrapment

1. 5% Solution of κ -carrageenan (low-gelling temperature) in water (5 mL).
2. Ice-cold solution of 0.3M potassium chloride and 0.01M calcium chloride (500 mL).
3. Appropriate growth medium.
4. Cell suspension between 2 and 32% (w/v) (5 mL).
5. Laboratory detergent (nonionic).

3. Method

3.1. Calcium Alginate Entrapment

1. Mix the cell suspension and sodium alginate solutions thoroughly, but gently. Any air bubbles present may be removed by gentle centrifugation, but take care that the cells are not separated out.
2. Put the calcium chloride solution in a large beaker (at least 1 L) on a magnetic stirrer set to stir slowly, and add a few drops of detergent.
3. Load the alginate/cell mixture into a 20-mL sterile syringe or peristaltic pump.
4. Drop or pump the alginate/cell mixture into the calcium chloride solution from a height of about 10 cm. Gel beads will form almost instantly (see Fig. 1 and section 4).
5. Leave the beads stirring in the calcium chloride solution for at least 90 min to allow for complete substitution of the calcium into the gel.
6. Wash beads in appropriate medium.



NOT TO SCALE

Fig 1. Apparatus for production of small calcium alginate-immobilized cell beads.

3.2. κ -Carrageenan Entrapment

1. Once the κ -carrageenan solution has been made, it should be kept a few degrees above its gelling temperature.
2. Gently heat the cell suspension to the same temperature as the κ -carrageenan solution.
3. Mix the cell suspension and κ -carrageenan gently.
4. Place the potassium/calcium chloride solution in a large beaker with a few drops of detergent, and stir gently with a magnetic stirrer.
5. Load the κ -carrageenan/cell mixture into a 20-mL syringe or peristaltic pump and add dropwise to the potassium/calcium chloride solution from a height of about 10 cm above the liquid surface.
6. Leave the beads so formed to cure for 90 min.
7. Wash with appropriate medium.

4. Notes

4.1. Calcium Alginate Entrapment

1. Alginic acid is a polysaccharide found in many species of macro alga composed of D-mannuronic (M) and D-guluronic (G) acids, arranged in linear chains as M-G-M-G sequences, interrupted by M-M-M and G-G-G blocks. The precise composition, chain length, and so on, vary not only with source organism, but also with the part of the plant and time of year of harvesting. Since it is the D-guluronic acid that is primarily responsible for binding to the metal ion, alginic acids high in this component are recommended in order to obtain a gel with high strength. The most common alginic acid for immobilization is thus isolated from *Laminara hy-*

perborea and can be supplied by various companies (e.g., Kelco-AIL, London).

2. Gel strength is dependent on initial alginate concentration, but is more or less independent of metal cation concentration or type. Metal cation type markedly influences the resistance of alginate gels to disruption/dissolution by chelating agents, and changing the gelling cation may prove useful in certain circumstances. Thus barium alginate gels tend to be less easily disrupted than calcium alginate gels.

It is not just the binding affinity of the metal cation to the alginate that is important in this respect, however, but the ratio of the metal cation binding affinities to both the alginate and the chelator. Thus, although Cu^{2+} ions bind more strongly to alginate than Ca^{2+} ions do, calcium alginate resists disruption by phosphate better than copper alginate. Generally, barium alginate gels are the least susceptible to disruption by chelators, with calcium or aluminum alginate gels next best. The moral of all this is that stability of the beads must be tested in the media to be used.

One common approach to this problem with calcium alginate gels is to reduce the concentration of chelators to a minimum and then add a slight molar excess of calcium chloride to the medium. This appears to enhance the stability of the alginate cell beads when studied in batch culture. The effect may be caused more by removal of the chelator by the calcium, however, rather than by supplementation of the gel's calcium content. If the chelator is essential to the growth or viability of the immobilized cells (often the case with phosphate), its removal is obviously not desirable. With continuous-

flow types of reactors, an additional problem is encountered, particularly in respect to phosphate chelation of calcium; that is, under the conditions often present, calcium phosphate is insoluble. Attempts to add calcium chloride to the medium reservoir will probably trap the phosphate there; without additional calcium chloride, even micromolar concentrations of phosphate will eventually chelate out most of the calcium from the alginate gel. Therefore, when phosphate or citrate are indispensable constituents of the medium, even at low concentration, it is probably better to evaluate the use of barium or aluminum alginate gels.

3. The best method for dissolving the sodium alginate prior to use is simply to autoclave a suspension of alginate in water (20 min at 15 psi). It should be noted, however, that autoclaving will probably shorten the polysaccharide chain lengths and thus lower the viscosity of the sodium alginate solution and the strength of the calcium alginate gel.
4. It is sometimes difficult to mix the cell suspension and sodium alginate solution without introducing bubbles into the mixture. Should this happen, these can be removed by gentle centrifugation.
5. The addition of detergent to the calcium chloride solution lowers the surface tension just sufficiently to allow the drops of sodium alginate/cell mixture to penetrate the surface. Without it, beads tend to float on the surface of the calcium chloride solution and become stuck together. The lessening of the impact on the falling drop may also help create a more spherical shape (particularly for small beads—see below).
6. Bead size is largely unaffected by the external diameter of the orifice from which the alginate/cell

mixture is dropped, because of the physical characteristics of the alginate in solution. Generally beads of between 2 and 4 mm diameter are formed. The mass transfer / diffusion limitation of gases and the effect of this on cell distribution within the gel-matrix has been discussed above (section 1), and for this or other reasons, smaller beads may be deemed desirable. A number of methods for producing small calcium alginate beads have been suggested, but perhaps the simplest and most reliable is the use of a stream of air flowing downward along the outside of the tubing from which the alginate/cell mixture is dropped (Fig 1). The drops are effectively blown off the tip of the aperture before they assume their usual size. By this means, beads of the order of 0.2 mm diameter are easily obtained. The apparatus for this purpose can be constructed easily from syringes, needles, and catheters of various sizes, and must be "tuned" by trial and error.

7. Metal ion alginate gels act as molecular sieves. The cut-off point will depend upon the precise composition of the gel, but in most cases it is around 100,000 dalton.
8. Although cell leakage from alginate gels is probably less problematic than from most other entrapment gels, it can still be significant (*see* section 1). One way around this, on a small scale, is to recoat preformed calcium alginate/cell beads with a thin layer of cell-less calcium alginate. This can be done by removing the formed calcium alginate beads from the calcium chloride solution and rinsing briefly in medium or distilled water. The beads are then placed carefully in a fresh solution of sodium alginate (of around 0.5–2%). Sufficient free calcium chloride will have been left in the gel beads to

diffuse out and precipitate the surrounding layer of sodium alginate. The thickness of this coating layer depends upon the initial calcium chloride concentration, rinsing efficiency, and concentration of, and time of exposure to, the fresh sodium alginate solution in a totally predictable manner. Beads produced thus show zero cell leakage, even under conditions of active cell growth.

4.2. κ -Carrageenan Entrapment

1. κ -Carrageenan is extracted from the red algae (*Rhodophyceae*). It is available from a number of suppliers. Since, however, a low-gelling-temperature κ -carrageenan is preferred in this application and this parameter varies with source and processing (see below), care must be exercised in its purchase. A suitable product is "Gelcarrin CIC," marketed by FMC Corporation, Marine Colloids Division, PO Box 70, Springfield, NY, 07081, USA.

κ -Carrageenan is a polysaccharide that characteristically contains repeating units of the disaccharide D-galactose-4-sulfate β 1:4,3,6-anhydro-D-galactose linked together by 1:3 glycosidic bonds. The gelling mechanism of κ -carrageenan is one of formation of a network of (aggregated) double helices. Aggregation of the double helices is believed to be caused by the presence of calcium ions and results in an increase in gel strength. The nonaggregated helical structure, itself a gel, is formed on cooling the κ -carrageenan to below its characteristic gelling temperature. The gelling temperature is affected by the source and processing of the κ -carrageenan and is markedly elevated by the presence of potassium ions. The presence of calcium ions and the

carrageenan concentration do not affect gelling temperature. Thus by following the method outlined in this text, κ -carrageenan/cell beads of fairly high melting temperature can be formed from a κ -carrageenan solution that will not gel spontaneously at room temperature. It should also be noted that even for a pure carrageenan solution, gelling and melting temperatures are not identical, the latter being about 20°C higher than the former. The gel structure will, however, eventually melt!

2. One problem that is often encountered is that caused by the presence of gelling cations, peptones, and so on, in the cell suspension. Mixing the cell suspension into the κ -carrageenan solution may therefore cause it to gel, unless it is kept well above its native gelling temperature.
3. Some workers (6) have proposed the use of polyethyleneimine or glutaraldehyde to increase the stability of the gel structure. Although this may have some utility with immobilized dead cells, however, these reagents are probably too toxic to be of much use with viable cells.
4. Although not subject to disruption by metal ion chelators as are alginate gels, κ -carrageenan gels are less strong and more porous and do break down in use.
5. The more porous open structure of the κ -carrageenan gel has two consequences: cell leakage can be a major problem; although higher plant and animal cells may be retained, algal and microbial cells are not well retained; the gel is rather squashy and spherical beads are difficult to form. Penetration of the gel structure by very high molecular weight compounds is, however, less of a problem than with alginate gels.

6. Compared to alginate, κ -carrageenan is expensive.
7. The easy dissolution of calcium alginate gels by the addition of phosphate or citrate may be a useful attribute if cell growth within the bead is to be measured directly (i.e., by counting cells) rather than indirectly (e.g., by pigment extraction). Purposeful dissolution of κ -carrageenan gels is more troublesome.

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